

# Stochastic Integral Reinforcement Learning for Distributed $H_\infty$ Optimal Formation Tracking of Multi-Agent Wheeled Mobile Robots

Khanh Thai<sup>1,\*</sup>, Huy Dao<sup>1,\*</sup>, Nga Thi-Thuy Vu<sup>1</sup>

<sup>1</sup>HUST, Vietnam.

<https://researchseerhust.github.io/SIRL/>

**Abstract:** In this paper a distributed stochastic  $H_\infty$  optimal tracking control problem for multi-agent wheeled mobile robot systems is proposed, under both Itô process noise in the agent dynamics and worst-case external disturbances. By reformulating the control task as a stochastic zero-sum game problem, a decentralized framework is established wherein each agent interacts locally with its neighbors and is influenced by disturbances. The solution involves a set of coupled stochastic Hamilton–Jacobi–Isaacs equations, which gain a genuine Itô-correction term over their deterministic counterpart and reduce to them exactly as the noise intensity vanishes. To derive an approximate optimal policy, a critic-only neural network scheme is developed. A historical data-based update law is employed to eliminate the need for persistence of excitation, thereby enhancing online learning efficiency. Furthermore, the proposed approach does not require knowledge of the drift dynamics of the augmented system. Simulation results are presented to demonstrate the effectiveness and real-time capability of the proposed method.

**Keywords:** Wheel mobile robot, Multi-agent system, Stochastic control, Reinforcement learning, Adaptive dynamic programming, Graphical games

## 1 Introduction

In recent years, cooperative control of multi-agent systems has attracted substantial attention from both the scientific and engineering communities, primarily due to its wide range of applications in areas such as traffic management, power grids, resource distribution, wireless sensor networks, and robotic swarms. Among these, formation control represents a fundamental challenge, involving the coordination of multiple autonomous agents to maintain predefined geometric patterns during motion. This problem has gained increasing importance owing to several inherent difficulties, including system heterogeneity, reliance on locally available information, model uncertainties, and the complexity of dynamic and unstructured environments<sup>[1,2,3]</sup>.

A variety of formation tracking control strategies have been developed for multi-robot systems, including adaptive control<sup>[4,5,6]</sup>, sliding mode control (SMC)<sup>[7,8,9]</sup>, and model predictive control (MPC), among others. In<sup>[7]</sup>, a high-order SMC approach is proposed for two-wheeled mobile robots, where a hyperbolic tangent-based sliding surface is adopted to mitigate the effects of external disturbances and alleviate chattering. Additionally, a radial basis function neural network (RBFNN) is employed to approximate the uncertain system dynamics. A robust fast fixed-time formation control method is presented in<sup>[8]</sup> for autonomous underwater vehicles (AUVs), using a leader–follower framework. The control design integrates fixed-time SMC and a sliding mode observer (SMO) to ensure tracking performance in the presence of disturbances and actuator faults. Similarly, a leader–follower SMC strategy is applied to quadrotor unmanned aerial vehicles (UAVs) in<sup>[9]</sup>, wherein a distributed estimator is designed to compensate for intermittent communication between the leader and followers. Fractional-order SMC is then utilized to enable each follower to maintain the desired formation configuration.

Alongside SMC-based approaches, adaptive control schemes have also been widely adopted to address model uncertainties in formation control of multi-robot systems. In<sup>[4]</sup>, an adaptive formation

tracking method using quantized communication is proposed for networked nonholonomic mobile robots, where unknown slippage constraints are handled through an adaptive estimator. Discontinuous control signals are generated via quantized state feedback and relative positioning. In [5], a finite-time adaptive formation control framework with event-triggered communication is designed, where the unknown nonlinear dynamics of follower robots are approximated by fuzzy logic systems. Adaptive control laws are then developed to ensure finite-time convergence to the leader’s trajectory. Furthermore, [6] presents an adaptive control architecture using the unscented Kalman filter (UKF) to estimate the target’s motion. The estimated information is used in a backstepping-based leader–follower controller, resulting in a smoother control input and improved tracking accuracy.

In recent years, reinforcement learning (RL) has considered as a powerful framework for solving nonlinear optimal control problems. Within the context of multi-agent system (MAS) control, RL techniques have been actively investigated to enhance system-wide performance. For example, game-theoretic backstepping controllers have been analyzed for nonlinear MAS in [10], where each performance index function incorporates interactions between pairs of agents. The game-based designs demonstrated superior convergence rates compared to conventional optimal controllers. To address the optimal consensus problem in discrete-time MAS, a model-free Q-learning approach is proposed in [11], leveraging system trajectory data instead of requiring accurate system models. A data-driven RL framework is further introduced in [12] to achieve optimal robust synchronization among heterogeneous agents. In [13], an off-policy integral reinforcement learning method is proposed to resolve the optimal synchronization task without needing prior knowledge of agent dynamics. To manage optimal synchronization in MAS with actuator constraints, [14,15] enhance the actor–critic neural network paradigm by incorporating a vanishing viscosity mechanism. The  $H_\infty$ /zero-sum-game route to disturbance-robust optimal control that this paper follows traces back to the classical dynamic-game formulation of [16], and the specific critic-only, synchronous-policy-iteration mechanism this paper’s Section 4 builds on was first developed for deterministic nonlinear zero-sum games in [17, 18].

Collectively, these studies emphasize optimal coordination strategies in MAS using RL techniques. However, most existing RL algorithms rely on either policy iteration or value iteration, and, with few exceptions, are posed for *deterministic* agent dynamics – leaving the practically important case of agents subject to genuine process noise (actuator ripple, unmodeled dynamics, terrain-induced excitation), rather than only bounded worst-case disturbance inputs, comparatively underexplored. While policy iteration typically necessitates an initially admissible policy, value iteration is known for its slower convergence. Furthermore, the dependency on the persistence of excitation (PE) condition during the early learning phase remains a significant barrier to the practical deployment of RL algorithms in real-time applications. Despite these advancements, challenges persist in achieving decentralized optimal tracking control for multi-agent systems under both stochastic process noise and worst-case external disturbances, without full knowledge of system dynamics. To address these issues, this paper introduces a distributed stochastic  $H_\infty$  optimal tracking control framework for multi-agent wheeled mobile robot systems, in which each agent’s dynamics are modeled as an Itô stochastic differential equation rather than an ordinary differential equation, in the spirit of the stochastic Hamilton–Jacobi–Bellman/Isaacs theory of [19] and the linear stochastic zero-sum-game solution of [20] that the recent model-free policy-iteration algorithms of [21, 22] build on. The latter is the closest antecedent of the present work on the algorithmic side: it solves the model-free stochastic  $H_\infty$  problem for a *single* linear Itô agent with state-dependent multiplicative noise by off-policy policy iteration on a generalized algebraic Riccati equation, and analyses the robustness of that iteration to errors incurred during learning. Unlike that linear, quadratic-cost setting – where the value function admits a closed-form (Riccati-type) solution – the nonholonomic wheeled-robot dynamics considered here are nonlinear, so no such closed form exists even after adding process noise; this is precisely why a critic-only neural-network solution to the resulting coupled stochastic Hamilton–Jacobi–Isaacs (HJI) equations remains necessary, and is the route this paper takes. By reformulating the control task as a stochastic zero-sum game and employing a critic-only neural network structure, the proposed approach effectively eliminates the need for persistence of excitation and prior knowledge of drift dynamics, while explicitly accounting for the diffusion term that

a purely deterministic formulation omits; an off-policy reformulation then removes the remaining dependence on the input and diffusion channels, yielding a completely model-free algorithm. This contribution aims to improve learning efficiency and real-time applicability in complex and uncertain environments.

The primary contributions of this study are summarized as follows

- The distributed  $H_\infty$  optimal formation tracking problem for multi-agent nonholonomic wheeled mobile robots is formulated under *stochastic*, rather than deterministic, agent dynamics: each agent’s strict-feedback dynamic subsystem is modeled as an Itô SDE with a Lipschitz, linear-growth diffusion term. The resulting coupled Hamilton–Jacobi–Isaacs (HJI) equations gain a genuine second-order (Itô-correction) term absent from the deterministic formulation, which we show reduces to the deterministic HJI equations exactly as the diffusion intensity vanishes.
- A reinforcement learning framework based on a critic-only neural network (NN) is introduced to approximate the solution of the coupled stochastic HJI equations. By integrating experience replay into an integral reinforcement learning process re-derived via Itô’s formula and conditional expectation, the proposed method reduces dependence on the conventional PE condition while preserving system stability during learning. This approach enables the design of a novel adaptive law for updating NN parameters under process noise.
- The scheme is then made *completely Smodel-free* through an off-policy reformulation of the stochastic IRL Bellman equation. By reading the stationarity conditions backwards, the unknown products  $\nabla V_i^T g_i$  and  $\nabla V_i^T k_i$  are lumped into the agent’s own policies weighted by the design matrices  $R_{ii}, T_{ii}$ , while the agent-coupled cross-channel terms  $\nabla V_i^T g_j$ , which admit no such identity in the graphical-game setting, are identified from data alongside the critic weights. The resulting algorithm requires no knowledge of the drift, the input channels, or the diffusion, and – because the diffusion, state-dependent though it is, does not depend on the control or disturbance channels – needs no diffusion-dependent parameter to be identified. We make precise that it is control-independence, and not additivity, that buys this: the same cancellation holds for the state-dependent multiplicative noise of [22], and fails only for the control-dependent diffusion of [21], which forces an extra lumped unknown. The extension amounts to a change of regressor rather than a change of algorithm, so the stability guarantee carries over unchanged.
- The mean-square stability of the resulting closed-loop system is rigorously established via a stochastic Lyapunov argument: the tracking and weight-estimation errors are uniformly ultimately bounded in the mean-square sense, with an explicit noise-dependent ultimate bound that reduces to its deterministic counterpart as the diffusion intensity vanishes. The learned policies correspondingly approach the Nash equilibrium of the graphical game to within a residual set, whose size is governed by the critic approximation error and therefore shrinks only as the basis is refined, not as the noise vanishes.
- Finally, the guarantee is shown to survive the fact that no implementation evaluates the learning quantities exactly. The idealized recursion is separated from the executed one, the difference – sampling error in the conditional expectation, quadrature error on Itô paths, identification error in the auxiliary regression, and communication error – is aggregated into a single exogenous signal, and the ultimate bound is shown to inflate by a class- $\mathcal{K}$  function of that signal, while the tracking-error bound is unaffected to first order and the nominal guarantee is recovered exactly as the error vanishes. This is the continuous-time, single-trajectory analogue of the input-to-state stability analysis of inexact policy iteration in [22], and, unlike that result, it requires no proximity of the initialization to the optimal solution.

The rest of the paper is organized as follows. Section 2 provides the required preliminaries and formulates the stochastic formation tracking problem. Section 3 reformulates the problem as a stochastic zero-sum differential graphical game and derives the coupled stochastic HJI equations. Section 4 presents the critic-only online integral reinforcement learning algorithm and its mean-square stability analysis. Section 5 illustrates the performance of the proposed algorithm through simulation. Conclusions and directions for future research are given in Section 6.

## 2 Preliminaries and problem formulation

**Notations:** In this paper, the notations  $\mathbb{R}$ ,  $\mathbb{R}^n$ , and  $\mathbb{R}^{m \times n}$  denote the set of real numbers, the  $n$ -dimensional Euclidean space, and the set of real-valued  $m \times n$  matrices, respectively. The symbol  $\mathbf{I}_n$  represents the  $n$ -dimensional identity matrix, while  $\mathbf{1}_N$  and  $\mathbf{0}$  denote the  $N$ -dimensional column vector of ones and the zero vector of appropriate dimension, respectively. The notation  $\|\cdot\|$  stands for the standard Euclidean norm when applied to vectors, and the induced matrix norm when applied to matrices. The  $\mathcal{L}_2$ -norm of a signal  $\beta(t) \in \mathcal{L}_2^n[0, \infty)$  is defined as  $\|\beta(t)\|_{\mathcal{L}_2} = \left( \int_0^\infty |\beta(t)|^2 d\tau \right)^{\frac{1}{2}} < +\infty$ ,  $\alpha(t) \in \mathbb{R}^n$ . The Kronecker product is denoted by  $\otimes$ . The notation  $\text{diag}\{x_1, x_2, \dots, x_n\}$  refers to a diagonal matrix whose diagonal entries are  $x_1, x_2, \dots, x_n$ . The symbol  $\lambda_{\min}(M)$  indicates the smallest eigenvalue of matrix  $M$ .

### 2.1 Graph theory

For a MAS consisting of  $N$  follower agents, the underlying communication topology can be modeled as a directed graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V} = \{v_1, \dots, v_N\}$  denotes the set of nodes (agents), and  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$  represents the set of directed edges, with  $|\mathcal{E}| = m$  being the total number of edges. The adjacency matrix associated with  $\mathcal{G}$  is defined as  $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ , where  $a_{ij} > 0$  if there exists a directed edge from node  $v_j$  to node  $v_i$ , i.e.,  $(v_j, v_i) \in \mathcal{E}$ , and  $a_{ij} = 0$  otherwise. Notably, self-loops are excluded, i.e.,  $a_{ii} = 0$  for all  $i$ . The neighbor set of agent  $i$  is denoted by  $\mathcal{N}_i = \{v_j \mid (v_j, v_i) \in \mathcal{E}\}$ , containing all agents that send information to agent  $i$ . The in-degree matrix is denoted as  $\mathcal{D} = \text{diag}(d_i)$ , where  $d_i = \sum_{j \in \mathcal{N}_i} a_{ij}$  represents the weighted in-degree of node  $v_i$ . The Laplacian matrix is then defined as  $\mathcal{L} = \mathcal{D} - \mathcal{A}$ , which satisfies the property that the sum of each row is zero. If the  $i$ -th follower agent receives information directly from the leader node, then  $a_{i0} > 0$ ; otherwise,  $a_{i0} = 0$ . It is assumed that at least one follower satisfies  $a_{i0} > 0$ . The entries  $b_{ij}$  correspond to the diagonal matrix  $\mathcal{B}$ , where each diagonal element is given by  $b_{ii} = a_{i0}$ . A directed path is a finite sequence of edges of the form  $(v_1, v_2), (v_2, v_3), \dots, (v_{k-1}, v_k)$ , where  $v_i \in \mathcal{V}$  for all  $i$ . The graph  $\mathcal{G}$  is said to be strongly connected if, for any pair of distinct nodes  $v_i, v_j \in \mathcal{V}$  with  $i \neq j$ , there exists a directed path from  $v_i$  to  $v_j$ . In this study, we consider a directed graph with a fixed and strongly connected topology.

### 2.2 Mathematical model of a nonholonomic wheeled mobile robot

Consider a multi-robot system comprising  $n$  nonholonomic wheeled mobile robots (NWMRs), each represented by a unicycle-type model with two actuated wheels and a passive caster. Let  $r_i$  and  $l_i$  denote the wheel radius and half the wheelbase of robot  $i$ , respectively. The mass center is assumed to coincide with the geometric center, located midway between the wheels in the global frame  $xOy$ . The configuration of robot  $i$  is described by the state vector  $(x_i, y_i, \theta_i)$ , where  $\theta_i$  denotes the heading angle. Following the standard Euler–Lagrange derivation for constrained mechanical systems [23], the dynamic model is given by

$$M_i(p_i)\ddot{p}_i + C_i(p_i, \dot{p}_i)\dot{p}_i + F_i(\dot{p}_i) + G_i(p_i) + \tau_{pi}(t) = B_i(p_i)\tau_i - A_i^T(p_i)\lambda_i, \quad i = 1, \dots, N, \quad (1)$$

where  $p_i \triangleq [x_i, y_i, \theta_i]^T \in \mathbb{R}^3$  denotes the position of the robot  $i$  in the earth frame,  $M_i : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$  denotes a symmetric positive-definite inertia matrix,  $C_i : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$  denotes the bounded centripetal and Coriolis matrix. It is noted that the Coriolis and centrifugal term vanishes, i.e.  $C_i(p_i, \dot{p}_i) = 0$ , when the mass center coincides with the geometric centroid of the robot.  $F_i : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  denotes surface friction forces,  $\tau_{pi} : [t_0, \infty) \rightarrow \mathbb{R}^3$  denotes bounded unknown disturbances including unstructured unmodeled dynamics,  $G_i : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is the gravitational vector,  $B_i : \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 2}$  denotes the input transformation matrix,  $\tau_i \triangleq [\tau_{i,\text{right}}, \tau_{i,\text{left}}]^T \in \mathbb{R}^2$  is the control vector with the actuation torques applied on right and left wheels, respectively,  $A_i^T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is the matrix associated with the constraints,  $\lambda_i \in \mathbb{R}$  is the Lagrange multiplier associated with the nonholonomic constraint. Moreover, the robot is assumed to move without lateral slip, thereby satisfying the nonholonomic constraint.

$$\dot{x}_i \sin \theta_i - \dot{y}_i \cos \theta_i = 0, \quad \forall i \in \mathbb{N}. \quad (2)$$

Wheeled mobile robots are renowned examples of nonholonomic mechanical systems. These robots adhere to the nonholonomic constraint 2, which dictates movement exclusively along directions normal to the axes of their driving wheels. This adherence ensures pure rolling and nonslipping motion. Consequently, the kinematic model of a mobile robot  $i$  can be expressed as:

$$\dot{p}_i = H_i(p_i)\eta_i, \quad (3)$$

where the vector  $\eta_i \in \mathbb{R}^2$  is composed of linear and angular velocities defined as  $\eta_i \triangleq [v_i, \omega_i]^T$ , and  $H_i(p_i) \in \mathbb{R}^{3 \times 2}$  is the full-rank velocity transformation matrix. Furthermore, the nonholonomic constraint, which can be written as  $A(p_i)\dot{p}_i = 0$ , can be associated with (3) to achieve that  $H^T(p_i)A^T(p_i) = 0$ . By applying Lagrange's formulation, we can obtain the determination of the system matrices in Eq. (1) and (2). We have  $M_i = \text{diag}(m_i, m_i, I_i)$ ,  $\lambda_i = -m_i(\dot{x}_i \cos \theta_i + \dot{y}_i \sin \theta_i)\dot{\theta}_i$ , where  $m_i$  denotes mass,  $I_i$  is the moment of inertia,  $2l_i$  represents the robot width, and  $R_i$  is the wheel radius, and other matrices as follows:

$$B_i(p_i) = \frac{1}{l_i} \begin{bmatrix} \cos \theta_i & \sin \theta_i \\ \cos \theta_i & \sin \theta_i \\ -R_i & -R_i \end{bmatrix}, \quad A_i^T(p_i) = \begin{bmatrix} -\sin \theta_i \\ \cos \theta_i \\ 0 \end{bmatrix}, \quad H_i(p_i) \triangleq \begin{bmatrix} \cos \theta_i & 0 \\ \sin \theta_i & 0 \\ 0 & 1 \end{bmatrix}.$$

To facilitate the subsequent development, the dynamic equations can be expressed in terms of the internal velocity  $\eta_i$  using (3), (1) and the fact  $H_i^T(p_i)A_i^T(p_i) = 0$  as follows:

$$\bar{M}_i(p_i)\dot{\eta}_i + \bar{C}_i(p_i, \dot{p}_i)\eta_i + \bar{F}_i(\dot{p}_i) = \bar{B}_i(p_i)\tau_i - \bar{\tau}_{di}, \quad (4)$$

where  $\bar{M}_i(p_i) \triangleq H_i^T(p_i)M_i(p_i)H_i(p_i)$ ,  $\bar{C}_i(p_i, \dot{p}_i) \triangleq \mathbf{0}$ ,  $\bar{F}_i(\dot{p}_i) \triangleq H_i^T(p_i)F_i(\dot{p}_i)$ ,  $\bar{B}_i(p_i) \triangleq H_i^T(p_i)B_i(p_i)$  and  $\bar{\tau}_{di} \triangleq H_i^T(p_i)\tau_{di}$ . All stochastic quantities in this paper are defined on a complete filtered probability space  $(\Omega, \mathcal{F}, \mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$  satisfying the usual conditions, on which  $N$  mutually independent standard scalar Brownian motions  $W_1(t), \dots, W_N(t)$  are defined, one per agent. We write  $\mathbb{E}[\cdot]$  and  $\mathbb{E}^{\mathcal{F}_t}[\cdot] \triangleq \mathbb{E}[\cdot | \mathcal{F}_t]$  for the expectation and the conditional expectation given  $\mathcal{F}_t$ , respectively.

Deriving from (3) and (4), a strict-feedback representation is adopted for agent  $i$  kinematic and dynamic behavior under nonholonomic constraints, with consideration of the uncertainty term  $f_{\eta_i}$  and disturbance  $\tau_{di}$ . Beyond this drift-level uncertainty, we account for unmodeled process noise entering the dynamic subsystem – e.g., wheel-torque ripple, unmodeled actuator dynamics, or terrain-induced excitation – as a diffusion term driven by  $W_i(t)$ , so that the strict-feedback representation is the following Itô stochastic differential equation (SDE):

$$\begin{aligned} dp_i &= g_{pi}(p_i)\eta_i dt \\ d\eta_i &= [f_{\eta_i}(p_i, \eta_i) + g_{\eta_i}(p_i, \eta_i)\tau_i + k_{\eta_i}(p_i, \eta_i)\tau_{di}]dt + \sigma_{\eta_i}(p_i, \eta_i) dW_i(t) \end{aligned} \quad (5)$$

where  $\sigma_{\eta_i} : \mathbb{R}^3 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is the diffusion coefficient of the dynamic subsystem. The kinematic subsystem is left noise-free:  $p_i$  is fully determined by the (noise-corrupted) velocity  $\eta_i$  through the exactly known kinematic relation (3), so any process noise entering the robot's motion does so through  $\eta_i$ , not directly through  $p_i$ .

**Assumption 1.**  $f_{\eta_i}(\cdot, \cdot)$ ,  $g_{\eta_i}(\cdot, \cdot)$ ,  $k_{\eta_i}(\cdot, \cdot)$ , and  $\sigma_{\eta_i}(\cdot, \cdot)$  are globally Lipschitz and of at most linear growth in  $(p_i, \eta_i)$ , uniformly in  $t$ . Consequently, for every admissible control  $\tau_i \in \mathcal{L}_{\mathbb{F}}^2(0, \infty; \mathbb{R}^2)$  and every  $\mathcal{F}_0$ -measurable initial condition with finite second moment, the SDE (5) admits a unique,  $\mathbb{F}$ -adapted strong solution.

Similarly, the geometric centroid  $(x_0, y_0)$  along with the desired heading  $\theta_0$  define the pose of a virtual leader (agent 0), which is assumed to produce a smooth and bounded time-varying trajectory  $q_0 = [x_0, y_0, \theta_0]^T \in \mathbb{R}^3$ , which satisfies  $\dot{x}_0 = v_0 \cos(\theta_0)$ ,  $\dot{y}_0 = v_0 \sin(\theta_0)$  and  $\dot{\theta}_0 = \omega_0$ , where  $v_0 \in \mathbb{R}$  and  $\omega_0 \in \mathbb{R}$  denote desired linear velocity and angular velocity, respectively.

**Assumption 2.** The  $\theta_i$  for  $0 \leq i \leq N$  is bounded,  $\omega_i$  for  $0 \leq i \leq N$  is persistently exciting, and  $|\omega_i| \leq \omega_{\max}$ .

**Remark 1.** The angular velocity  $\omega_i$  is assumed to be persistently exciting, implying that it does not converge to zero. This assumption arises from the nonholonomic nature of the wheeled mobile robot system. It should not be confused with the persistence-of-excitation condition that the learning scheme of Section 4 dispenses with, and the two are neither substitutes nor consequences of one another. The condition assumed here is a property of the plant trajectory: a nonholonomic vehicle whose angular velocity vanishes loses the directions along which the lateral error is controllable, so the tracking problem itself degenerates, independently of how the policy is computed. What the learning scheme removes is the separate requirement that the critic regressor be persistently exciting along the trajectory, which is what conventional adaptive-critic designs need for the weight estimates to converge; that requirement is replaced by Condition 1, a rank test on a finite stack of recorded samples, verifiable offline at any instant rather than asserted over an infinite horizon.

**Assumption 3.** At least one agent in the formation is capable of receiving information from the virtual leader to ensure effective coordination.

### 2.3 Problem formulation

The dynamic model of each NWMR is regarded as a strict-feedback nonlinear stochastic system subject to both external disturbances and process noise, formulated as the Itô SDE (5). The target trajectory is generated by the dynamics of an uncontrolled virtual leader, represented as  $\dot{p}_0 = g_p(p_0) \eta_0$ , where  $\eta_0 \triangleq [v_0, \omega_0]^T \in \mathbb{R}^2$  denotes the desired velocity. The local neighborhood kinematic and dynamic tracking error  $\delta_{pi} \in \mathbb{R}^3$  and  $\delta_{\eta i} \in \mathbb{R}^3$ , for all  $i \in \mathcal{V}$ , respectively, are defined as

$$\begin{aligned}\delta_{pi} &\triangleq \sum_{j \in \mathcal{N}_i} a_{ij}(p_i + \Delta_i - p_j - \Delta_j) + b_i(p_i + \Delta_i - p_0), \\ \delta_{\eta i} &\triangleq \sum_{j \in \mathcal{N}_i} a_{ij}(\eta_i - \eta_j) + b_i(\eta_i - \eta_0),\end{aligned}\tag{6}$$

where  $\Delta_i \triangleq [d_{i,x}, d_{i,y}]^T \in \mathbb{R}^2$  specifies the target offset between agent  $i$  and the leader to maintain the desired formation. Since  $\delta_{pi}$  is a linear combination of the (noise-free) positions  $p_i$ , its temporal derivative retains the deterministic form  $\dot{\delta}_{pi} = (b_i + d_i)g_{pi}\eta_i - \sum_{j \in \mathcal{N}_i} a_{ij}g_{pj}\eta_j - b_i g_p(p_0)\eta_0$ . Let  $\dot{\delta}_p \triangleq [\dot{\delta}_{p1}^T, \dots, \dot{\delta}_{pn}^T]^T$ ,  $\dot{p}_0 \triangleq (\bar{1}_3 \otimes I_3)\dot{p}_0$ ,  $p \triangleq [p_1^T, \dots, p_n^T]^T$ ,  $g_p \triangleq \text{diag}(g_{p1}, \dots, g_{pn})$ , and  $\eta \triangleq [\eta_1^T, \dots, \eta_n^T]^T$ , the collective kinematic tracking error vector of the entire multi-agent system is expressed as

$$\dot{\delta}_p = -(\mathcal{B} \otimes I_3)\dot{p}_0 + ((\mathcal{L} + \mathcal{B}) \otimes I_3)g_p\eta.\tag{7}$$

In contrast,  $\delta_{\eta i}$  is a linear combination of the noise-corrupted velocities  $\eta_i, \eta_j$ , so its stochastic differential inherits a diffusion term from (5):  $d\delta_{\eta i} = [f_i(t) + (b_i + d_i)(g_{\eta i}\tau_i + k_{\eta i}\tau_{di}) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_{\eta j}\tau_j + k_{\eta j}\tau_{dj})]dt + (b_i + d_i)\sigma_{\eta i}(p_i, \eta_i)dW_i(t) - \sum_{j \in \mathcal{N}_i} a_{ij}\sigma_{\eta j}(p_j, \eta_j)dW_j(t)$ , where  $f_i(t) \triangleq \sum_{j \in \mathcal{N}_i} a_{ij}(f_{\eta i} - f_{\eta j}) + b_i(f_{\eta i} - f_{\eta_0})$ . Accordingly, let  $\dot{\delta}_\eta \triangleq [\dot{\delta}_{\eta 1}^T, \dots, \dot{\delta}_{\eta n}^T]^T$ ,  $f(t) \triangleq [f_1^T, \dots, f_n^T]^T$ ,  $g_\eta \triangleq \text{diag}(g_{\eta 1}, \dots, g_{\eta n})$ ,  $k_\eta \triangleq \text{diag}(k_{\eta 1}, \dots, k_{\eta n})$ ,  $\sigma_\eta \triangleq \text{diag}(\sigma_{\eta 1}, \dots, \sigma_{\eta n})$ ,  $\tau \triangleq [\tau_1^T, \dots, \tau_n^T]^T$ ,  $\tau_d \triangleq [\tau_{d1}^T, \dots, \tau_{dn}^T]^T$ , and  $W(t) \triangleq [W_1(t), \dots, W_n(t)]^T$ , the collective dynamic tracking error of all agents now can be expressed as the SDE

$$d\delta_\eta = [f(t) + ((\mathcal{L} + \mathcal{B}) \otimes I_2)(g_\eta\tau + k_\eta\tau_d)]dt + ((\mathcal{L} + \mathcal{B}) \otimes I_2)\sigma_\eta(x)dW(t).\tag{8}$$

To design distributed optimal control for both dynamics and kinematics, the backstepping approach will be used. Let us define  $\eta_a \triangleq [\eta_{a1}^T, \dots, \eta_{an}^T]^T$  and  $\tau_a \triangleq [\tau_{a1}^T \dots \tau_{an}^T]^T$  be the feed-forward control inputs designed as

$$\begin{aligned}\eta_a &\triangleq (((\mathcal{L} + \mathcal{B}) \otimes I_3)g_p)^{-1}((\mathcal{B} \otimes I_3)\dot{p}_0 + ((\mathcal{L} + \mathcal{B}) \otimes I_3)g_p e_\eta), \\ \tau_a &\triangleq (((\mathcal{L} + \mathcal{B}) \otimes I_2)g_\eta)^{-1}(-((\mathcal{L} + \mathcal{B}) \otimes I_2)g_p^T e_p).\end{aligned}\tag{9}$$

Let  $u_a \triangleq [\eta_a^T, \tau_a^T]^T$  denote auxiliary inputs, and  $u^* \triangleq [\eta^{*T}, \tau^{*T}]^T$  denote the distributed optimal feedback control inputs, in which  $\eta^*$  and  $\tau^*$  will be developed in the subsequent sections.

We now define the actual input  $u \triangleq [\eta^T, \tau^T]^T$  as  $u \triangleq u^* + u_a$ . Let  $\delta \triangleq [\delta_1^T, \dots, \delta_n^T]^T$ , where  $\delta_i \triangleq [\delta_{pi}^T, \delta_{\eta i}^T]^T$ ,  $x \triangleq [x_1^T, \dots, x_n^T]^T$ , where  $x_i \triangleq [p_i^T, \eta_i^T]^T$ ,  $\vartheta \triangleq [\vartheta_1^T, \dots, \vartheta_n^T]^T$ , where  $\vartheta_i \triangleq [0, \tau_{di}^T]^T$ ,  $f_e(t) \triangleq [f_{e1}^T, \dots, f_{en}^T]^T$ , where  $f_{ei}(t) \triangleq [0, f_i^T]^T$ ,  $k(x) \triangleq \text{diag}(k_1, \dots, k_n)$ , where  $k_i(x) \triangleq \text{diag}(0, k_{\eta i})$ , and  $g(x) \triangleq \text{diag}(g_1, \dots, g_n)$ , where  $g_i(x) = \text{diag}(g_{pi}, g_{\eta i})$ , and  $\sigma(x) \triangleq \text{diag}(\sigma_1, \dots, \sigma_n)$ , where  $\sigma_i(x_i) \triangleq \text{diag}(0, \sigma_{\eta i})$  places the diffusion coefficient in the same (dynamic) channel as  $k_i(x)$  and  $g_i(x)$ , we now can obtain the SDE governing the variable  $\delta$  as follows:

$$d\delta = \left[ f_e(t) + ((\mathcal{L} + \mathcal{B}) \otimes I_5)(g(x)u^* + k(x)\vartheta) \right] dt + ((\mathcal{L} + \mathcal{B}) \otimes I_5)\sigma(x) dW(t). \quad (10)$$

Let us consider the tracking error dynamics of agent  $i$ ,  $i = 1, \dots, N$  which can be obtained from equation (10):

$$\begin{aligned} d\delta_i = & \left[ f_{ei}(t) + (b_i + d_i)(g_i(x_i)u_i + k_i(x_i)\vartheta_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j(x_j)u_j + k_j(x_j)\vartheta_j) \right] dt \\ & + (b_i + d_i)\sigma_i(x_i) dW_i(t) - \sum_{j \in \mathcal{N}_i} a_{ij}\sigma_j(x_j) dW_j(t) \end{aligned} \quad (11)$$

For later use in deriving the stochastic HJI equation (Section 3), it is convenient to collect the  $1 + |\mathcal{N}_i|$  mutually independent Brownian motions driving (11) into the vector  $\mathcal{W}_i(t) \triangleq [W_i(t), \{W_j(t)\}_{j \in \mathcal{N}_i}]^T$  and the corresponding diffusion coefficient into the matrix

$$\Sigma_i(x) \triangleq \left[ (b_i + d_i)\sigma_i(x_i), \{-a_{ij}\sigma_j(x_j)\}_{j \in \mathcal{N}_i} \right], \quad (12)$$

so that (11) is written compactly as  $d\delta_i = F_i(x, u, \vartheta) dt + \Sigma_i(x) d\mathcal{W}_i(t)$ , with  $F_i(x, u, \vartheta)$  denoting the drift term in the brackets of (11).

**Lemma 1.** *The distributed  $H_\infty$  optimal formation tracking problem, in the mean-square sense, for multi-agent systems governed by the stochastic dynamics in equation (5) is equivalent to solving the distributed  $H_\infty$  optimal control problem for the system given in equation (10).*

### 3 Bounded $L_2$ -gain optimal synchronization problem

Suppose the output measurements are given by  $y_i = h_i(e_i) \in \mathbb{R}^{n_i}$ , where  $h_i(e_i)$  is a continuously differentiable and left-invertible mapping. This ensures that  $h_i(e_i)$  is directly accessible. Under nonzero disturbances, i.e.,  $\vartheta_i \neq 0$  and  $\vartheta_j \neq 0$ , we define the generalized disturbance vector as  $\varpi_i \triangleq [\vartheta_i^T \quad \vartheta_{-i}^T]^T$ , where  $\vartheta_{-i} \triangleq \{\vartheta_j, j \in \mathcal{N}_i\}$ . The performance output vector is defined as  $\varphi_i \triangleq [\delta_i^T(t) \quad u_i^T(t) \quad u_{-i}^T(t)]^T$ , where  $u_{-i} \triangleq \{u_j, j \in \mathcal{N}_i\}$ . Since  $\delta_i$  now evolves according to the Itô SDE (11), a pathwise integral cost is no longer well-defined as a deterministic quantity; the local performance indices are accordingly defined as an expectation over sample paths,

$$J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) \triangleq \mathbb{E} \left[ \int_0^\infty r_i(\delta_i(\rho), u_i(\rho), u_{-i}(\rho), \vartheta_i(\rho), \vartheta_{-i}(\rho)) d\rho \right], \quad (13)$$

where  $r_i : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$  denotes the instantaneous cost defined as  $r_i \triangleq Q_{ii}(e_i) + u_i^T R_{ii} u_i + \sum_{j \in \mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii} \vartheta_i - \gamma^2 \sum_{j \in \mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j$ , with  $Q_{ii} : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$  being continuously differentiable function satisfying  $\underline{q}\|x\|^2 \leq Q_{ii}(x) \leq \bar{q}\|x\|^2$  for  $\underline{q}, \bar{q} \in \mathbb{R}_{>0}$ ,  $R_{ii}$  and  $R_{ij}$  are positive definite matrices, and  $\gamma$  represents the amount of attenuation from the disturbance input  $\vartheta(t)$  to the defined performance output variable  $\varphi(t)$ .

**Definition 1.** *The multi-agent control protocols  $u_i$ ,  $i = 1, \dots, N$ , for (5) is said to be admissible on the set  $\Omega$  denoted by  $u_i \in \Psi(\Omega)$ , if  $u_i$  is continuous with  $u_i(0) = 0$ ,  $u_i$  stabilizes (5) on  $\Omega$  in the mean-square sense (i.e.,  $\mathbb{E}\|\delta_i(t)\|^2 \rightarrow 0$  as  $t \rightarrow \infty$  for every  $\mathcal{F}_0$ -measurable  $\delta_i(0)$  with finite second moment), and  $J_i(e_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$  is finite.*

**Definition 2.** (Bounded  $L_2$ -Gain or Disturbance Attenuation)[[15]], [[24]], [[25]] *The objective is to identify admissible control inputs that bound the local errors in (7) and (8) by  $L_2$ -bounded in the mean-square sense, minimize the cost functional in (13), and meet the disturbance attenuation condition stated below:*

$$\mathbb{E} \left[ \int_0^T \|z_i\|^2 dt \right] \leq \gamma^2 \mathbb{E} \left[ \int_0^T \|\omega_i\|^2 dt \right] + \beta_i(e_i(0)), \quad (14)$$

where  $z_i \triangleq Q_{ii}(e_i) + u_i^T R_{ii} u_i + \sum_{j \in \mathcal{N}_i} u_j^T R_{ij} u_j$ , and  $\omega_i \triangleq \vartheta_i^T T_{ii} \vartheta_i + \sum_{j \in \mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j$ . This is the direct stochastic analogue of the mean-square stabilizability condition used to pose the linear stochastic  $H_\infty$  problem in [21], specialized to our nonlinear, agent-coupled setting.

**Remark 2.** The disturbance attenuation inequality given in (14) indicates that the effect of external disturbances on the system's expected output performance is constrained within a specified bound determined by the attenuation constant  $\gamma$ . The minimal value of  $\gamma$  that satisfies this condition is known as the optimal robust control gain. Nevertheless, due to the complexity of general nonlinear stochastic systems, there is no closed-form solution to compute this optimal value. As a result, a conservative (i.e., sufficiently large) value of  $\gamma$  is usually selected beforehand to ensure that the control design remains realizable.

According to [25], the previously described problem can be reformulated as the following zero-sum differential game  $J_i(\delta_i(0)) = \min_{u_i} \max_{\vartheta_i} J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$ , where  $J_i$  is defined in (13). In this setting, the control player minimizes the cost functional  $J_i$ , whereas the disturbance player maximizes it. The problem defines a two-player zero-sum differential game, admitting a unique solution if a game-theoretic saddle point  $(u_i^*, v_i^*)$  exists such that

$$\begin{aligned} J_i^*(\delta_i(0)) &= \min_{u_i} \max_{\vartheta_i} J_i(\delta_i(0), u_i, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*) \\ &= \max_{\vartheta_i} \min_{u_i} J_i(\delta_i(0), u_i, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*), \end{aligned} \quad (15)$$

where  $J_i^*$  denotes the Nash equilibrium of the zero-sum differential graphical game. The corresponding performance satisfies

$$\begin{aligned} J_i(u_i^*, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*) &\leq J_i(u_i^*, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*) \\ &\leq J_i(u_i, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*). \end{aligned} \quad (16)$$

The distributed local value function for agent  $i$  is defined as the conditional expectation

$$V_i(\delta_i(t), u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) = \mathbb{E}^{\mathcal{F}_t} \left[ \int_t^\infty r_i(\delta_i(\rho), u_i(\rho), u_{-i}(\rho), \vartheta_i(\rho), \vartheta_{-i}(\rho)) d\rho \right], \quad (17)$$

which is well-defined by the tower property since  $\delta_i$  is  $\mathbb{F}$ -adapted (Assumption 1). Because  $\delta_i$  now follows the Itô SDE (11) rather than an ordinary differential equation, Leibniz's rule no longer applies to differentiate  $V_i$  along trajectories; instead we apply Itô's formula to  $V_i(\delta_i(t))$  and invoke Dynkin's formula on the resulting martingale representation. This introduces the infinitesimal generator of (11),

$$\begin{aligned} \mathcal{A}_i V_i &\triangleq \nabla V_i^T \left( f_{ei} + (b_i + d_i)(g_i(x_i)u_i + k_i(x_i)\vartheta_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j(x_j)u_j + k_j(x_j)\vartheta_j) \right) \\ &\quad + \frac{1}{2} \text{tr} \left( \Sigma_i(x)^T \nabla^2 V_i(\delta_i) \Sigma_i(x) \right), \end{aligned} \quad (18)$$

where  $\nabla V_i \triangleq \frac{\partial V_i}{\partial \delta_i} \in \mathbb{R}^n$  is the gradient,  $\nabla^2 V_i \triangleq \frac{\partial^2 V_i}{\partial \delta_i \partial \delta_i^T} \in \mathbb{R}^{n \times n}$  is the Hessian of  $V_i$  with respect to  $\delta_i$ , and  $\Sigma_i(x)$  is the stacked diffusion coefficient of (11) defined in (12). The trace term  $\frac{1}{2} \text{tr}(\Sigma_i^T \nabla^2 V_i \Sigma_i)$  is the genuine stochastic (Itô) correction with no deterministic counterpart; setting  $\Sigma_i \equiv 0$  collapses  $\mathcal{A}_i$  back to the ordinary drift-only directional derivative used in the deterministic formulation. The Hamilton–Jacobi–Isaacs (HJI) equation, also known as the Bellman equation, is then

$$H_i(\delta_i, \nabla V_i, \nabla^2 V_i, u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) \triangleq \mathcal{A}_i V_i + r_i(\delta_i, u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) = 0. \quad (19)$$

Since  $\Sigma_i(x)$  does not depend on  $u_i$  or  $\vartheta_i$  under the process-noise model of (5) (the diffusion is state-dependent but control-independent; see Remark 3), the trace term in  $\mathcal{A}_i V_i$  is unaffected by

the stationary conditions below; the optimal control signals  $u_i^*$  and the optimal disturbance signals  $\vartheta_i^*$  are obtained exactly as in the deterministic case, from  $\frac{\partial H_i}{\partial u_i} = 0$  and  $\frac{\partial H_i}{\partial \vartheta_i} = 0$ , respectively, as follows

$$u_i^*(\delta_i) = -\frac{1}{2}(b_i + d_i)R_{ii}^{-1}g_i^T(x_i)\nabla V_i^*, \quad (20)$$

$$\vartheta_i^*(\delta_i) = \frac{1}{2\gamma^2}(b_i + d_i)T_{ii}^{-1}k_i^T(x_i)\nabla V_i^*. \quad (21)$$

Based on the substitution of the optimal control and disturbance strategies from (20) and (21) into (19), one can derive the associated coupled stochastic Hamilton–Jacobi–Isaacs (HJI) equations governing the optimal value functions.

$$\begin{aligned} Q_{ii}(\delta_i) + \nabla V_i F_i + \frac{1}{4}(b_i + d_i)^2 \nabla V_i^T g_i R_{ii}^{-1} g_i^T \nabla V_i \\ + \frac{1}{4} \sum_{j \in \mathcal{N}_i} (b_j + d_j)^2 \nabla V_j^T g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \nabla V_j \\ - \frac{1}{4\gamma^2} \sum_{j \in \mathcal{N}_i} (b_j + d_j)^2 \nabla V_j^T k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \nabla V_j \\ - \frac{1}{4\gamma^2} (b_i + d_i)^2 \nabla V_i^T k_i T_{ii}^{-1} k_i^T \nabla V_i \\ + \frac{1}{2} \text{tr} \left( \Sigma_i(x)^T \nabla^2 V_i(\delta_i) \Sigma_i(x) \right) = 0. \end{aligned} \quad (22)$$

where  $F_i \triangleq f_{ei}(t) - \frac{1}{2}(b_i + d_i)^2 g_i R_{ii}^{-1} g_i^T \nabla V_i + \frac{1}{2} \sum_{j \in \mathcal{N}_i} a_{ij}(b_j + d_j) g_j R_{jj}^{-1} g_j^T \nabla V_j + \frac{1}{2\gamma^2} (b_i + d_i)^2 k_i T_{ii}^{-1} k_i^T \nabla V_i - \frac{1}{2\gamma^2} \sum_{j \in \mathcal{N}_i} a_{ij}(b_j + d_j) k_j T_{jj}^{-1} k_j^T \nabla V_j$ , exactly as in the deterministic case, since the trace term does not involve  $u_i$  or  $\vartheta_i$  and therefore does not enter  $F_i$ . The added trace term is the nonlinear, agent-coupled analogue of the extra  $C^T P C$  and  $D^T P D$  diffusion terms that distinguish the Generalized Algebraic Riccati Equation (GARE) of [21] from the deterministic Riccati equation: there,  $V(x) = x^T P x$  and  $\Sigma(x) = Cx + Du$  are both quadratic/linear-in-state, so  $\frac{1}{2} \text{tr}(\Sigma^T \nabla^2 V \Sigma)$  reduces exactly to the algebraic terms  $x^T (C^T P C) x$  and  $u^T (D^T P D) u$  appearing in the GARE; here  $V_i$  and  $\Sigma_i$  are general nonlinear functions of  $\delta_i$ , so the correction survives only as a trace of a state-dependent Hessian rather than as a constant coefficient matrix, and admits no closed-form (algebraic) solution – which is precisely why the reinforcement-learning approach developed in the remainder of this paper is needed. The closed-loop system corresponds to agent  $i$ . For a given solution  $V_i$ , define  $u_i^* = u_i(V_i)$ ,  $u_{-i}^* = u_{-i}(V_i)$ ,  $\vartheta_i^* = \vartheta_i(V_i)$ , and  $\vartheta_{-i}^* = \vartheta_{-i}(V_i)$  in terms of  $V_i$ , then (22) can be written as

$$H_i(e_i, \nabla V_i, \nabla^2 V_i, u_i^*, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*) = 0, \quad (23)$$

with  $V_i(0) = 0$ . The optimal tracking control problem for multiple nonholonomic WMRs formulated as a stochastic zero-sum game leads to the challenge of solving a set of coupled stochastic HJI equations corresponding to the nonlinear dynamics in (11). The intrinsic nonlinearity of these systems renders an analytical solution intractable regardless of whether the dynamics are deterministic or stochastic, and the added trace term rules out the closed-form (GARE-based) route available in the linear stochastic case. Consequently, the remainder of this paper is devoted to addressing this issue via a reinforcement learning-based approach.

## 4 Online integral reinforcement learning algorithm for multi-agent cooperative games

### 4.1 IRL and value function approximation

Leveraging the IRL concept proposed in [ref], we aim to reformulate the Bellman equation in a way that bypasses explicit use of system dynamics. Since  $\delta_i$  is now an Itô process governed by

(11) and  $V_i$  is defined as the conditional expectation (17), the deterministic IRL identity no longer holds pathwise; instead, applying Itô's formula to  $V_i(\delta_i(\tau))$  on  $[t - T, t]$ , using  $\mathcal{A}_i V_i + r_i = 0$  (19), and taking the conditional expectation  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$  to annihilate the resulting local-martingale term (the standard Dynkin's-formula argument, mirroring [21, Eq. (ito\_2)]) gives, for any time  $t$  over a finite interval  $[t - T, t]$  of length  $T > 0$ ,

$$V_i(\delta_i(t - T)) = \mathbb{E}^{\mathcal{F}_{t-T}} \left[ \int_{t-T}^t \left( Q_{ii}(\delta_i) + u_i^T R_{ii} u_i + \sum_{j \in \mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii} \vartheta_i - \gamma^2 \sum_{j \in \mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j \right) d\tau + V_i(\delta_i(t)) \right]. \quad (24)$$

Notably, the IRL-based representation of the Bellman equation eliminates the explicit dependency on system dynamics, exactly as in the deterministic case; the only change is that (24) holds in conditional expectation rather than pathwise, since a single realized trajectory carries an extra zero-conditional-mean fluctuation contributed by the diffusion term of (11) that only cancels on average. To facilitate solving the value function  $V_i(\delta_i)$  in the IRL-based Bellman equation (24), we consider approximating it using function approximation techniques. Based on the Weierstrass higher-order approximation theorem [], for each  $i \in N$ , both the value function  $V_i$  and its gradient  $\nabla V_i$  can be approximated with arbitrary accuracy as follows:

$$V_i(\delta_i) = W_i^T \phi_i(\delta_i) + \epsilon_i \quad (25)$$

$$\nabla V_i(\delta_i) = \nabla \phi_i^T(\delta_i) W_i + \nabla \epsilon_i \quad (26)$$

Where  $\phi_i(\delta_i) \in \mathbb{R}^L$  is the critic NNs activation function vectors,  $W_i \in \mathbb{R}^L$  are the ideal weight parameter vectors, with  $L$  denotes the number of neurons in the critic NNs hidden layer.  $\epsilon_i(\delta_i)$  is the NN approximation error. The gradients are given by  $\nabla \phi_i(\delta_i) = \frac{\partial \phi_i}{\partial \delta_i}$  and  $\nabla \epsilon_i(\delta_i) = \frac{\partial \epsilon_i}{\partial \delta_i}$ . It is worth mentioning that  $\epsilon_i(\delta_i)$  converges to zero uniformly as  $L \rightarrow \infty$ .

**Assumption 4.**  $\phi_i(\delta_i)$ ,  $\nabla \phi_i(\delta_i)$ ,  $\nabla^2 \phi_i(\delta_i)$ ,  $\epsilon_i$ ,  $\nabla \epsilon_i$ ,  $\nabla^2 \epsilon_i$ , and the diffusion coefficient  $\Sigma_i(x)$  of (11) are bounded on the compact set  $\Omega$  such that  $\|\phi_i\| \leq b_{\phi_i}$ ,  $\|\nabla \phi_i\| \leq b_{\nabla \phi_i}$ ,  $\|\nabla^2 \phi_i\| \leq b_{\nabla^2 \phi_i}$ ,  $\|\epsilon_i\| \leq b_{\epsilon_i}$ ,  $\|\nabla \epsilon_i\| \leq b_{\nabla \epsilon_i}$ ,  $\|\nabla^2 \epsilon_i\| \leq b_{\nabla^2 \epsilon_i}$ , and  $\|\Sigma_i(x)\| \leq b_{\Sigma_i}$ .

Then, the Bellman equation (24) can be approximated at each step as

$$\begin{aligned} \mathbb{E}^{\mathcal{F}_{t-T}} \left[ \int_{t-T}^t \left( Q_{ii}(\delta_i) + u_i^T R_{ii} u_i + \sum_{j \in \mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii} \vartheta_i - \gamma^2 \sum_{j \in \mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j \right) d\tau \right] \\ = -W_i^T \mathbb{E}^{\mathcal{F}_{t-T}} [\Delta \phi_i(\delta_i(t))] + \bar{\epsilon}_{B_i} \end{aligned} \quad (27)$$

where  $\Delta \phi_i(\delta_i(t)) \triangleq \phi_i(\delta_i(t)) - \phi_i(\delta_i(t - T))$  and  $\bar{\epsilon}_{B_i} \triangleq -\mathbb{E}^{\mathcal{F}_{t-T}} [\epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t - T))]$ . Since  $\delta_i(\tau)$  is now an Itô process,  $\Delta \phi_i(\delta_i(t))$  and  $\epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t - T))$  must be expanded by Itô's formula rather than the ordinary chain rule: writing  $F_i(\tau) \triangleq f_{ei} + (b_i + d_i)(g_i u_i + k_i \vartheta_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j u_j + k_j \vartheta_j)$  for the drift of (11),

$$\begin{aligned} \Delta \phi_i(\delta_i(t)) &= \int_{t-T}^t \left[ \nabla \phi_i^T F_i + \frac{1}{2} \text{tr}(\Sigma_i^T \nabla^2 \phi_i \Sigma_i) \right] d\tau + \int_{t-T}^t \nabla \phi_i^T \Sigma_i d\mathcal{W}_i(\tau), \\ \epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t - T)) &= \int_{t-T}^t \left[ \nabla \epsilon_i^T F_i + \frac{1}{2} \text{tr}(\Sigma_i^T \nabla^2 \epsilon_i \Sigma_i) \right] d\tau + \int_{t-T}^t \nabla \epsilon_i^T \Sigma_i d\mathcal{W}_i(\tau), \end{aligned} \quad (28)$$

each with a drift-plus-Itô-correction integral (deterministic, given the trajectory) and a local-martingale term that is  $\mathcal{F}_{t-T}$ -conditionally mean-zero. Taking  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$  removes both martingale terms from (27) itself, leaving  $\bar{\epsilon}_{B_i} = - \int_{t-T}^t \mathbb{E}^{\mathcal{F}_{t-T}} \left[ \nabla \epsilon_i^T F_i + \frac{1}{2} \text{tr}(\Sigma_i^T \nabla^2 \epsilon_i \Sigma_i) \right] d\tau$ , which reduces exactly to the deterministic  $\epsilon_{B_i}$  of the noise-free case when  $\Sigma_i \equiv 0$ . In the online, single-trajectory implementation used by the adaptive law below, the realized (non-averaged) residual is used directly rather than its conditional expectation; write

$$\epsilon_{B_i}(t) \triangleq \bar{\epsilon}_{B_i} - \int_{t-T}^t \nabla \epsilon_i^T \Sigma_i d\mathcal{W}_i(\tau) \quad (29)$$

for the realized value, so that  $\mathbb{E}^{\mathcal{F}_{t-T}}[\epsilon_{B_i}(t)] = \bar{\epsilon}_{B_i}$ : using  $\epsilon_{B_i}(t)$  in place of  $\bar{\epsilon}_{B_i}$  in (27) and in the adaptive law of Section 4.2 introduces an additional,  $\mathcal{F}_{t-T}$ -conditionally-mean-zero perturbation relative to the noise-free case – it does not bias the residual in expectation, but it does increase its variance, which is why Theorem 1 below concludes boundedness in the mean-square sense rather than pathwise. Under Assumption 4,  $\epsilon_{B_i}(t)$  is bounded in mean square on the compact set  $\Omega$ : by the triangle inequality on the drift and Itô-correction integrals and the Itô isometry on the martingale term (which gives  $\mathbb{E}^{\mathcal{F}_{t-T}} \left\| \int_{t-T}^t \nabla \epsilon_i^T \Sigma_i d\mathcal{W}_i \right\|^2 = \int_{t-T}^t \mathbb{E}^{\mathcal{F}_{t-T}} \left\| \nabla \epsilon_i^T \Sigma_i \right\|^2 d\tau \leq T b_{\nabla \epsilon_i}^2 b_{\Sigma_i}^2$ ), there exists a bound  $\epsilon_{max}$  such that  $\mathbb{E}^{\mathcal{F}_{t-T}} \|\epsilon_{B_i}(t)\|^2 \leq \epsilon_{max}^2$  (the deterministic case,  $\|\epsilon_{B_i}\| \leq \epsilon_{max}$  pathwise, per [17], is recovered exactly when  $\Sigma_i \equiv 0$ ). Analogously to (28),  $\Delta \phi_i(\delta_i(t))$  used operationally in the adaptive law below is its realized (pathwise) value, i.e. the full right-hand side of the first line of (28), not its conditional expectation.

## 4.2 Critic network weight convergence and tuning via experience replay

An online integral reinforcement learning (IRL) approach is introduced, in which a single neural network—the critic—is employed to approximate both the value function and the optimal policy derived from the IRL-based Bellman formulation. Unlike traditional methods, this framework enables the reuse of collected experience data, leading to a practical and verifiable condition for the convergence of the critic’s weight parameters. Since the true ideal weight vectors  $W_i$  are not available, we utilize current estimates  $\hat{W}_i$  during the learning process. Based on these estimates, the value function  $V_i$  and its gradient  $\nabla V_i$  can be approximated accordingly.

$$\hat{V}_i(\delta_i) = \hat{W}_i^T \phi_i(\delta_i) \quad (30)$$

$$\nabla \hat{V}_i(\delta_i) = \nabla \phi_i^T(\delta_i) \hat{W}_i \quad (31)$$

Consequently, the  $H_\infty$  optimal control policy and the corresponding worst-case disturbance strategy can be expressed in terms of the approximated value function and its gradient as follows:

$$\hat{u}_i = -\frac{1}{2}(b_i + d_i)R_{ii}^{-1}g_i^T \nabla \phi_i(\delta_i)^T \hat{W}_i, \quad (32)$$

$$\hat{v}_i = \frac{1}{2\gamma^2}(b_i + d_i)T_{ii}^{-1}k_i^T \nabla \phi_i(\delta_i)^T \hat{W}_i, \quad (33)$$

Using the value-function approximation (30) together with the IRL-based Bellman equation (27), the approximate Bellman equation becomes  $e_i(t) = \hat{W}_i^T \Delta \phi_i(\delta_i(t)) + \varrho_i(t)$ , where  $\varrho_i(t) \triangleq \int_{t-T}^t \left( Q_{ii}(\delta_i) + \hat{u}_i^T R_{ii} \hat{u}_i + \sum_{j \in \mathcal{N}_i} \hat{u}_j^T R_{ij} \hat{u}_j - \gamma^2 \hat{v}_i^T T_{ii} \hat{v}_i - \gamma^2 \sum_{j \in \mathcal{N}_i} \hat{v}_j^T T_{ij} \hat{v}_j \right) d\tau$ . Substituting (30)-(33) into (22), we can obtain

$$\begin{aligned}
& \hat{H}_i(\delta_i, \hat{W}_i, \hat{u}_i, \hat{u}_{-i}, \hat{\vartheta}_i, \hat{\vartheta}_{-i}) \\
&= Q_{ii}(\delta_i) + \hat{W}_i^T \nabla \phi_i(\delta_i) f_{ei}(t) \\
&\quad - \frac{1}{4} (b_i + d_i)^2 \hat{W}_i^T \nabla \phi_i(\delta_i) g_i R_{ii}^{-1} g_i^T \nabla \phi_i(\delta_i)^T \hat{W}_i \\
&\quad + \frac{1}{4\gamma^2} (b_i + d_i)^2 \hat{W}_i^T \nabla \phi_i(\delta_i) k_i T_{ii}^{-1} k_i^T \nabla \phi_i(\delta_i)^T \hat{W}_i \\
&\quad + \frac{1}{4} \sum_{j \in \mathcal{N}_i} (b_j + d_j)^2 \hat{W}_j^T \nabla \phi_j(\delta_j) g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \nabla \phi_j(\delta_j)^T \hat{W}_j \\
&\quad - \frac{1}{4\gamma^2} \sum_{j \in \mathcal{N}_i} (b_j + d_j)^2 \hat{W}_j^T \nabla \phi_j(\delta_j) k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \nabla \phi_j(\delta_j)^T \hat{W}_j \\
&\quad + \frac{1}{2} \hat{W}_i^T \nabla \phi_i(\delta_i) \sum_{j \in \mathcal{N}_i} a_{ij} (b_j + d_j) g_j R_{jj}^{-1} g_j^T \nabla \phi_j(\delta_j)^T \hat{W}_j \\
&\quad - \frac{1}{2\gamma^2} \hat{W}_i^T \nabla \phi_i(\delta_i) \sum_{j \in \mathcal{N}_i} a_{ij} (b_j + d_j) k_j T_{jj}^{-1} k_j^T \nabla \phi_j(\delta_j)^T \hat{W}_j
\end{aligned} \tag{34}$$

The Itô-correction trace term of the stochastic coupled HJI equation (22) is omitted from (34): it depends only on  $\Sigma_i(x)$  and  $\nabla^2 \phi_i(\delta_i)$ , not on  $\hat{W}_i$ ,  $\hat{u}_i$ , or  $\hat{\vartheta}_i$ , so it contributes no gradient with respect to the weights being adapted and drops out of the weight-update law derived below in exactly the same way it dropped out of the stationary conditions (20)–(21); it is instead accounted for through  $\epsilon_{B_i}(t)$  in (29), which is where the IRL-integral route below actually uses it.

To facilitate the convergence of the estimated weight vector  $\hat{W}_i$ , a concurrent learning approach incorporating experience replay is employed. This method leverages both previously stored historical data and real-time data to simultaneously update  $\hat{W}_i$ . As in (29), every quantity below ( $\Delta \phi_i$ ,  $\Psi_i$ ,  $\Theta_i$ , and  $\varrho_i$ ) is computed from the single realized trajectory observed online, not from a conditional expectation over sample paths – consistent with treating (35) as a stochastic-approximation recursion driven by noisy (single-path) Bellman residuals rather than a batch regression over multiple rollouts. The following presents the proposed gradient-descent-based learning algorithm for the critic neural network using a novel experience replay mechanism:

$$\dot{\hat{W}}_i = -\alpha_i \frac{\Delta \phi_i(t)}{(\Delta \phi_i(t)^T \Delta \phi_i(t) + 1)^2} \Psi_i(t) + \xi_i(t) \Theta_i(t) + \tag{35}$$

$$\sum_{k=1}^l \left( -\alpha_i \frac{\Delta \phi_i(t_k)}{(\Delta \phi_i(t_k)^T \Delta \phi_i(t_k) + 1)^2} \Psi_i(t_k) + \xi_i(t_k) \Theta_i(t_k) \right) \tag{36}$$

where  $\Psi_i(t) \triangleq \Delta \phi_i(t)^T \hat{W}_i + \varrho_i(t)$ . Stability of the closed-loop system is ensured by the second term in (35), formulated as  $\xi_i(t) = 0$ , if  $\delta_i^T(t) \delta_i(t) \leq \delta_i^T(t-T) \delta_i(t-T)$ , and  $\xi_i(t) = 1$ , otherwise; and the term  $\Theta_i$  is defined as

$$\Theta_i(t) = -\beta_i \int_{t-T}^t \left( \frac{1}{2} (b_i + d_i) \nabla \phi_i g_i R_{ii}^{-1} g_i^T \delta_i - \frac{1}{2\gamma^2} (b_i + d_i) \nabla \phi_i k_i T_{ii}^{-1} k_i^T \delta_i \right. \tag{37}$$

$$\left. + \sum_{j \in \mathcal{N}_i} \frac{1}{2} a_{ij} (b_j + d_j) \nabla \phi_j g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \delta_j \right. \tag{38}$$

$$\left. - \sum_{j \in \mathcal{N}_i} \frac{1}{2} a_{ij} (b_j + d_j) \nabla \phi_j k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \delta_j \right) d\tau \tag{39}$$

Unlike  $\Psi_i(t)$ , the stabilizing term  $\Theta_i(t)$  does not depend on  $\hat{W}_i$ : it is built only from the known regressor gradients  $\nabla \phi_i$ ,  $\nabla \phi_j$ , the fixed design matrices  $g_i$ ,  $k_i$ ,  $R_{ii}$ ,  $T_{ii}$  (and their neighbor counterparts), and the measured tracking errors  $\delta_i$ ,  $\delta_j$ . Consequently, under Assumption 4 and boundedness

of  $\delta_i, \delta_j$  on the compact set  $\Omega$ ,  $\Theta_i(t)$  is a bounded known signal rather than an estimation error: there exists a constant  $\theta_{max_i} > 0$ , independent of  $\hat{W}_i$ , such that  $\|\Theta_i(t)\| \leq \theta_{max_i}$  for all  $t$ . This bound is used below to account for the contribution of  $\Theta_i$  to the weight-error dynamics (43) and to the convergence thresholds of Theorem 1.

The sample data are accumulated in a history stack, which serves as a memory buffer for experience replay. To populate this history stack, data are recorded during the system's online operation, capturing state trajectories and corresponding temporal difference errors under the current policy. Consider  $\Delta\phi_i(t_k)$  and  $\varrho_i(t_k)$  as evaluated values of  $\Delta\phi_i(t)$  and  $\varrho_i(t)$  at the recorded time  $t_k$ , defined as  $\Delta\phi_i(\delta_i(t_k)) \triangleq \phi_i(\delta_i(t_k)) - \phi_i(\delta_i(t_k - T))$ .

**Condition 1.** Let  $Z_i = \{\Delta\bar{\phi}_i^1, \dots, \Delta\bar{\phi}_i^l\}$  denote the recorded history of regressor data,  $\text{rank}(Z_i) = L$ . This implies that the recorded set contains  $L$  linearly independent elements, which matches the number of neurons used in the neural network structure in (25). The history stack is assumed to store a fixed number of samples  $l$ , satisfying  $l > L$ .

Let the estimation error for the adaptive weights  $\hat{W}_i$  be defined as  $\tilde{W}_i = W_i - \hat{W}_i$ . Differentiating both sides yields  $\dot{\tilde{W}}_i = -\dot{\hat{W}}_i$ . Since (35) also carries the stabilizing term  $\xi_i(t)\Theta_i(t) + \sum_{k=1}^l \xi_i(t_k)\Theta_i(t_k)$ , this term must be carried through with the opposite sign as well. For brevity, collect the regressor Gram matrix and the two bounded perturbation contributions to (35) as

$$\Xi_i \triangleq \Delta\bar{\phi}_i(t)\Delta\bar{\phi}_i(t)^T + \sum_{k=1}^l \Delta\bar{\phi}_i(t_k)\Delta\bar{\phi}_i(t_k)^T, \quad (40)$$

$$\mathcal{E}_i(t) \triangleq \frac{\Delta\bar{\phi}_i(t)}{m_i} \epsilon_{B_i}(t) + \sum_{k=1}^l \frac{\Delta\bar{\phi}_i(t_k)}{m_i(t_k)} \epsilon_{B_i}(t_k), \quad (41)$$

$$\mathcal{T}_i(t) \triangleq \xi_i(t)\Theta_i(t) + \sum_{k=1}^l \xi_i(t_k)\Theta_i(t_k), \quad (42)$$

where  $\Delta\bar{\phi}_i(t) = \frac{\Delta\phi_i(t)}{\Delta\phi_i(t)^T \Delta\phi_i(t) + 1}$ ,  $m_i = \Delta\phi_i(t)^T \Delta\phi_i(t) + 1$ ,  $\Delta\bar{\phi}_i(t_k) = \frac{\Delta\phi_i(t_k)}{\Delta\phi_i(t_k)^T \Delta\phi_i(t_k) + 1}$ , and  $m_i(t_k) = \Delta\phi_i(t_k)^T \Delta\phi_i(t_k) + 1$ . The error dynamics can then be expressed compactly, based on equation (35), as

$$\dot{\tilde{W}}_i(t) = -\alpha_i \Xi_i \tilde{W}_i(t) + \alpha_i \mathcal{E}_i(t) - \mathcal{T}_i(t). \quad (43)$$

Since  $\xi_i \in \{0, 1\}$ ,  $\mathbb{E}\|\epsilon_{B_i}(t)\|^2 \leq \epsilon_{max}^2$  (established after (27)), and  $\|\Theta_i(t)\| \leq \theta_{max_i}$  (established after (37)), Cauchy–Schwarz gives  $\alpha_i \mathbb{E}[\tilde{W}_i^T \mathcal{E}_i(t)] \leq \alpha_i(l+1)\epsilon_{max} \mathbb{E}\|\tilde{W}_i\|$  and  $-\mathbb{E}[\tilde{W}_i^T \mathcal{T}_i(t)] \leq (l+1)\theta_{max_i} \mathbb{E}\|\tilde{W}_i\|$  (with at most  $l$  terms whenever  $\xi_i(t) = 0$ , since  $\mathcal{T}_i(t)$  then reduces to the historical sum alone); these two bounds are used throughout the proof below.

The complete procedure is summarized in Algorithm 1. Note that steps 3–7 are the only ones that evaluate the input matrices  $g_i, k_i$ ; the drift  $f_{ei}$  and the diffusion  $\Sigma_i$  are never needed, the former because the IRL representation (24) eliminates it and the latter because it is absorbed into the generator identity. Section 4.3 removes the remaining dependence on  $g_i, k_i$  as well.

### 4.3 Model-free extension via off-policy integral reinforcement learning

The formulation developed so far is already model-free *in the drift*: the IRL representation (24) removes  $f_{ei}$  from the Bellman equation entirely, and the diffusion  $\Sigma_i$  never has to be evaluated – it is absorbed into the generator identity  $\mathcal{A}_i V_i + r_i = 0$  and enters the analysis only through the residual bound  $\epsilon_{max}$  and the constant  $c_{\sigma_i}$ . What is *not* yet model-free is the input channel: the policies (32)–(33) and the stabilizing term (37) evaluate  $g_i(x_i)$  and  $k_i(x_i)$  explicitly. This subsection removes that last requirement, so that no component of (11) – neither  $f_{ei}$ , nor  $g_i, k_i$ , nor  $\Sigma_i$  – need be known. The construction is the nonlinear, agent-coupled counterpart of the off-policy reformulation that makes

---

**Algorithm 1** Model-Based Stochastic IRL for the  $H_\infty$  Graphical Game
 

---

**Require:**  $Q_{ii} \succeq 0$ ,  $R_{ij}, T_{ij} \succ 0$ ,  $\gamma > 0$ ,  $\alpha_i, \beta_i > 0$ ,  $T > 0$ ; critic basis  $\phi_i(\delta_i) \in \mathbb{R}^L$ ; admissible  $\hat{W}_i(0)$ ; stack size  $l > L$ . Known:  $g_i, k_i$ .

```

1: loop for each  $t$ 
2:   Measure  $\delta_i(t)$ ; receive  $\{\delta_j, \hat{W}_j\}_{j \in \mathcal{N}_i}$ .
3:    $\hat{u}_i \leftarrow -\frac{1}{2}(b_i + d_i)R_{ii}^{-1}g_i^T \nabla \phi_i^T \hat{W}_i$ ,  $\hat{v}_i \leftarrow \frac{1}{2\gamma^2}(b_i + d_i)T_{ii}^{-1}k_i^T \nabla \phi_i^T \hat{W}_i$ 
4:    $\Delta \phi_i(t) \leftarrow \phi_i(\delta_i(t)) - \phi_i(\delta_i(t - T))$ ,  $\varrho_i(t) \leftarrow \int_{t-T}^t r_i d\tau$ 
5:    $\Psi_i(t) \leftarrow \Delta \phi_i(t)^T \hat{W}_i + \varrho_i(t)$ 
6:    $\xi_i(t) \leftarrow \mathbb{1}\{\delta_i^T(t)\delta_i(t) > \delta_i^T(t - T)\delta_i(t - T)\}$ 
7:    $\Theta_i(t) \leftarrow (37)$   $\triangleright$  requires  $g_i, k_i, g_j, k_j$ 
8:    $Z_i \leftarrow Z_i \cup \{\Delta \bar{\phi}_i(t)\}$  while  $\text{rank}(Z_i) < L$   $\triangleright$  Condition 1
9:    $\dot{\hat{W}}_i \leftarrow -\alpha_i \frac{\Delta \phi_i(t)\Psi_i(t)}{(\Delta \phi_i^T(t)\Delta \phi_i(t) + 1)^2} + \xi_i(t)\Theta_i(t) + \sum_{k=1}^l \left( -\alpha_i \frac{\Delta \phi_i(t_k)\Psi_i(t_k)}{(\Delta \phi_i^T(t_k)\Delta \phi_i(t_k) + 1)^2} + \xi_i(t_k)\Theta_i(t_k) \right)$ 
10: end loop

```

**Ensure:**  $\hat{W}_i$  mean-square UUB in a residual set about  $W_i$ ;  $\hat{V}_i = \hat{W}_i^T \phi_i$ ,  $\hat{u}_i, \hat{v}_i$  near-optimal in mean square (Theorem 1).

---

policy iteration model-free for the *linear* stochastic zero-sum game in [21]; there, the unknown matrices  $A, B_1, B_2, C, D$  survive only inside the lumped products  $B_1^T P + D^T P C$ ,  $B_2^T P$  and  $D^T P D$ , which are then identified jointly with  $P$  by least squares. The same mechanism is available here, with the ideal weights  $W_i$  playing the role of  $P$  and the stationarity conditions (20)–(21) supplying the lumping. The same lumping is what makes the off-policy scheme of [22] model-free for a single-agent linear Itô system with state-dependent multiplicative noise, where the regression vector is exactly  $[\text{vecs}(P)^T, \text{vec}(B^T P)^T, \text{vec}(E^T P)^T]^T$  – the value parameter together with *two* lumped blocks, one per channel of the agent’s own input. It is worth being explicit about what the graphical-game setting adds to that picture: because agent  $i$ ’s error  $\delta_i$  is driven by its neighbours’ inputs as well as its own, the mismatch (45) contains terms  $g_j, k_j$  that no stationarity condition of agent  $i$  can absorb, so a *third* family of lumped unknowns – one pair per neighbour – has to be carried in the regressor. This family has no counterpart in the single-agent problem of [21, 22] and is the reason the excitation requirement below scales with  $|\mathcal{N}_i|$  rather than with the state dimension alone.

**Behaviour–target separation.** Let  $u_j, \vartheta_j$  denote the signals *actually* applied to and acting on the plant (the behaviour data, generated as  $u_j = \hat{u}_j + e_j$  with a bounded exploration signal  $e_j$ ), and let  $\hat{u}_j, \hat{\vartheta}_j$  denote the current *target* policies (32)–(33) being evaluated and improved. Adding and subtracting the target policies in (11) gives the off-policy form

$$d\delta_i = \hat{F}_i dt + \Delta_i dt + \Sigma_i(x) dW_i(t), \quad (44)$$

where  $\hat{F}_i \triangleq f_{ei} + (b_i + d_i)(g_i \hat{u}_i + k_i \hat{\vartheta}_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j \hat{u}_j + k_j \hat{\vartheta}_j)$  is the drift under the target policies and

$$\Delta_i \triangleq (b_i + d_i)[g_i(u_i - \hat{u}_i) + k_i(\vartheta_i - \hat{\vartheta}_i)] - \sum_{j \in \mathcal{N}_i} a_{ij}[g_j(u_j - \hat{u}_j) + k_j(\vartheta_j - \hat{\vartheta}_j)] \quad (45)$$

collects the behaviour–target mismatch. Only  $\hat{F}_i$  carries the unknown drift  $f_{ei}$ , and it is exactly  $\hat{F}_i$  that the Bellman identity eliminates.

**Model-free lumping of the input channels.** Applying Itô’s formula to  $V_i(\delta_i)$  along (44), using  $\nabla V_i^T \hat{F}_i + \frac{1}{2} \text{tr}(\Sigma_i^T \nabla^2 V_i \Sigma_i) = \mathcal{A}_i V_i = -r_i$  from (19) evaluated at the target policies, and taking  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$  to annihilate the martingale exactly as in (24), leaves only  $\nabla V_i^T \Delta_i$  to be dealt with. The decisive observation is that the stationarity conditions (20)–(21) can be read *backwards*: rather than producing the policy from  $\nabla V_i$ , they express the unknown products  $\nabla V_i^T g_i$  and  $\nabla V_i^T k_i$  in terms of

the policy itself,

$$(b_i + d_i) \nabla V_i^T g_i(x_i) = -2 \hat{u}_i^T R_{ii}, \quad (b_i + d_i) \nabla V_i^T k_i(x_i) = 2\gamma^2 \hat{\vartheta}_i^T T_{ii}, \quad (46)$$

in which  $R_{ii}$  and  $T_{ii}$  are design weights and  $\hat{u}_i, \hat{\vartheta}_i$  are the signals the agent itself generates. Both sides of (46) are therefore computable online without  $g_i$  or  $k_i$ . This is precisely the step by which  $2x^\top PB(\cdot)$  becomes  $2x^\top \hat{K}^\top R(\cdot)$  in [21]. The neighbour terms of (45) admit no such identity, since no stationarity condition relates agent  $i$ 's gradient  $\nabla V_i$  to agent  $j$ 's input matrix  $g_j$ ; they are instead retained as unknown *cross-channel* maps

$$\Xi_{ij}^u(\delta) \triangleq a_{ij} g_j^T(x_j) \nabla V_i(\delta_i) \in \mathbb{R}^{m_j}, \quad \Xi_{ij}^\vartheta(\delta) \triangleq a_{ij} k_j^T(x_j) \nabla V_i(\delta_i) \in \mathbb{R}^{q_j}, \quad (47)$$

to be identified from data alongside  $W_i$ . Substituting (46)–(47) into  $\nabla V_i^T \Delta_i$  and using the NN representation (25) yields the *off-policy stochastic IRL Bellman equation*

$$\begin{aligned} W_i^T \Delta \phi_i(\delta_i(t)) + \mathbb{E}^{\mathcal{F}_{t-T}} \left[ \int_{t-T}^t \left( 2\hat{u}_i^T R_{ii}(u_i - \hat{u}_i) - 2\gamma^2 \hat{\vartheta}_i^T T_{ii}(\vartheta_i - \hat{\vartheta}_i) \right. \right. \\ \left. \left. + \sum_{j \in \mathcal{N}_i} (\Xi_{ij}^{uT}(u_j - \hat{u}_j) + \Xi_{ij}^{\vartheta T}(\vartheta_j - \hat{\vartheta}_j)) \right) d\tau \right] = -\mathbb{E}^{\mathcal{F}_{t-T}} \left[ \int_{t-T}^t r_i d\tau \right] + \bar{\epsilon}_{B_i}, \quad (48) \end{aligned}$$

in which every quantity other than the unknowns  $W_i, \Xi_{ij}^u, \Xi_{ij}^\vartheta$  is either measured  $(\delta_i, u_j, \vartheta_j)$ , self-generated  $(\hat{u}_i, \hat{\vartheta}_i)$ , communicated by neighbours  $(\hat{u}_j, \hat{\vartheta}_j)$ , already required by  $\varrho_i$  in the model-based case), or a design choice  $(Q_{ii}, R_{ii}, T_{ii}, \gamma)$ . Setting  $u_j \equiv \hat{u}_j$  and  $\vartheta_j \equiv \hat{\vartheta}_j$  (no exploration, on-policy data) collapses (48) back to (27) exactly.

**Actor parameterisation and the enlarged regressor.** Because the policies can no longer be evaluated in the closed form (32)–(33), they are represented by their own approximators, as are the cross-channel maps (47):

$$\hat{u}_i = \hat{W}_{u_i}^T \psi_{u_i}(\delta_i, x_i), \quad \hat{\vartheta}_i = \hat{W}_{\vartheta_i}^T \psi_{\vartheta_i}(\delta_i, x_i), \quad \Xi_{ij}^u = \hat{W}_{ij}^{uT} \psi_{ij}^u(\delta, x), \quad \Xi_{ij}^\vartheta = \hat{W}_{ij}^{\vartheta T} \psi_{ij}^\vartheta(\delta, x). \quad (49)$$

The bases are allowed to depend on the agent's own measured configuration  $x_i$  (and, for the cross-channel maps, on the neighbour configuration  $x_j$  already communicated), not on  $\delta_i$  alone. This is not a cosmetic generalisation: the exact policy (32) is  $-\frac{1}{2}(b_i + d_i)R_{ii}^{-1}g_i^T(x_i)\nabla\phi_i^T W_i$ , so whenever the input channel is configuration-dependent – as it is for a nonholonomic vehicle, where  $g_i$  carries the heading through  $S(\theta_i)$  – no basis in  $\delta_i$  alone can represent the ideal actor exactly, and the residual would enter as a permanent approximation error rather than a vanishing one. With  $x_i$  admitted, a basis of the form  $\{\delta_i, \delta_i \cos \theta_i, \delta_i \sin \theta_i\}$  spans the ideal actor exactly for a quadratic critic. Nothing in the construction requires the two cross-channel families to share a basis, so  $\psi_{ij}^u$  and  $\psi_{ij}^\vartheta$  are kept distinct; they may be taken equal when convenient. Every unknown in (48) then enters *linearly*, so the entire estimation problem retains the structure of Section 4.2. Collecting

$$\mathcal{Z}_i \triangleq [\hat{W}_i^T, \text{vec}(\hat{W}_{u_i})^T, \text{vec}(\hat{W}_{\vartheta_i})^T, \{\text{vec}(\hat{W}_{ij}^u)^T\}_{j \in \mathcal{N}_i}, \{\text{vec}(\hat{W}_{ij}^\vartheta)^T\}_{j \in \mathcal{N}_i}]^T \quad (50)$$

and, using  $a^T X b = (b^T \otimes a^T) \text{vec}(X)$ , the matching regressor

$$\begin{aligned} \Lambda_i(t) \triangleq \left[ \Delta \phi_i(t)^T, 2 \int_{t-T}^t ((u_i - \hat{u}_i)^T R_{ii} \otimes \psi_{u_i}^T) d\tau, -2\gamma^2 \int_{t-T}^t ((\vartheta_i - \hat{\vartheta}_i)^T T_{ii} \otimes \psi_{\vartheta_i}^T) d\tau, \right. \\ \left. \left\{ \int_{t-T}^t ((u_j - \hat{u}_j)^T \otimes \psi_{ij}^{uT}) d\tau \right\}_{j \in \mathcal{N}_i}, \left\{ \int_{t-T}^t ((\vartheta_j - \hat{\vartheta}_j)^T \otimes \psi_{ij}^{\vartheta T}) d\tau \right\}_{j \in \mathcal{N}_i} \right]^T, \quad (51) \end{aligned}$$

equation (48) becomes  $\Lambda_i(t)^T \mathcal{Z}_i + \varrho_i(t) = \epsilon_{B_i}(t)$  along the realized trajectory, with  $\varrho_i$  unchanged. Consequently the model-free Bellman residual is

$$\Psi_i^{\text{mf}}(t) \triangleq \Lambda_i(t)^T \mathcal{Z}_i + \varrho_i(t), \quad (52)$$

and the adaptive law (35), the experience-replay mechanism, and the error dynamics (43) all carry over *verbatim* under the substitution

$$\Delta\phi_i \mapsto \Lambda_i, \quad \hat{W}_i \mapsto \mathcal{Z}_i, \quad \Psi_i \mapsto \Psi_i^{\text{mf}}, \quad \Theta_i \mapsto \Theta_i^{\text{mf}}, \quad (53)$$

with  $\Theta_i^{\text{mf}}$  given by (57) below. In particular  $\Xi_i$ ,  $\mathcal{E}_i$  and  $\mathcal{T}_i$  of (40)–(42) keep their definitions with  $\Delta\bar{\phi}_i$  built from  $\Lambda_i$  instead of  $\Delta\phi_i$ . *The model-free extension is thus a change of regressor, not a change of algorithm* – which is why the mean-square UUB result of Theorem 1 applies to it unchanged.

**Model-free stabilizing term.** The term  $\Theta_i$  of (37) is the gradient, with respect to the weights being adapted, of the control-channel contribution to  $\delta_i^T \dot{\delta}_i$ ; in the parameterisation (49) that gradient is taken with respect to  $\text{vec}(\hat{W}_{u_i})$  and  $\text{vec}(\hat{W}_{\vartheta_i})$  rather than  $\hat{W}_i$ , and therefore requires the input channel only through the *vector-valued* quantities

$$\Upsilon_i \triangleq (b_i + d_i)g_i^T(x_i)\delta_i \in \mathbb{R}^{m_i}, \quad \Upsilon_i^\vartheta \triangleq (b_i + d_i)k_i^T(x_i)\delta_i \in \mathbb{R}^{q_i}, \quad (54)$$

together with their neighbour counterparts  $\Upsilon_{ij} \triangleq a_{ij}g_j^T\delta_i$ ,  $\Upsilon_{ij}^\vartheta \triangleq a_{ij}k_j^T\delta_i$ . These are identified from the *same* off-policy data set by applying the behaviour–target split (44) to  $S_i(\delta_i) \triangleq \frac{1}{2}\delta_i^T\delta_i$  instead of to  $V_i$ . One point deserves care here, because  $S_i$  – unlike  $V_i$  – satisfies no Bellman identity that could annihilate the target drift. Itô’s formula applied to  $S_i$  along (44) and integrated over the window gives, after taking  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$  to remove the martingale,

$$\frac{1}{2}\delta_i^T\delta_i \Big|_{t-T}^t = \mathbb{E}^{\mathcal{F}_{t-T}} \left[ \int_{t-T}^t \left( \mu_i(\delta_i, x_i) + \Upsilon_i^T(u_i - \hat{u}_i) + \Upsilon_i^{\vartheta T}(\vartheta_i - \hat{\vartheta}_i) - \sum_{j \in \mathcal{N}_i} \left( \Upsilon_{ij}^T(u_j - \hat{u}_j) + \Upsilon_{ij}^{\vartheta T}(\vartheta_j - \hat{\vartheta}_j) \right) \right) d\tau \right], \quad (55)$$

in which the mismatch contribution  $\delta_i^T \Delta_i$  appears exactly as the linear combination of the unknowns (54) claimed above, but is accompanied by the *nuisance term*

$$\mu_i(\delta_i, x_i) \triangleq \delta_i^T \hat{F}_i + \frac{1}{2} \text{tr}(\Sigma_i^T \Sigma_i), \quad (56)$$

the target-policy drift and Itô correction of  $S_i$ , which is unknown and does *not* cancel. Dropping it would bias every estimate of  $\Upsilon$ . It is instead carried as a further linear-in-parameters unknown,  $\mu_i = \hat{W}_{\mu_i}^T \psi_{\mu_i}(\delta_i, x_i)$ , so that (55) becomes one more scalar regression of exactly the form (48) – linear in  $(\hat{W}_{\mu_i}, \Upsilon_i, \Upsilon_i^\vartheta, \{\Upsilon_{ij}, \Upsilon_{ij}^\vartheta\})$ , driven by the same measured mismatch signals, and solvable by the same concurrent-learning machinery on the same history stack. The two regressions are separately identifiable because the nuisance regressor  $\psi_{\mu_i}$  does not depend on the exploration signals whereas the  $\Upsilon$ -regressors are proportional to them; enriching  $e_j$  therefore excites the  $\Upsilon$ -block without exciting the  $\mu$ -block, which is the same separation that makes (48) identifiable. Note that the  $\Upsilon$ -block of this auxiliary problem is small –  $\Upsilon_i$  has the dimension of the input, not of the critic basis – precisely because  $\nabla S_i$  is known whereas  $\nabla \phi_i$  is not; the nuisance block is what carries the cost of  $S_i$  having no Bellman identity of its own. The model-free stabilizing term is then

$$\Theta_i^{\text{mf}}(t) = -\beta_i \int_{t-T}^t \left[ \mathbf{0}_L; \frac{1}{2} \Upsilon_i \otimes \psi_{u_i}; -\frac{1}{2\gamma^2} \Upsilon_i^\vartheta \otimes \psi_{\vartheta_i}; \left\{ \frac{1}{2} \Upsilon_{ij} \otimes \psi_{ij}^u \right\}_j; \left\{ -\frac{1}{2\gamma^2} \Upsilon_{ij}^\vartheta \otimes \psi_{ij}^\vartheta \right\}_j \right] d\tau, \quad (57)$$

which is block-conformable with  $\mathcal{Z}_i$  in (50) and reduces to (37) whenever the actor bases are chosen as  $\psi_{u_i} = R_{ii}^{-1} g_i^T \nabla \phi_i^T$ , i.e. when the model is in fact known. Its leading block is zero because, under (49), the critic weights  $\hat{W}_i$  no longer enter the control channel directly – the actors do. Since  $\Upsilon_i, \Upsilon_i^\vartheta$  are bounded on the compact set  $\Omega$  whenever  $\delta_i$  is and the activation functions are bounded by Assumption 4,  $\Theta_i^{\text{mf}}$  satisfies the same bound  $\|\Theta_i^{\text{mf}}\| \leq \theta_{\max_i}$  used throughout the proof of Theorem 1.

**Excitation requirement.** Condition 1 must be restated for the enlarged regressor: the history stack  $\mathcal{Z}_i = \{\Lambda_i(t_1), \dots, \Lambda_i(t_l)\}$  is required to satisfy  $\text{rank}(\mathcal{Z}_i) = \dim(\mathcal{Z}_i) = L + m_i L_{u_i} + q_i L_{\vartheta_i} + \sum_{j \in \mathcal{N}_i} (m_j L_{ij}^u + q_j L_{ij}^\vartheta)$ , with  $l > \dim(\mathcal{Z}_i)$ , which is the concurrent-learning analogue of the rank condition on  $[I_{xx}, I_{xu}, I_{xv}, \delta_{uu}]$  imposed in [21, Lemma 5], and of the condition

$\text{rank}([\mathcal{I}_{\bar{x}}, \mathcal{I}_{xu}, \mathcal{I}_{xv}]) = \frac{n(n+1)}{2} + nm + np$  of [22, Lemma 5], which counts unknowns in precisely the same way – one block per lumped product – for the linear single-agent case. As in the model-based case this is a verifiable condition on recorded data rather than an a priori PE requirement on the trajectory; the price of removing the model is that the exploration signals  $e_j$  must be rich enough to excite the enlarged regressor, so more data are required than in Condition 1.

The resulting procedure is summarized in Algorithm 2. Comparing it with Algorithm 1 makes the nature of the extension explicit: the two differ only in how the regressor is built (steps 3–8), while the learning mechanism – residual, history stack, concurrent-learning update – is untouched.

---

**Algorithm 2** Model-Free Stochastic IRL for the  $H_\infty$  Graphical Game

---

**Require:**  $Q_{ii} \succeq 0$ ,  $R_{ij}, T_{ij} \succ 0$ ,  $\gamma > 0$ ,  $\alpha_i, \beta_i > 0$ ,  $T > 0$ ; bases  $\phi_i, \psi_{u_i}, \psi_{\vartheta_i}, \psi_{\mu_i}, \{\psi_{ij}^u, \psi_{ij}^\vartheta\}_{j \in \mathcal{N}_i}$ ; admissible  $\mathcal{Z}_i(0)$  (Remark 4); stack size  $l > \dim(\mathcal{Z}_i)$ .

**Data collection (off-policy).** Apply  $u_j = \hat{u}_j + e_j$ ,  $j \in \{i\} \cup \mathcal{N}_i$ , with  $e_j$  bounded and satisfying the rank condition of Section 4.3.

- 1: **loop** for each  $t$
- 2:   Measure  $\delta_i(t), u_i(t), \vartheta_i(t)$ ; receive  $\{\delta_j, u_j, \vartheta_j, \hat{u}_j, \hat{\vartheta}_j\}_{j \in \mathcal{N}_i}$ .
- 3:    $\hat{u}_i \leftarrow \hat{W}_{u_i}^T \psi_{u_i}(\delta_i)$ ,  $\hat{\vartheta}_i \leftarrow \hat{W}_{\vartheta_i}^T \psi_{\vartheta_i}(\delta_i)$   $\triangleright$  cf. (49); no  $g_i, k_i$
- 4:    $\Delta\phi_i(t) \leftarrow \phi_i(\delta_i(t)) - \phi_i(\delta_i(t-T))$ ,  $\varrho_i(t) \leftarrow \int_{t-T}^t r_i d\tau$
- 5:    $\Lambda_i(t) \leftarrow$  (51), assembled from  $\Delta\phi_i(t)$  and the mismatches  $u_j - \hat{u}_j$ ,  $\vartheta_j - \hat{\vartheta}_j$
- 6:    $\Psi_i^{\text{mf}}(t) \leftarrow \Lambda_i(t)^T \mathcal{Z}_i + \varrho_i(t)$
- 7:    $\xi_i(t) \leftarrow \mathbb{K}\{\delta_i^T(t)\delta_i(t) > \delta_i^T(t-T)\delta_i(t-T)\}$
- 8:    $\{\hat{W}_{\mu_i}, \Upsilon_i, \Upsilon_i^\vartheta, \Upsilon_{ij}, \Upsilon_{ij}^\vartheta\} \leftarrow$  least squares on (55);  $\Theta_i^{\text{mf}}(t) \leftarrow$  (57)
- 9:    $Z_i \leftarrow Z_i \cup \{\Lambda_i(t)\}$  while  $\text{rank}(Z_i) < \dim(\mathcal{Z}_i)$
- 10:    $\dot{Z}_i \leftarrow -\alpha_i \frac{\Lambda_i(t)\Psi_i^{\text{mf}}(t)}{(\Lambda_i^T(t)\Lambda_i(t) + 1)^2} + \xi_i(t)\Theta_i^{\text{mf}}(t)$   
 $\quad + \sum_{k=1}^l \left( -\alpha_i \frac{\Lambda_i(t_k)\Psi_i^{\text{mf}}(t_k)}{(\Lambda_i^T(t_k)\Lambda_i(t_k) + 1)^2} + \xi_i(t_k)\Theta_i^{\text{mf}}(t_k) \right)$
- 11: **end loop**

**Ensure:**  $Z_i$  mean-square UUB in a residual set about  $\mathcal{Z}_i^*$ ;  $\hat{V}_i = \hat{W}_i^T \phi_i$ ,  $\hat{u}_i, \hat{\vartheta}_i$  near-optimal in mean square (Corollary 1), with  $f_{ei}, g_i, k_i, \Sigma_i$  all unknown.

---

**Remark 3** (What the diffusion channel costs). *It is tempting to attribute the fact that Algorithm 2 identifies no diffusion-dependent parameter to the process noise being additive. That is not the operative property, and the distinction matters, since the diffusion  $\Sigma_i(x)$  of (12) is itself state-dependent. The correct dichotomy is whether the diffusion depends on the decision variables, and it separates three cases:*

- (a)  $\Sigma_i$  independent of  $x$  (purely additive). *The Itô correction  $\frac{1}{2}\text{tr}(\Sigma_i^T \nabla^2 V_i \Sigma_i)$  is a known constant offset and is absorbed into the running cost.*
- (b)  $\Sigma_i$  state-dependent but independent of  $u_i, \vartheta_i$  – the present case, and the case  $A_1 x dW$  of [22]. *The correction is no longer constant, but it is the same function of the state under the behaviour and the target policies, so it is common to  $\hat{F}_i$  in (44) and cancels against the Bellman identity  $A_i V_i = -r_i$  exactly as the drift does. It never appears in the mismatch  $\Delta_i$  of (45) and therefore never enters the regressor. The confirmation is structural rather than incidental: the regression vector of [22] for a system with genuinely multiplicative noise contains no block built from  $A_1$ , because  $A_1^T P A_1$  sits inside the generalized Lyapunov equation that the Bellman identity annihilates – the exact linear counterpart of the cancellation used here.*
- (c)  $\Sigma_i$  control-dependent, as in the  $Cx + Du$  model of [21]. *The correction now differs between behaviour and target policies, survives in  $\Delta_i$ , and forces the extra lumped unknown  $D^T P D$  into the least-squares problem.*

The present scheme is model-free in the diffusion channel for free because it sits in case (b), not because the noise is additive; and only case (c) would break the free lunch. Whether a stationarity-type identity can be found that absorbs a control-dependent diffusion into the policies themselves, in the manner of (46), is left open.

**Remark 4** (Obtaining an admissible initialization). Both algorithms require an initial parameter –  $\hat{W}_i(0)$  in Algorithm 1,  $\mathcal{Z}_i(0)$  in Algorithm 2 – whose induced policy is admissible in the sense of Definition 1, and neither can certify admissibility from the model, which is precisely what is assumed unknown. In the model-free case the certificate must therefore be obtained from data. We use the elementary test employed for the same purpose in [22]: candidate gains are drawn at random, applied to the plant over a short horizon, and retained if the realized trajectories enter and remain in a neighbourhood of the origin – an empirical, single-run surrogate for the mean-square stabilizability requirement, which is all that is available without a model. Two remarks are in order. First, the test is a necessary condition checked on finitely many sample paths, not a proof of mean-square stability; the guarantee it supports is the conditional one already stated in Theorem 1, namely that the conclusions hold provided the initialization is admissible. Second, unlike the local convergence results of [22, 21], no requirement is imposed here that the initialization lie in a prescribed neighbourhood  $\mathcal{B}_{\delta_0}(P^*)$  of the optimal solution: the Lyapunov argument of Theorem 1 is driven by the rank condition and Assumption 5, neither of which refers to the distance from  $\hat{W}_i(0)$  to  $W_i$ . The price is that the conclusion is uniform ultimate boundedness rather than convergence.

**Remark 5** (Realizing the conditional expectation). Equations (24) and (48) hold in conditional expectation, and any implementation must decide how  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$  is realized. Two routes exist, and they lead to genuinely different algorithms. The batch route replaces the conditional expectation by an average over  $\bar{L}$  independently generated sample paths sharing the initial condition  $\delta_i(t-T)$ , with the integrals approximated by Riemann sums on a mesh  $h$ ; this is the route taken in [22], where  $\bar{L} = 500$  rollouts per data window are used. It yields a low-variance regression problem, but it requires the plant to be re-initialized and re-run  $\bar{L}$  times per window, which is available in simulation and generally not on hardware. The single-trajectory route, adopted here and described after (29), uses the realized value along the one trajectory the robots actually execute; by construction it is conditionally unbiased,  $\mathbb{E}^{\mathcal{F}_{t-T}}[\epsilon_{B_i}(t)] = \bar{\epsilon}_{B_i}$ , but has strictly larger variance, and it is this variance that is responsible for the conclusion of Theorem 1 being mean-square uniform ultimate boundedness rather than convergence. The trade is deliberate: the scheme is intended to run online, on one physical run, which forbids the batch route. Section 4.5 quantifies what either choice costs in the guarantee.

#### 4.4 Stability and convergence analysis

One constant used below must be introduced explicitly, since it does not follow from Assumption 4.

**Assumption 5.** Let  $F_i \triangleq f_{ei} + (b_i + d_i)(g_i \hat{u}_i + k_i \hat{v}_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j \hat{u}_j + k_j \hat{v}_j)$  denote the closed-loop drift of (11) under the approximate policies actually applied. On the compact set  $\Omega$  there exists a constant  $\zeta_i > 0$  such that, whenever  $\delta_i^T(t)\delta_i(t) \leq \delta_i^T(t-T)\delta_i(t-T)$  (strict decrease on the contracting regime),

$$\mathbb{E} \left[ \int_{t-T}^t \delta_i^T F_i d\tau \mid \xi_i(t) = 0 \right] \leq -\zeta_i \mathbb{E} \left[ \int_{t-T}^t \|\delta_i(\tau)\| d\tau \mid \xi_i(t) = 0 \right]. \quad (58)$$

**Remark 6.** This is a genuine assumption, not a consequence of  $\xi_i = 0$ : as (65) makes explicit, the switching rule alone – without further hypotheses – yields only that the drift term and an uncontrolled, conditioned martingale term sum to something non-positive, not a bound on the drift term in isolation. Assumption 5 instead posits directly, as a hypothesis on the closed-loop dynamics, that a strict decrease proportional to  $\mathbb{E} \int \|\delta_i\|$  holds on this event; this is the continuous-time analogue of the detectability-type conditions imposed on the policy-evaluation step in [21], and holds, for instance, whenever  $\hat{u}_i$  is admissible in the sense of Definition 1 with a uniform decay rate on  $\Omega \setminus \{0\}$ . Unlike an earlier draft of this result, no separate “closed-loop linear growth” bound on

$\delta_i^T F_i$  is required: the  $\xi_i(t) = 1$  regime below bounds the corresponding window term directly from the elementary non-negativity of a squared norm, without any reference to  $F_i$ .

**Theorem 1.** *Given the stochastic system dynamics in (11) and the coupled stochastic HJI equations in (22), suppose the control law and disturbance policy are generated by (32) and (33), respectively. Assume that the update law for the critic NN weights  $W_i$  follows (35), using the realized single-trajectory residual  $\epsilon_{B_i}(t)$  of (29). If **Condition 1** and Assumption 5 hold, then the closed-loop tracking error  $\delta_i$  and the weight estimation error  $\tilde{W}_i$  are ensured to be ultimately uniformly bounded in the mean-square sense (mean-square UUB):  $\mathbb{E}\|\delta_i(t)\|^2$  and  $\mathbb{E}\|\tilde{W}_i(t)\|^2$  are ultimately bounded, with an ultimate bound that reduces exactly to the pathwise UUB bound of the deterministic case as the diffusion intensity  $\Sigma_i \rightarrow 0$ . Furthermore, the approximated value function  $\hat{V}_i$ , the control policy  $\hat{u}_i$ , and the worst-case disturbance approach their corresponding optimal solutions in mean square:  $\mathbb{E}\|V_i^* - \hat{V}_i\|^2 < \epsilon_{V_i}$ ,  $\mathbb{E}\|u_i^* - \hat{u}_i\|^2 < \epsilon_{u_i}$ , and  $\mathbb{E}\|\vartheta_i^* - \hat{\vartheta}_i\|^2 < \epsilon_{\vartheta_i}$ , where  $\epsilon_{V_i}$ ,  $\epsilon_{u_i}$ , and  $\epsilon_{\vartheta_i}$  are the small positive constants (78)–(79). These constants decrease to their deterministic values as  $\Sigma_i \rightarrow 0$  and to zero only as the critic basis is refined,  $L \rightarrow \infty$ ; they do not vanish with the noise alone, since the neural-network approximation error persists in the noise-free limit.*

*Proof.* Two departures from the deterministic proof must be made at the outset, both caused by the switching signal  $\xi_i$  becoming a random variable once  $\delta_i$  is an Itô process.

First, the deterministic argument uses one Lyapunov function on  $\{\xi_i = 0\}$  and a different one on  $\{\xi_i = 1\}$ . Switching between two Lyapunov functions is legitimate only under a multiple-Lyapunov-function (dwell-time) argument that accounts for the jump  $L_i^{(1)} - L_i^{(0)}$  at each switching instant. We avoid that difficulty altogether by using the *single* function

$$L_i(t) \triangleq \frac{1}{2} \int_{t-T}^t \delta_i^T \delta_i d\tau + \frac{1}{2} \tilde{W}_i^T \tilde{W}_i + V_i(\delta_i(t)) \quad (59)$$

in *both* regimes, where  $V_i \geq 0$  is a local smooth solution of the coupled stochastic HJI (22). Since  $V_i \geq 0$ , (59) dominates the windowed function used in the deterministic  $\xi_i = 0$  analysis, so no conclusion is weakened; the two regimes now differ only in how  $\mathbb{E}[\dot{L}_i]$  is *bounded*, not in which function is being bounded.

Second, the events  $\{\xi_i(t) = 0\}$  and  $\{\xi_i(t) = 1\}$  are  $\mathcal{F}_t$ -measurable events of positive probability rather than a partition of the time axis: at a fixed  $t$ , some sample paths contract over  $[t-T, t]$  and others do not. Taking an *unconditional* expectation of an inequality that holds only on one of these events is therefore invalid. All expectations in the two regimes below are accordingly conditional, and we abbreviate  $\mathbb{E}_0[\cdot] \triangleq \mathbb{E}[\cdot | \xi_i(t) = 0]$  and  $\mathbb{E}_1[\cdot] \triangleq \mathbb{E}[\cdot | \xi_i(t) = 1]$ . They are recombined at the end by the tower property

$$\mathbb{E}[\dot{L}_i(t)] = \mathbb{P}(\xi_i(t) = 0) \mathbb{E}[\dot{L}_i(t) | \xi_i(t) = 0] + \mathbb{P}(\xi_i(t) = 1) \mathbb{E}[\dot{L}_i(t) | \xi_i(t) = 1], \quad (60)$$

so that a bound established in each regime yields a bound on  $\mathbb{E}[\dot{L}_i(t)]$  as a convex combination, with no additional hypothesis on the switching statistics.

With (59) fixed, two further facts must be kept separate. First, the window integral  $\frac{1}{2} \int_{t-T}^t \delta_i^T \delta_i d\tau$  is differentiated in  $t$  by the ordinary Leibniz rule for a moving window, which requires only a.s. continuity of the (now stochastic) integrand  $\delta_i^T(\tau) \delta_i(\tau)$  – true since (11) has continuous sample paths – and gives, pathwise,  $\frac{d}{dt} [\frac{1}{2} \int_{t-T}^t \delta_i^T \delta_i d\tau] = \frac{1}{2} [\delta_i^T(t) \delta_i(t) - \delta_i^T(t-T) \delta_i(t-T)]$ , exactly as in the deterministic case. Second, unlike the deterministic case, this boundary term can no longer be rewritten as  $\int_{t-T}^t \delta_i^T \dot{\delta}_i d\tau$  by the ordinary fundamental theorem of calculus, since  $\delta_i(\tau)$  has no pathwise derivative; Itô's formula must be used instead. Applying Itô's formula to  $\delta_i^T(\tau) \delta_i(\tau)$  along (11) gives

$$\frac{1}{2} [\delta_i^T(t) \delta_i(t) - \delta_i^T(t-T) \delta_i(t-T)] = \int_{t-T}^t \delta_i^T F_i d\tau + \frac{1}{2} \int_{t-T}^t \text{tr}(\Sigma_i^T \Sigma_i) d\tau + \int_{t-T}^t \delta_i^T \Sigma_i d\mathcal{W}_i(\tau), \quad (61)$$

where  $F_i$  is the closed-loop drift defined in Assumption 5, the middle term is the new Itô correction, and the last term is a local martingale that is conditionally mean-zero and vanishes upon taking expectations. The third summand of (59) is handled by Itô's formula applied to  $V_i(\delta_i(t))$  along (11), giving  $dV_i = \mathcal{A}_i^{\hat{u}} V_i d\tau + \nabla V_i^T \Sigma_i d\mathcal{W}_i$ , in which  $\mathcal{A}_i^{\hat{u}}$  is the generator (18) evaluated at the *approximate* policies actually applied. The HJI equation (19) certifies  $\mathcal{A}_i^{u^*} V_i = -r_i$  at the *optimal* policies, so the two differ by a policy-mismatch term,

$$\begin{aligned} \mathcal{A}_i^{\hat{u}} V_i = & -r_i + \nabla V_i^T \left[ (b_i + d_i)(g_i(\hat{u}_i - u_i^*) + k_i(\hat{\vartheta}_i - \vartheta_i^*)) \right. \\ & \left. - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j(\hat{u}_j - u_j^*) + k_j(\hat{\vartheta}_j - \vartheta_j^*)) \right], \end{aligned} \quad (62)$$

which is precisely the source of the  $\Gamma$ - and  $\Lambda$ -dependent terms collected below; the Itô trace term is common to both generators and cancels from the difference. Combining with  $\mathbb{E}[\tilde{W}_i^T \dot{\tilde{W}}_i]$  (unaffected, since  $\hat{W}_i$  solves the ordinary differential recursion (35) driven by the realized trajectory), the exact mean rate of (59) is

$$\mathbb{E}[\dot{L}_i(t)] = \mathbb{E} \left[ \int_{t-T}^t \delta_i^T F_i d\tau + \tilde{W}_i^T \dot{\tilde{W}}_i + \mathcal{A}_i^{\hat{u}} V_i \right] + \frac{1}{2} \mathbb{E} \left[ \int_{t-T}^t \text{tr}(\Sigma_i^T \Sigma_i) d\tau \right], \quad (63)$$

which we bound above by

$$\mathbb{E}[\dot{L}_i(t)] \leq \mathbb{E} \left[ \int_{t-T}^t \delta_i^T F_i d\tau + \tilde{W}_i^T \dot{\tilde{W}}_i + \mathcal{A}_i^{\hat{u}} V_i \right] + c_{\sigma_i} T, \quad (64)$$

where  $c_{\sigma_i} \triangleq \frac{1}{2} \sup_{x \in \Omega} \text{tr}(\Sigma_i(x)^T \Sigma_i(x))$  is bounded by Assumption 4, so that  $c_{\sigma_i} T \geq \frac{1}{2} \mathbb{E} \left[ \int_{t-T}^t \text{tr}(\Sigma_i^T \Sigma_i) d\tau \right]$  for every window of length  $T$ . Note  $c_{\sigma_i} \rightarrow 0$  exactly as  $\Sigma_i \rightarrow 0$ , recovering the deterministic  $\dot{L}_i(t)$  in that limit. Every expectation from here until (60) is conditional on the regime under discussion.

*Regime*  $\xi_i(t) = 0$ . On this event  $\delta_i^T(t) \delta_i(t) \leq \delta_i^T(t-T) \delta_i(t-T)$ , so the left-hand side of (61) is non-positive. Taking  $\mathbb{E}[\cdot | \xi_i(t) = 0]$  of (61) and dropping the non-negative trace term gives

$$\mathbb{E}_0 \left[ \int_{t-T}^t \delta_i^T F_i d\tau \right] + \mathbb{E}_0 \left[ \int_{t-T}^t \delta_i^T \Sigma_i d\mathcal{W}_i(\tau) \right] \leq 0. \quad (65)$$

Unlike the deterministic case, the martingale term here need *not* vanish upon conditioning:  $\{\xi_i(t) = 0\}$  is an  $\mathcal{F}_t$ -measurable event that depends on  $\delta_i(t)$ , hence on the very martingale increment being conditioned on, so the usual "martingale  $\Rightarrow$  zero conditional mean" property – valid for conditioning on an  $\mathcal{F}_{t-T}$ -measurable event – does not apply here. Consequently (65) only certifies that the drift term and the (generally nonzero, sign-uncontrolled) conditioned martingale term *together* sum to something non-positive, not that the drift term does so on its own; it is retained here only as motivation for the switching rule, not as a derivation step. The strict, quantified decrease actually used below is instead supplied directly by Assumption 5, i.e. by (58), which is posited exactly in the conditional form needed and does not rely on (65).

Two bounds are needed before substituting. First, from (62) and the definition of  $r_i$ , the generator along the applied policies obeys

$$\mathcal{A}_i^{\hat{u}} V_i \leq -Q_{ii}(\delta_i) + \varepsilon_{\pi_i} \leq -\lambda_{\min}(Q_{ii}) \|\delta_i\|^2 + \varepsilon_{\pi_i}, \quad \varepsilon_{\pi_i} \triangleq \sup_{\Omega} \left[ \mathcal{A}_i^{\hat{u}} V_i + Q_{ii}(\delta_i) \right], \quad (66)$$

which is finite on the compact set  $\Omega$  by Assumption 4. The constant  $\varepsilon_{\pi_i}$  absorbs both the policy-mismatch bracket of (62) and the sign-indefinite disturbance-cost terms  $+\gamma^2 \vartheta_i^T T_{ii} \vartheta_i + \gamma^2 \sum_j \vartheta_j^T T_{ij} \vartheta_j$  contained in  $-r_i$ ; it vanishes when  $\hat{u}_i \rightarrow u_i^*$ ,  $\hat{\vartheta}_i \rightarrow \vartheta_i^*$  and  $\vartheta_i \rightarrow 0$ . Second,

$-\alpha_i \mathbb{E}_0[\tilde{W}_i^T \Xi_i \tilde{W}_i] \leq -\alpha_i \lambda_{\min}(\Xi_i) \mathbb{E}_0 \|\tilde{W}_i\|^2$ , with  $\lambda_{\min}(\Xi_i) > 0$  by Condition 1. Substituting these together with (43) and (58),

$$\begin{aligned} \mathbb{E}_0[\dot{L}_i(t)] &\leq -\zeta_i \mathbb{E}_0 \left[ \int_{t-T}^t \|\delta_i(\tau)\| d\tau \right] - \lambda_{\min}(Q_{ii}) \mathbb{E}_0 \|\delta_i\|^2 - \alpha_i \lambda_{\min}(\Xi_i) \mathbb{E}_0 \|\tilde{W}_i\|^2 \\ &\quad + [\alpha_i(l+1)\epsilon_{max} + l\theta_{max_i}] \mathbb{E}_0 \|\tilde{W}_i\| + \varepsilon_{\pi_i} + c_{\sigma_i} T, \end{aligned} \quad (67)$$

where, since  $\xi_i(t) = 0$  on this event,  $\mathcal{T}_i(t) = \sum_{k=1}^l \xi_i(t_k) \Theta_i(t_k)$  contains only the historical sum, so the Cauchy–Schwarz bound after (43) sharpens to  $-\mathbb{E}_0[\tilde{W}_i^T \mathcal{T}_i(t)] \leq l\theta_{max_i} \mathbb{E}_0 \|\tilde{W}_i\|$  (one fewer term than the general  $(l+1)\theta_{max_i}$  bound), while  $\alpha_i \mathbb{E}_0[\tilde{W}_i^T \mathcal{E}_i(t)] \leq \alpha_i(l+1)\epsilon_{max} \mathbb{E}_0 \|\tilde{W}_i\|$  as before. The first three terms of (67) are non-positive and act on  $\mathbb{E}_0 \int \|\delta_i\|$ ,  $\mathbb{E}_0 \|\delta_i\|^2$  and  $\mathbb{E}_0 \|\tilde{W}_i\|^2$  respectively, while the remaining terms are the perturbation. Hence  $\mathbb{E}_0[\dot{L}_i(t)] < 0$  holds as soon as *any one* of the three dominates that perturbation, in particular whenever

$$\mathbb{E}_0 \left[ \int_{t-T}^t \|\delta_i(\tau)\| d\tau \right] \geq \frac{[\alpha_i(l+1)\epsilon_{max} + l\theta_{max_i}] \mathbb{E}_0 \|\tilde{W}_i\| + \varepsilon_{\pi_i} + c_{\sigma_i} T}{\zeta_i} \triangleq \hat{\chi}_{1i} \quad (68)$$

or, completing the square in  $\mathbb{E}_0 \|\tilde{W}_i\|$  in the third and fourth terms,

$$\mathbb{E}_0 \|\tilde{W}_i\| \geq \frac{\alpha_i(l+1)\epsilon_{max} + l\theta_{max_i}}{\alpha_i \lambda_{\min}(\Xi_i)} + \sqrt{\frac{\varepsilon_{\pi_i} + c_{\sigma_i} T}{\alpha_i \lambda_{\min}(\Xi_i)}} \triangleq \chi_{1i} \quad (69)$$

where  $\lambda_{\min}(\Xi_i)$  is the smallest eigenvalue of  $\Xi_i$ . Both thresholds reduce to their deterministic counterparts as  $c_{\sigma_i} \rightarrow 0$ ; the  $\theta_{max_i}$ ,  $\epsilon_{max}$  and  $\varepsilon_{\pi_i}$  contributions are approximation-level terms that persist in the deterministic limit.

*Regime  $\xi_i(t) = 1$ .* Here the boundary term of (61) is positive, so the contraction (65) is unavailable and the  $\delta_i$ -negativity must instead come from the  $V_i$  summand of the *same* Lyapunov function (59) – this is where carrying  $V_i \geq 0$  in both regimes pays off, since no change of Lyapunov function, and hence no jump term, is incurred at a switching instant.

The window term is bounded without reference to  $F_i$ , Assumption 5, or any window/ $T$ -dependent quantity at all: since  $\delta_i^T(t-T)\delta_i(t-T) \geq 0$  always, the exact (Leibniz) identity for this term gives, pathwise,

$$\frac{d}{dt} \left[ \frac{1}{2} \int_{t-T}^t \delta_i^T \delta_i d\tau \right] = \frac{1}{2} \delta_i^T(t) \delta_i(t) - \frac{1}{2} \delta_i^T(t-T) \delta_i(t-T) \leq \frac{1}{2} \|\delta_i(t)\|^2. \quad (70)$$

For the  $V_i$  summand, expanding  $\mathcal{A}_i^{\hat{u}} V_i$  by (62) and writing  $r_i \triangleq Q_{ii}(\delta_i) + u_i^T R_{ii} u_i + \sum_{j \in \mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii} \vartheta_i - \gamma^2 \sum_{j \in \mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j$  for the instantaneous cost of (13), and  $\Gamma_i \triangleq g_i R_{ii}^{-1} g_i^T$ ,  $\Gamma_j \triangleq \sum_{j \in \mathcal{N}_i} g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T$ ,  $\Lambda_i \triangleq k_i T_{ii}^{-1} k_i^T$ ,  $\Lambda_j \triangleq \sum_{j \in \mathcal{N}_i} k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T$  with  $\Gamma_{i_{\min}} \leq \|\Gamma_i\| \leq \Gamma_{i_{\max}}$ ,  $\Gamma_{j_{\min}} \leq \|\Gamma_j\| \leq \Gamma_{j_{\max}}$ ,  $\Lambda_{i_{\min}} \leq \|\Lambda_i\| \leq \Lambda_{i_{\max}}$ ,  $\Lambda_{j_{\min}} \leq \|\Lambda_j\| \leq \Lambda_{j_{\max}}$ , using  $-Q_{ii}(\delta_i) \leq -\lambda_{\min}(Q_{ii}) \mathbb{E} \|\delta_i\|^2$  and the worst-case bounds on  $u_i^T R_{ii} u_i + \sum_j u_j^T R_{ij} u_j$  and  $\vartheta_i^T T_{ii} \vartheta_i + \sum_j \vartheta_j^T T_{ij} \vartheta_j$  implied by the feedback forms (32)–(33), the generator obeys

$$\mathcal{A}_i^{\hat{u}} V_i \leq -\lambda_{\min}(Q_{ii}) \|\delta_i\|^2 + \frac{1}{2} b_{\nabla \phi_i} (\gamma^2 \Lambda_{i_{\max}} - \Gamma_{i_{\min}}) \|\delta_i\| + \frac{1}{2} b_{\nabla \phi_j} (\gamma^2 \Lambda_{j_{\max}} - \Gamma_{j_{\min}}) \|\delta_j\| + c_{V_i}, \quad (71)$$

where  $c_{V_i} \triangleq \frac{1}{2} \sup_{x \in \Omega} \text{tr}(\Sigma_i(x)^T \nabla^2 V_i(\delta_i) \Sigma_i(x))$  is the Itô-correction term of the generator (18) itself, evaluated at the applied policies – finite on  $\Omega$  by Assumption 4 (which bounds  $\nabla^2 \phi_i$ ,  $\nabla^2 \epsilon_i$ ,  $\Sigma_i$ , and hence  $\nabla^2 V_i = W_i^T \nabla^2 \phi_i + \nabla^2 \epsilon_i$ ) – and *distinct* from  $c_{\sigma_i}$  of (64):  $c_{\sigma_i}$  bounds the unrelated Hessian-of- $\delta_i^T \delta_i$  trace term  $\text{tr}(\Sigma_i^T \Sigma_i)$  used only in the windowed identity (61) of Regime  $\xi_i(t) = 0$ , whereas  $c_{V_i}$  bounds the trace term intrinsic to  $\mathcal{A}_i^{\hat{u}} V_i$  via  $\nabla^2 V_i$ . Like  $c_{\sigma_i}$ ,  $c_{V_i} \rightarrow 0$  exactly as  $\Sigma_i \rightarrow 0$ ; and since (71) is the exact pointwise generator evaluated at  $t$  (not an integral over  $[t-T, t]$ ), no

window/ $T$ -dependent bound is needed here, unlike in Regime  $\xi_i(t) = 0$ . Combining (70) and (71) with (43),

$$\begin{aligned}\mathbb{E}_1[\dot{L}_i(t)] &\leq \left(\frac{1}{2} - \lambda_{\min}(Q_{ii})\right)\mathbb{E}_1\|\delta_i\|^2 + \frac{1}{2}b_{\nabla\phi_i}(\gamma^2\Lambda_{i_{\max}} - \Gamma_{i_{\min}})\mathbb{E}_1\|\delta_i\| \\ &\quad + \frac{1}{2}b_{\nabla\phi_j}(\gamma^2\Lambda_{j_{\max}} - \Gamma_{j_{\min}})\mathbb{E}_1\|\delta_j\| - \alpha_i\mathbb{E}_1[\tilde{W}_i^T\Xi_i\tilde{W}_i] \\ &\quad + \alpha_i\mathbb{E}_1[\tilde{W}_i^T\mathcal{E}_i(t)] - \mathbb{E}_1[\tilde{W}_i^T\mathcal{T}_i(t)] + c_{V_i}\end{aligned}\quad (72)$$

Applying the identity  $x = (x + \frac{1}{2})^2 - x^2 - \frac{1}{4}$  to the  $\mathbb{E}\|\delta_i\|$ -linear term, so that it combines with  $\mathbb{E}\|\delta_i\|^2$  into a single square, and bounding the  $\tilde{W}_i$ -cross terms by the Cauchy–Schwarz inequalities established after (43), gives the fully collected bound

$$\begin{aligned}\mathbb{E}_1[\dot{L}_i(t)] &\leq \left(\frac{1}{2} - \lambda_{\min}(Q_{ii}) + \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{\min}} - \frac{1}{2}\gamma^2b_{\nabla\phi_i}\Lambda_{i_{\max}}\right)\mathbb{E}_1\|\delta_i\|^2 \\ &\quad + \left(\frac{1}{2}\gamma^2b_{\nabla\phi_i}\Lambda_{i_{\max}} - \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{\min}}\right)\left(\mathbb{E}_1\|\delta_i\| + \frac{1}{2}\right)^2 + \frac{1}{8}b_{\nabla\phi_i}(\Gamma_{i_{\min}} - \gamma^2\Lambda_{i_{\max}}) \\ &\quad + \frac{1}{2}b_{\nabla\phi_j}(\gamma^2\Lambda_{j_{\max}} - \Gamma_{j_{\min}})\mathbb{E}_1\|\delta_j\| - \alpha_i\lambda_{\min}(\Xi_i)\mathbb{E}_1\|\tilde{W}_i\|^2 \\ &\quad + [\alpha_i(l+1)\epsilon_{\max} + (l+1)\theta_{\max i}]\mathbb{E}_1\|\tilde{W}_i\| + c_{V_i}\end{aligned}\quad (73)$$

To ensure the negativity of  $\mathbb{E}_1[\dot{L}_i(t)]$ , writing  $D_i \triangleq 2\lambda_{\min}(Q_{ii}) - 1 + \gamma^2b_{\nabla\phi_i}\Lambda_{i_{\max}} - b_{\nabla\phi_i}\Gamma_{i_{\min}} > 0$  for twice the magnitude of the (negative)  $\mathbb{E}\|\delta_i\|^2$  coefficient in (73), it is required that

$$\mathbb{E}_1\|\delta_i\|^2 \geq \frac{b_{\nabla\phi_i}(\Gamma_{i_{\min}} - \gamma^2\Lambda_{i_{\max}})}{4D_i} + \frac{2c_{V_i}}{D_i} \triangleq \hat{\chi}_{2i} \quad (74)$$

or

$$\mathbb{E}_1\|\tilde{W}_i\| \geq \frac{\alpha_i(l+1)\epsilon_{\max} + (l+1)\theta_{\max i}}{\alpha_i\lambda_{\min}(\Xi_i)} + \sqrt{\frac{c_{V_i} + \frac{1}{8}b_{\nabla\phi_i}(\Gamma_{i_{\min}} - \gamma^2\Lambda_{i_{\max}})}{\alpha_i\lambda_{\min}(\Xi_i)}} \triangleq \chi_{2i} \quad (75)$$

The first term of (74) is the constant  $\frac{1}{8}b_{\nabla\phi_i}(\Gamma_{i_{\min}} - \gamma^2\Lambda_{i_{\max}})$  of (73) divided by  $\frac{1}{2}D_i$ , and the second is  $c_{V_i}$  divided by the same quantity; the numerator carries the full factor  $(\Gamma_{i_{\min}} - \gamma^2\Lambda_{i_{\max}})$ , positive under the sign conditions imposed next. Those conditions, which also let the sign-indefinite  $(\mathbb{E}_1\|\delta_i\| + \frac{1}{2})^2$  term and the linear  $\mathbb{E}_1\|\delta_j\|$  term of (73) be dropped as non-positive, are

$$\begin{aligned}\left(\frac{1}{2} - \lambda_{\min}(Q_{ii}) + \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{\min}} - \frac{1}{2}\gamma^2b_{\nabla\phi_i}\Lambda_{i_{\max}}\right) &< 0, \\ \left(\frac{1}{2}\gamma^2b_{\nabla\phi_i}\Lambda_{i_{\max}} - \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{\min}}\right) &< 0, \quad \left(\frac{1}{2}\gamma^2b_{\nabla\phi_j}\Lambda_{j_{\max}} - \frac{1}{2}b_{\nabla\phi_j}\Gamma_{j_{\min}}\right) < 0\end{aligned}$$

*Recombination.* Both regimes bound the mean rate of the *same* function (59), so by the tower property (60) the unconditional rate  $\mathbb{E}[\dot{L}_i(t)]$  is a convex combination of the two conditional rates and is therefore negative whenever both are. Collecting (68)–(69) and (74)–(75),  $\mathbb{E}[\dot{L}_i(t)] < 0$  holds outside the set on which

$$\mathbb{E}_1\|\delta_i\|^2 < \hat{\chi}_{2i}, \quad \mathbb{E}_0\left[\int_{t-T}^t \|\delta_i(\tau)\| d\tau\right] < \hat{\chi}_{1i}, \quad \mathbb{E}_0\|\tilde{W}_i\| < \chi_{1i}, \quad \mathbb{E}_1\|\tilde{W}_i\| < \chi_{2i} \quad (76)$$

all hold simultaneously. Since  $\mathbb{E}\|\tilde{W}_i\| \leq \max\{\mathbb{E}_0\|\tilde{W}_i\|, \mathbb{E}_1\|\tilde{W}_i\|\}$  by (60), the weight-error bound aggregates to the unconditional statement  $\mathbb{E}\|\tilde{W}_i\| \leq \max\{\chi_{1i}, \chi_{2i}\}$ .

For the tracking error the two thresholds are stated on *different functionals* and must not be merged:  $\hat{\chi}_{2i}$  bounds the pointwise second moment, whereas  $\hat{\chi}_{1i}$  bounds the windowed first moment  $\mathbb{E}_0\int_{t-T}^t \|\delta_i\| d\tau$ . They are related in one direction only: by Jensen's and Cauchy–Schwarz inequalities,

$$\left(\mathbb{E}\int_{t-T}^t \|\delta_i(\tau)\| d\tau\right)^2 \leq T \int_{t-T}^t \mathbb{E}\|\delta_i(\tau)\|^2 d\tau, \quad (77)$$

so a bound on the windowed first moment does not by itself bound  $\mathbb{E}\|\delta_i(t)\|^2$  at the endpoint. The ultimate bound on the tracking error is therefore the pointwise one,  $\hat{\chi}_{2i}$ , obtained on  $\{\xi_i = 1\}$  – the regime in which  $\|\delta_i\|$  is growing, and hence the regime that determines the ultimate bound – while  $\hat{\chi}_{1i}$  certifies that the windowed error cannot remain large on the contracting regime. Because  $\{\xi_i(t) = 1\}$  is precisely the event on which  $\|\delta_i\|$  increases over the window,  $\mathbb{E}\|\delta_i(t)\|^2$  can exceed  $\hat{\chi}_{2i}$  only on  $\{\xi_i(t) = 0\}$ , where it is non-increasing over the window; hence  $\hat{\chi}_{2i}$  is an ultimate bound for the unconditional second moment as well.

Mean-square uniform ultimate boundedness of the augmented state and the weight estimation error is thus established:  $\mathbb{E}\|\delta_i\|^2$  is ultimately bounded by  $\hat{\chi}_{2i}$  and  $\mathbb{E}\|\tilde{W}_i\|$  by  $\max\{\chi_{1i}, \chi_{2i}\}$ , thresholds that reduce to their deterministic counterparts as  $c_{\sigma_i}, c_{V_i} \rightarrow 0$ , i.e. as  $\Sigma_i \rightarrow 0$ .

*Near-optimality.* The remaining claims of the theorem follow from the bound on  $\tilde{W}_i$  and Assumption 4. From (25) and (30),  $V_i^* - \hat{V}_i = \tilde{W}_i^T \phi_i + \epsilon_i$ , so

$$\mathbb{E}\|V_i^* - \hat{V}_i\|^2 \leq 2b_{\phi_i}^2 \mathbb{E}\|\tilde{W}_i\|^2 + 2b_{\epsilon_i}^2 \triangleq \epsilon_{V_i}. \quad (78)$$

Likewise, subtracting (32) from (20) and (33) from (21) and using  $\nabla V_i^* = \nabla \phi_i^T W_i + \nabla \epsilon_i$ ,

$$\mathbb{E}\|u_i^* - \hat{u}_i\|^2 \leq \frac{1}{2}(b_i + d_i)^2 \|R_{ii}^{-1}\| \Gamma_{i_{max}} (b_{\nabla \phi_i}^2 \mathbb{E}\|\tilde{W}_i\|^2 + b_{\nabla \epsilon_i}^2) \triangleq \epsilon_{u_i}, \quad (79)$$

using  $\|R_{ii}^{-1} g_i^T v\|^2 \leq \|R_{ii}^{-1}\| v^T \Gamma_i v \leq \|R_{ii}^{-1}\| \Gamma_{i_{max}} \|v\|^2$ ; analogously  $\mathbb{E}\|\vartheta_i^* - \hat{\vartheta}_i\|^2 \leq \epsilon_{\vartheta_i}$  with  $R_{ii}^{-1}, \Gamma_{i_{max}}, \frac{1}{2}$  replaced by  $T_{ii}^{-1}, \Lambda_{i_{max}}, \frac{1}{2\gamma^2}$ . Each of  $\epsilon_{V_i}, \epsilon_{u_i}, \epsilon_{\vartheta_i}$  is therefore small whenever  $\max\{\chi_{1i}, \chi_{2i}\}$  and the NN approximation errors are small; they shrink to the deterministic residual, not to zero, as  $\Sigma_i \rightarrow 0$ , since the approximation errors  $b_{\epsilon_i}, b_{\nabla \epsilon_i}$  persist and vanish only as  $L \rightarrow \infty$ . The proof is thus complete. ■

□

**Corollary 1** (Model-free case). *Let the dynamics (11) be completely unknown: neither the drift  $f_{ei}$ , nor the input channels  $g_i, k_i$ , nor the diffusion  $\Sigma_i$  is available. Suppose the behaviour signals  $u_j = \hat{u}_j + e_j$  are applied with bounded exploration  $e_j$ , the policies and cross-channel maps are parameterised as in (49), and the weights  $\mathcal{Z}_i$  of (50) are updated by the adaptive law (35) under the substitution (53). If the enlarged rank condition of Section 4.3 holds, then  $\mathbb{E}\|\delta_i(t)\|^2$  and  $\mathbb{E}\|\tilde{\mathcal{Z}}_i(t)\|^2$ ,  $\tilde{\mathcal{Z}}_i \triangleq \mathcal{Z}_i^* - \mathcal{Z}_i$ , are mean-square UUB with the thresholds  $\hat{\chi}_{1i}, \hat{\chi}_{2i}, \chi_{1i}, \chi_{2i}$  of Theorem 1, in which  $\epsilon_{max}$  is replaced by the corresponding bound on the off-policy residual of (48) and  $\lambda_{min}(\Xi_i), \lambda_{max}(\Xi_i)$  by the extreme eigenvalues of the enlarged Gram matrix built from  $\Lambda_i$ .*

*Proof.* The proof of Theorem 1 uses the adaptive law (35) only through the error dynamics (43), and (43) only through three properties: (i) the Gram matrix  $\Xi_i$  is positive semidefinite, with  $\lambda_{min}(\Xi_i) > 0$  guaranteed by the rank condition; (ii) the residual term satisfies  $\mathbb{E}\|\epsilon_{B_i}\|^2 \leq \epsilon_{max}^2$ ; and (iii) the stabilizing term is bounded,  $\|\Theta_i\| \leq \theta_{max_i}$ . None of the three refers to the *contents* of the regressor. Under (53), (i) holds by the enlarged rank condition, (ii) holds because (48) differs from (27) only by terms that are exact under (46)–(47), so its residual is again the NN approximation error of (29) augmented by the actor and cross-channel approximation errors – all bounded in mean square on  $\Omega$  by Assumption 4 applied to the additional bases  $\psi_{u_i}, \psi_{\vartheta_i}, \psi_{ij}$  – and (iii) holds as established after (57). The Lyapunov argument of Theorem 1 therefore applies verbatim with  $\tilde{W}_i$  replaced by  $\tilde{\mathcal{Z}}_i$ . ■

□

#### 4.5 Robustness of the learning process to estimation errors

Theorem 1 and Corollary 1 analyse the *idealized* recursion, in which  $\Delta \phi_i$  (resp.  $\Lambda_i$ ),  $\varrho_i$  and  $\Theta_i$  are the exact quantities appearing in (27) and (48). No implementation computes them exactly. This subsection separates the idealized algorithm from what is actually executed, treats the difference as an exogenous input, and shows that the guarantee degrades gracefully – linearly in the size of that input, and not at all in the tracking channel to first order. The construction is the continuous-time, single-trajectory counterpart of the inexact-policy-evaluation analysis carried out for iteration-indexed policy iteration in [22, 26].

**Remark 7** (Where the error comes from). *Four mechanisms perturb the quantities entering (35), and it is worth keeping them apart because they scale differently with the designer's choices.*

- (E1) Sampling error in the conditional expectation. Both (24) and (48) hold under  $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$ , which is realized either from one trajectory or from  $\bar{L}$  rollouts (Remark 5). The single-trajectory choice is conditionally unbiased but contributes a zero-mean fluctuation of order  $\sqrt{T} b_{\nabla\phi_i} b_{\Sigma_i}$  by the Itô isometry; the batch choice contributes a residual of order  $\bar{L}^{-1/2}$ .
- (E2) Quadrature error. The window integrals  $\int_{t-T}^t(\cdot) d\tau$  are evaluated by summation on a mesh  $h$ . Because  $\delta_i$  is an Itô process, its sample paths are Hölder continuous of every order below  $\frac{1}{2}$  rather than differentiable, so the left-endpoint sum incurs an error of order  $Th^{\frac{1}{2}-\varepsilon}$  for any  $\varepsilon > 0$ , not the  $O(Th)$  of the deterministic case.
- (E3) Identification error in the auxiliary quantities. In the model-free scheme,  $\Upsilon_i, \Upsilon_i^\theta, \Upsilon_{ij}, \Upsilon_{ij}^\theta$  of (54) are obtained by least squares from the same data, so  $\Theta_i^{\text{mf}}$  is itself an estimate; its error is governed by the conditioning of that auxiliary regression.
- (E4) Communication and measurement error. The neighbour signals  $\{\delta_j, u_j, \vartheta_j, \hat{u}_j, \hat{\vartheta}_j\}_{j \in \mathcal{N}_i}$  required by  $\varrho_i$  and by (51) arrive quantized and delayed over the graph.

Only (E1) is intrinsic to the stochastic setting; (E2)–(E4) have deterministic counterparts. Crucially, (E1) and (E2) are the two the designer controls directly, through  $\bar{L}$  and  $h$  – which is what makes part (iii) of Proposition 1 actionable rather than merely reassuring.

**The inexact recursion.** Write  $\mathcal{Z}_i$  for the parameter being adapted, which is the critic weight  $\hat{W}_i$  in the model-based case and the enlarged vector (50) in the model-free one, and let  $\Lambda_i$  denote its regressor under the correspondence (53). What the implementation actually forms is

$$\hat{\Lambda}_i(t) = \Lambda_i(t) + \Delta_i^\Lambda(t), \quad \hat{\varrho}_i(t) = \varrho_i(t) + \Delta_i^\varrho(t), \quad \hat{\Theta}_i(t) = \Theta_i(t) + \Delta_i^\Theta(t), \quad (80)$$

with the three perturbations produced by the mechanisms of Remark 7. Collect them into the aggregated error signal and its magnitude

$$\Delta_i(t) \triangleq [\Delta_i^\Lambda(t)^T, \Delta_i^\varrho(t), \Delta_i^\Theta(t)^T]^T, \quad \|\Delta_i\|_\infty \triangleq \sup_{t \geq 0} (\mathbb{E} \|\Delta_i(t)\|^2)^{1/2}, \quad (81)$$

which plays here exactly the role that the aggregated policy-evaluation error  $\Delta M^i$ , and its  $l_\infty$ -norm, play in the inexact policy-evaluation step of [22]. Running (35) on (80) in place of the exact quantities perturbs the error dynamics (43) into

$$\dot{\tilde{\mathcal{Z}}}_i(t) = -\alpha_i \hat{\Xi}_i \tilde{\mathcal{Z}}_i(t) + \alpha_i \hat{\mathcal{E}}_i(t) - \hat{\mathcal{T}}_i(t), \quad (82)$$

in which  $\hat{\Xi}_i, \hat{\mathcal{E}}_i, \hat{\mathcal{T}}_i$  are (40)–(42) built from the perturbed quantities. Comparing (82) with (43) shows that the whole effect of the implementation error is confined to the three objects on which the proof of Theorem 1 actually depends – which is what makes the following statement short.

**Proposition 1** (Mean-square robustness of the learning process). *Let the hypotheses of Theorem 1 hold in the model-based case, or those of Corollary 1 in the model-free case, and let the algorithm be executed on the inexact quantities (80) with  $\mathcal{Z}_i$  remaining in the compact set  $\Omega_{\mathcal{Z}}$  on which Assumption 4 is posed,  $\|\mathcal{Z}_i\| \leq z_{\max}$ . Define*

$$\delta_{1i} \triangleq \frac{\lambda_{\min}(\Xi_i)}{2(l+1)}. \quad (83)$$

*Then, provided  $\|\Delta_i\|_\infty < \delta_{1i}$ :*

- (i) (Well-posedness of the concurrent-learning term.)  $\lambda_{\min}(\hat{\Xi}_i) \geq \lambda_{\min}(\Xi_i) - (l+1)\|\Delta_i\|_\infty > \frac{1}{2}\lambda_{\min}(\Xi_i) > 0$  for all  $t$ , so the Gram matrix of the history stack remains positive definite and (82) remains strictly contracting in its nominal part.

- (ii) (Ultimate bound, inflated by a class- $\mathcal{K}$  function of the error.)  $\mathbb{E}\|\tilde{Z}_i(t)\|$  is ultimately bounded by  $\max\{\chi_{1i}, \chi_{2i}\} + \beta_i(\|\Delta_i\|_\infty)$ , where

$$\beta_i(s) \triangleq \frac{2(l+1)[\alpha_i(z_{max}+1)+1]}{\alpha_i\lambda_{min}(\Xi_i)}s + \frac{2(l+1)[\alpha_i\epsilon_{max}+\theta_{max_i}]}{\alpha_i\lambda_{min}(\Xi_i)} \cdot \frac{(l+1)s}{\lambda_{min}(\Xi_i) - (l+1)s}, \quad (84)$$

with  $\beta_i \in \mathcal{K}$  on  $[0, \delta_{1i})$ ,  $\beta_i(0) = 0$ , and  $\chi_{1i}, \chi_{2i}$  the nominal thresholds of Theorem 1. The tracking-error threshold  $\hat{\chi}_{2i}$  of (74) is unchanged, since it involves neither  $\epsilon_{max}$ , nor  $\theta_{max_i}$ , nor  $\lambda_{min}(\Xi_i)$ : implementation error degrades the parameter estimate directly and the tracking error only indirectly, through the near-optimality constants (78)–(79).

- (iii) (Consistency.)  $\|\Delta_i\|_\infty \rightarrow 0$  recovers the conclusions of Theorem 1 and Corollary 1 exactly. More generally, if  $\mathbb{E}\|\Delta_i(t)\|^2 \rightarrow 0$  as  $t \rightarrow \infty$ , then  $\limsup_{t \rightarrow \infty} \mathbb{E}\|\tilde{Z}_i(t)\| \leq \max\{\chi_{1i}, \chi_{2i}\}$ : an implementation whose accuracy is refined over time inherits the nominal guarantee asymptotically.

*Proof.* The proof of Theorem 1 uses the adaptive law only through the three properties recalled in the proof of Corollary 1: positive definiteness of  $\Xi_i$ , the residual bound  $\mathbb{E}\|\epsilon_{B_i}\|^2 \leq \epsilon_{max}^2$ , and  $\|\Theta_i\| \leq \theta_{max_i}$ . It therefore suffices to propagate (80) through those three.

(i) By construction  $\|\Delta_i\| = \|\Lambda_i\|/(\Lambda_i^T \Lambda_i + 1) \leq \frac{1}{2}$ , so the normalized regressors are bounded by  $\frac{1}{2}$  irrespective of the data, and the same holds for the perturbed ones. Hence each of the  $l+1$  rank-one terms of  $\hat{\Xi}_i - \Xi_i$  has spectral norm at most  $\|\Delta_i(t_k)\|$ , giving  $\|\hat{\Xi}_i - \Xi_i\|_2 \leq (l+1)\|\Delta_i\|_\infty$  in mean square. Weyl's inequality then yields the stated lower bound on  $\lambda_{min}(\hat{\Xi}_i)$ , which is at least  $\frac{1}{2}\lambda_{min}(\Xi_i)$  under  $\|\Delta_i\|_\infty < \delta_{1i}$  by (83).

(ii) The realized residual satisfies  $\hat{\Psi}_i = \hat{\Lambda}_i^T Z_i + \hat{\varrho}_i = \Psi_i + \Delta_i^{\Lambda^T} Z_i + \Delta_i^{\varrho}$ , so on  $\Omega_Z$

$$(\mathbb{E}\|\hat{\epsilon}_{B_i}(t)\|^2)^{1/2} \leq \epsilon_{max} + (z_{max}+1)\|\Delta_i\|_\infty, \quad \|\hat{\Theta}_i(t)\| \leq \theta_{max_i} + \|\Delta_i\|_\infty,$$

whence the two Cauchy–Schwarz bounds stated after (43) hold with  $\epsilon_{max}$  and  $\theta_{max_i}$  replaced by these inflated constants. Substituting them, together with the bound of part (i) on  $\lambda_{min}(\hat{\Xi}_i)$ , into (75) and into the corresponding threshold (69) of the contracting regime gives

$$\chi_{2i}(\Delta_i) = \frac{(l+1)[\alpha_i\epsilon_{max}+\theta_{max_i}] + (l+1)\|\Delta_i\|_\infty[\alpha_i(z_{max}+1)+1]}{\alpha_i(\lambda_{min}(\Xi_i) - (l+1)\|\Delta_i\|_\infty)} + \sqrt{\frac{cV_i + \frac{1}{8}b_{\nabla\phi_i}(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}})}{\alpha_i(\lambda_{min}(\Xi_i) - (l+1)\|\Delta_i\|_\infty)}}.$$

Splitting the first fraction about its nominal value, and using  $\lambda_{min}(\Xi_i) - (l+1)\|\Delta_i\|_\infty \geq \frac{1}{2}\lambda_{min}(\Xi_i)$  in the perturbation part, gives  $\chi_{2i}(\Delta_i) \leq \chi_{2i} + \beta_i(\|\Delta_i\|_\infty)$  with  $\beta_i$  as in (84);  $\beta_i$  is continuous, strictly increasing and vanishes at the origin on  $[0, \delta_{1i})$ , hence is of class  $\mathcal{K}$  there. The same substitution leaves (74) untouched, since none of the three perturbed constants appears in it. The remainder of the argument – the two regimes, their recombination by the tower property, and the near-optimality step – is verbatim that of Theorem 1.

- (iii) The first claim is  $\beta_i(0) = 0$ . For the second, note that on the event  $\{\lambda_{min}(\hat{\Xi}_i) \geq \frac{1}{2}\lambda_{min}(\Xi_i)\}$  guaranteed by (i), the nominal part of (82) is uniformly exponentially stable and the driving input has mean-square magnitude tending to zero; the converging-input converging-state property of such a system, applied to the mean rate of (59), gives the stated  $\limsup$ . ■

**Remark 8** (Reading the bound). *Proposition 1 is the counterpart, for the present continuous-time scheme, of the local input-to-state stability result established for inexact policy iteration on the generalized algebraic Riccati equation in [22, Thm. 3], and the differences are as informative as the similarity. There, the state is the Riccati iterate  $\hat{P}^i$ , the input is the policy-evaluation error  $\Delta M^i$ , and the bound  $\|\hat{P}^i - P^*\|_F \leq \alpha(\|\hat{P}^0 - P^*\|_F, i) + \beta(\|\Delta M\|_{l_\infty})$  holds locally, for initializations  $\hat{P}^0$  already inside a ball around the optimum – a restriction the authors identify as an open problem. Here the state is the parameter error  $\tilde{Z}_i$  in continuous time, and no such proximity to the optimum is required: admissibility of the initialization and the rank condition suffice, which is what one would expect from a Lyapunov argument rather than a Newton–Kantorovich one. Conversely, the price is that (84) is an ultimate bound and not a convergence rate, whereas the Newton structure available in*

the linear-quadratic problem delivers a local quadratic rate. Substituting the orders of Remark 7 into (84) gives the practical reading: the excess over the nominal ultimate bound is  $O(h^{\frac{1}{2}-\varepsilon} + \bar{L}^{-1/2})$ , so halving it costs a four-fold increase in the number of rollouts, or a mesh refinement by a factor slightly worse than four – the usual Monte-Carlo and strong-order- $\frac{1}{2}$  scalings, arriving here through the ultimate bound rather than through a pathwise error estimate.

## 5 Simulation results

In this section, simulations are conducted to validate the performance of the proposed control strategy. The scenario involves a multi-WMR system, where a virtual leader WMR (indexed by 0) guides the formation. The information flow among WMRs is depicted in the directed graph shown in Fig. 1a.

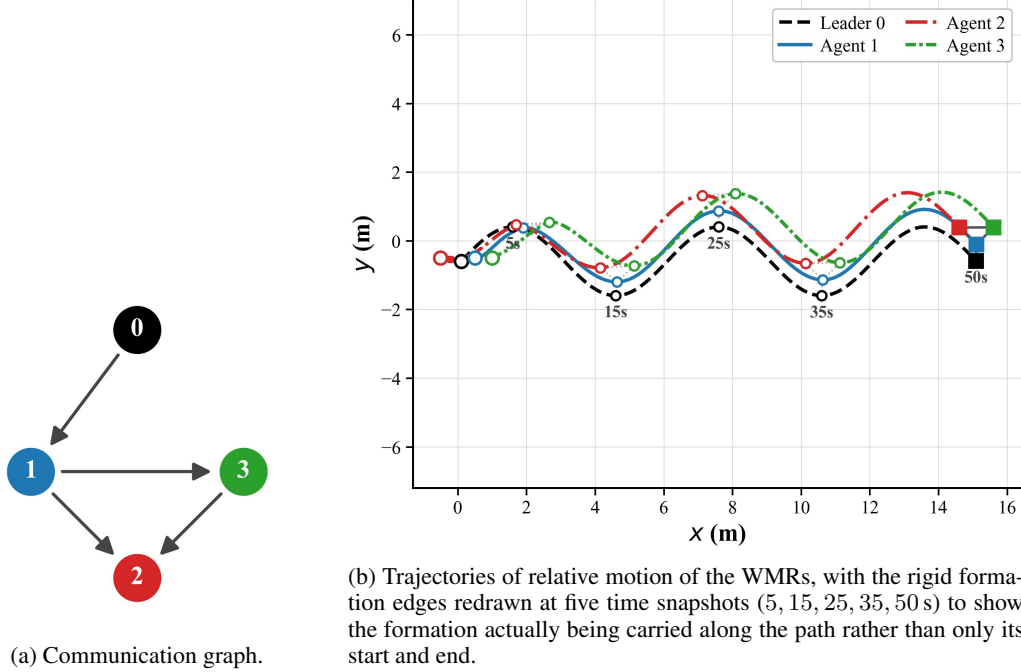


Figure 1: Communication topology and the resulting formation trajectory.

The desired trajectory deviation are set to  $\Delta_1 = [0, -0.5, 0]^T$ ,  $\Delta_2 = [0.5, -1, 0]^T$ ,  $\Delta_3 = [-0.5, -1, 0]^T$ . The initial position of WMR are respectively  $q_0 = [0.1, -0.6, 0.808]^T$ ,  $q_1 = [0.5, -0.5, 0]^T$ ,  $q_2 = [-0.5, -0.5, 0]^T$ ,  $q_3 = [1, -0.5, \pi/6]^T$ . The initial velocity vectors of the WMRs are set to  $\eta_1 = \eta_2 = \eta_3 = [0, 0]^T$ , and the simulation horizon is 50 s with step size 0.01 s.

The virtual leader traces a *serpentine* (sinusoidal S-curve) reference path

$$x_0(t) = x_0(0) + v_x t, \quad y_0(t) = y_0(0) + A \sin(\omega t), \quad (85)$$

which alternates straight and curved segments and reverses its curvature twice per period, so that the reference is genuinely time-varying rather than a steady-state circular motion. The corresponding unicycle velocity reference is obtained from  $v_0 = \sqrt{\dot{x}_0^2 + \dot{y}_0^2}$  and  $\omega_0 = (\ddot{y}_0 \dot{x}_0 - \ddot{x}_0 \dot{y}_0) / (\dot{x}_0^2 + \dot{y}_0^2)$ , giving

$$\eta_0 = \begin{bmatrix} v_0 \\ \omega_0 \end{bmatrix} = \begin{bmatrix} \sqrt{v_x^2 + (A\omega \cos(\omega t))^2} \\ \frac{-A\omega^2 v_x \sin(\omega t)}{v_x^2 + (A\omega \cos(\omega t))^2} \end{bmatrix}$$

where  $v_x = 0.3$  m/s,  $A = 1$  m and  $\omega = \pi/10$  rad/s, yielding 2.5 complete periods over the simulation horizon,  $v_0 \in [0.300, 0.434]$  m/s and  $\omega_0 \in [-0.329, 0.329]$  rad/s. The leader head-

ing is initialised on the path tangent,  $\theta_0(0) = \arctan(A\omega/v_x) \approx 0.808$  rad, so that integrating  $\dot{q}_0 = H_0(q_0)\eta_0$  reproduces (85) exactly. Note that  $\omega_0$  here oscillates about zero instead of being constant, which excites the critic regressor more richly than a circular reference and makes the rank requirement of Condition 1 easier to meet: the recorded history stacks reach  $\text{rank}(Z_i) = L = 15$  for all three agents.

The NNs activation functions are selected as

$$\phi_i(\delta_i) = [\delta_{xi}^2, \delta_{xi}\delta_{yi}, \delta_{xi}\delta_{\theta_i}, \delta_{xi}\delta_{vi}, \delta_{xi}\delta_{\omega_i}, \delta_{yi}^2, \delta_{yi}\delta_{\theta_i}, \delta_{yi}\delta_{vi}, \delta_{yi}\delta_{\omega_i}, \delta_{\theta_i}^2, \delta_{\theta_i}\delta_{vi}, \delta_{\theta_i}\delta_{\omega_i}, \delta_{vi}^2, \delta_{vi}\delta_{\omega_i}, \delta_{\omega_i}^2].$$

and the initial NNs weight vectors are set to  $\hat{W}_i(0) = 0.5 \cdot \mathbf{1}_{15}$  for all three agents. The learning rates are  $\alpha_i = 0.01$  and  $\beta_i = 0.01$ ,  $Q_{ii}(e_i) = e_i^T Q_i e_i$ ,  $Q_i = \text{diag}[1, 1, 1, 1, 1]$ ,  $T_{ii} = T_{ij} = I$ , and  $R_{ii} = R_{ij} = 0.3I$  – recall that the control and disturbance laws (32)–(33) act through  $R_{ii}^{-1}$ ,  $T_{ii}^{-1}$ , so this weight corresponds to a feedback gain  $R_{ii}^{-1} \approx 3.33I$ , not  $0.3I$ . A sweep of the control-penalty weight over  $R_{ii} = R_{ij} = cI$ ,  $c \in \{1, 0.7, 0.5, 0.4, 0.3\}$  – a *smaller*  $c$  giving a *larger* gain  $R_{ii}^{-1} = c^{-1}I$  – showed the post-transient tracking error and cost shrinking as  $c$  is reduced from 1 toward 0.4, after which the trend is no longer monotone and the closed loop becomes numerically unstable below  $c \approx 0.28$ . We keep  $c = 0.3$ , close enough to this margin that the two safety mechanisms of (35) are actually exercised: removing either the stabilizing term  $\xi_i \Theta_i$  or the weight adaptation destabilizes the closed loop within about three seconds (Sec. 5.3), whereas at looser gains (e.g.  $c = 0.4$ – $0.5$ ) both ablated variants remain stable and that demonstration is not visible. Varying  $Q_i$  or  $T_{ii}, T_{ij}$  over the same range left the closed loop virtually unchanged, since only  $R_{ii}, R_{ij}$  enter the control law  $u_i^r$  directly. The selection of physical parameters of each WMR model is chosen as  $r_i = 0.05$  m,  $l_i = 0.25$  m,  $m_1 = 10$  kg,  $m_2 = m_3 = 5$  kg. The history stack of Condition 1 retains  $l = 20 > L$  samples per agent, admitted at a randomized interval of 0.15–0.35 s (mean 0.25 s) and evicted first-in-first-out once full, so that the stored regressors are drawn from a rolling recent window rather than clustering in one transient; the mean admission rate is chosen so the stack finishes filling within the persistent-excitation window below, rather than on an unrelated, independent timescale.  $\text{rank}(Z_i) = L = 15$  is maintained throughout for all three agents.

The evolution of the position states  $x_i$ ,  $y_i$ , and  $\theta_i$  for robots 1, 2, 3, along with the virtual leader 0, are illustrated in Figs. 2a–2c. The learning progress of the critic neural network weights for each agent is presented in Figs. 3a–3c: all fifteen weights of each agent move together, since they are updated from the same regressor stack, and settle to constant values once that stack fills, by  $t \approx 4.7$  s, coinciding with the end of the persistent-excitation window; they remain flat for the rest of the horizon. Furthermore, the 2D trajectories of both the leader and the followers are depicted in Fig. 1b, confirming that the proposed online learning-based control scheme enables the multi-agent system to achieve optimal synchronization.

Quantitatively, the formation tracking errors  $\|[\delta_{xi}, \delta_{yi}]\|$  decay from 0.57, 1.12 and 0.50 m at  $t = 0$ , though not all agents settle at the same rate: agents 1 and 3 are down to 0.03 m and 0.01 m by  $t = 25$  s, while agent 2 – whose initial error is more than twice the others’ and which sits in the middle of the communication graph, coupled to both – traverses a slower, extended transient and does not reach its own residual band until  $t \approx 28$ – $30$  s. From  $t = 30$  s onward, all three agents remain within a common residual band of roughly 0.00–0.05 m rather than converging to zero. This residual is the expected signature of the mean-square UUB result of Theorem 1: under persistent process noise the errors cannot vanish, and the ultimate bound  $\hat{\chi}_{2i}$  of (74) is non-zero for  $\Sigma_i \neq 0$ . Consistent with this bound trading off against the feedback gain through  $\Gamma_{i,max}$ , the sweep above shows the residual growing monotonically as  $c$  increases toward 1 (i.e. as the gain  $R_{ii}^{-1}$  weakens), and at the operating point  $c = 0.3$  the residual shows only a mild, low-amplitude wobble with no growth over the horizon. These results demonstrate that the proposed approach drives the system states to track a genuinely time-varying reference under process noise, thereby validating the efficacy of the developed method.

The individual neighborhood tracking errors  $e_x$ ,  $e_y$ , and  $e_\theta$  that feed the critic NN are shown in Figs. 4a–4c, confirming the same transient decay and bounded residual behavior discussed above

at the component level; each signed component still fluctuates about zero at this small amplitude rather than converging to it exactly, consistent with a non-zero (if now much smaller) mean-square ultimate bound. The corresponding position-error norm  $\|\delta_i\| = \sqrt{e_x^2 + e_y^2}$ , shown in Fig. 4d, makes this UUB behavior easier to read at a glance: it is non-negative by construction, decays over the transient (monotonically for agents 1 and 3, with agent 2’s slower shoulder visible until  $t \approx 30$  s as discussed above), and then settles into the 0.00–0.05 m residual band quoted above. The estimated value functions  $\hat{V}_i = \hat{W}_i^T \phi_i(\delta_i)$ , shown in Fig. 5a, start from the transient peak induced by the initial tracking error and decay to a small residual as the critic weights converge, consistent with the near-optimality bound  $\mathbb{E}\|V_i^* - \hat{V}_i\|^2 < \epsilon_{V_i}$  of Theorem 1. The resulting control inputs  $v_i$  and  $\omega_i$  are shown in Figs. 5b–5c: the large-amplitude fluctuations over  $[0, 5]$  s are the injected persistent-excitation exploration noise used to populate the history stack (Condition 1), after which the control smoothly tracks the leader’s serpentine reference velocity without chattering.

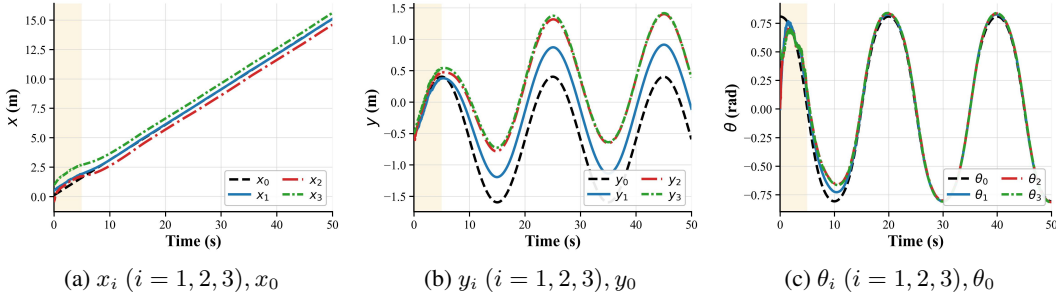


Figure 2: Position and heading trajectories of the leader and the three followers.

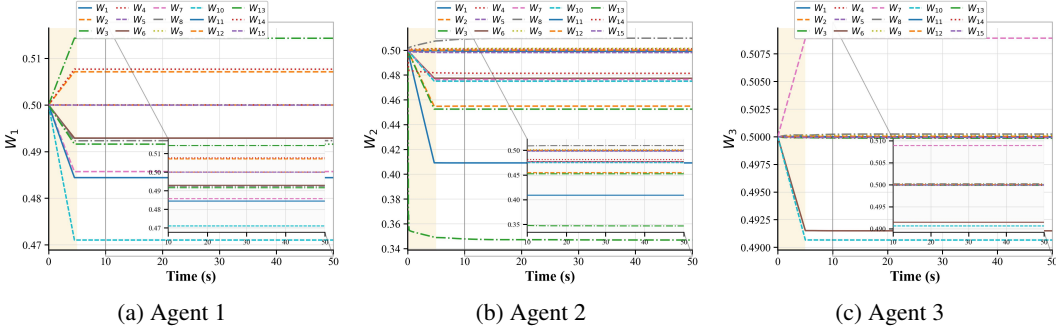


Figure 3: Weight update trajectories of the critic NNs, agents 1–3.

## 5.1 Sensitivity to the process-noise intensity

Theorem 1 claims that the mean-square ultimate bound  $\hat{\chi}_{2i}$  reduces exactly to the deterministic (pathwise) UUB bound as  $\Sigma_i \rightarrow 0$ , and grows with the noise intensity otherwise. To check this directly rather than only asserting it, we repeat the full experiment above – same topology, reference, initial conditions, learning rates, and  $R_{ii} = 0.3I$  – at  $\sigma_\eta \in \{0, 0.03, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$  (recall  $\sigma_\eta = 0.03$  is the value used in every other figure in this section), keeping the seed and hence the realized Brownian path fixed up to the rescaling by  $\sigma_\eta$ . Figure 7 reports the settled residual band, mean and max of  $\|\delta_i\|$  over  $t > 30$  s, against  $\sigma_\eta$ , together with the full worst-agent trajectories.

Three observations follow. First,  $\sigma_\eta = 0$  does *not* give zero residual: the settled band is mean = 0.022 m, consistent with the theorem’s own remark that the near-optimality constants  $\epsilon_{V_i}, \epsilon_{u_i}, \epsilon_{\partial_i}$  persist in the deterministic limit and vanish only as the critic basis is refined ( $L \rightarrow \infty$ ), not as  $\Sigma_i \rightarrow 0$  alone. Second, the residual grows with  $\sigma_\eta$  essentially monotonically – from mean = 0.022 m at  $\sigma_\eta = 0$  to 0.057 m at  $\sigma_\eta = 0.6$ , roughly a  $2.6\times$  increase, and more sharply so in the max statistic (0.049 m  $\rightarrow$  0.241 m) – exactly the qualitative trend  $\hat{\chi}_{2i}$  predicts through its  $c_{V_i}$  term, which is itself

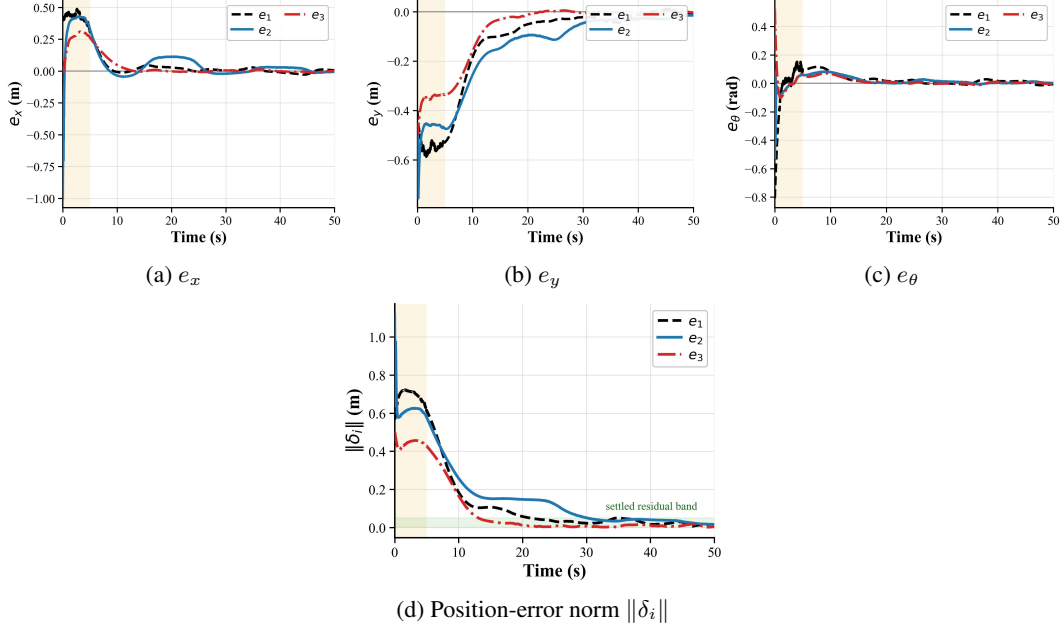


Figure 4: Neighborhood tracking errors  $e_x$ ,  $e_y$ ,  $e_\theta$  ( $i = 1, 2, 3$ ), and the resulting position-error norm.

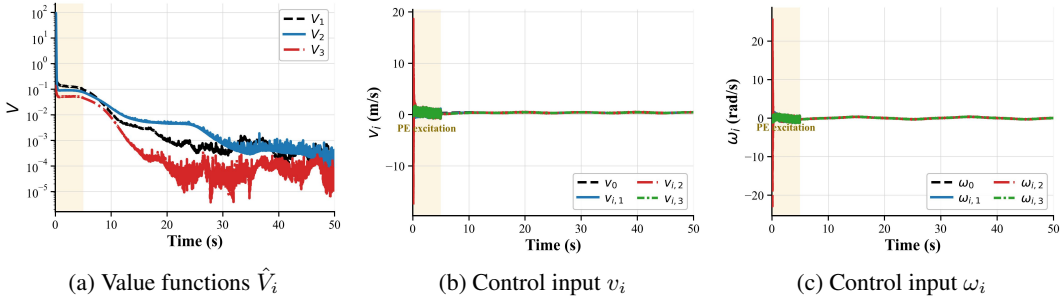


Figure 5: Estimated value functions and the resulting control inputs  $v_i$ ,  $\omega_i$  ( $i = 1, 2, 3$ ).

increasing in  $\Sigma_i$ . Third, and visible in Fig. 7(right), the *transient* (dominated by the initial tracking error, not the noise) is essentially unaffected by  $\sigma_\eta$ : all eight traces overlap almost exactly for  $t < 25$  s and only separate in the settled regime, confirming that  $\sigma_\eta$  acts on the ultimate bound and not on the transient decay rate, as the theory requires. Beyond  $\sigma_\eta \approx 0.65$  the closed loop becomes numerically unstable, consistent with Assumptions 1 and 4 requiring the diffusion to remain within the regime over which the SDE is well-posed and the NN bases and their derivatives stay bounded.

## 5.2 Robustness to an exogenous disturbance

The results above already exercise the disturbance channel, since the worst-case policy  $\hat{\vartheta}_i$  of (33) is applied to the plant throughout as the self-play adversary (Fig. 6). This is not, however, a test against a disturbance the controller did not itself generate. To check the disturbance-attenuation condition (14) against a genuinely exogenous signal, we inject a bounded torque disturbance  $d_{\text{ext}}(t)$  – a low-frequency oscillation (0.3 rad/s, amplitude 1.5 N·m and half that on the two wheel channels) plus a 1 N·m step at  $t = 20$  s – into the dynamic channel of every agent, identically, on top of (not instead of) the existing self-play disturbance, and repeat the experiment with every other setting unchanged. Figure 8 compares the resulting worst-agent tracking error against the undisturbed baseline and shows the applied disturbance itself.

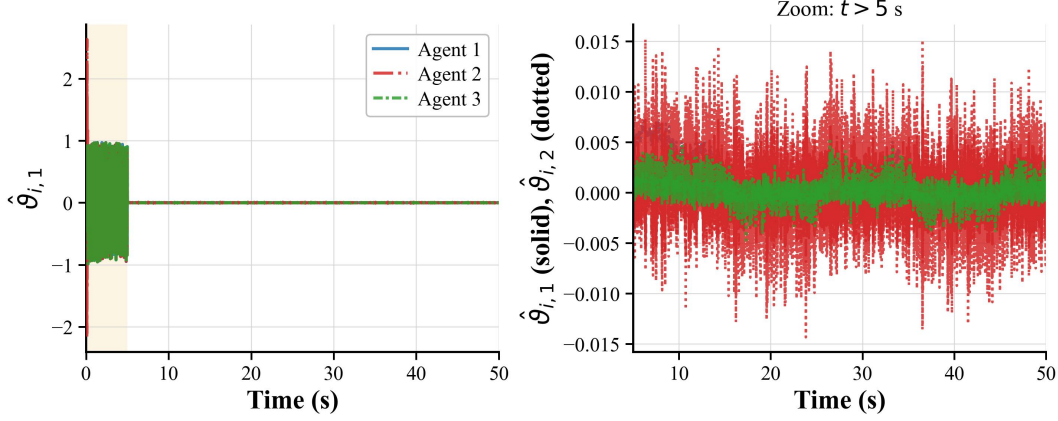


Figure 6: Estimated worst-case disturbance policy  $\hat{\vartheta}_i$  (33), dynamic-channel component 1. Left: full horizon, dominated over  $[0, 5]$  s by the same injected persistent-excitation noise seen in Figs. 5b–5c – shared identically across the three agents, which is why agents 1 and 3 (of comparable magnitude) are visually indistinguishable here. Right: zoom on  $t > 5$  s, both dynamic-channel components, showing the converged worst-case policy settling to a small, noise-driven fluctuation about zero rather than a fixed value, consistent with  $\vartheta_i \rightarrow 0$  as  $\delta_i \rightarrow 0$ .

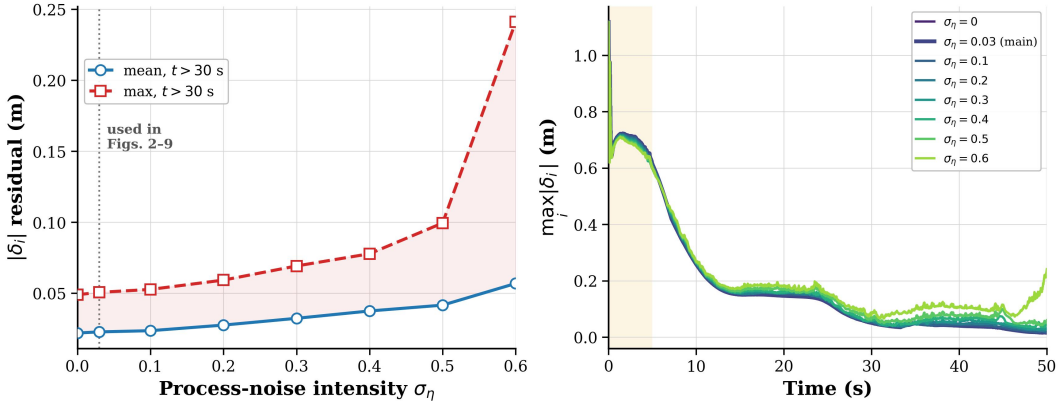


Figure 7: Sensitivity of the settled residual to the process-noise intensity  $\sigma_\eta$ . Left: mean and max of  $\|\delta_i\|$  over  $t > 30$  s vs.  $\sigma_\eta$ ; the dotted line marks  $\sigma_\eta = 0.03$ , the value used in every other figure in this section. Right: the full worst-agent trajectory  $\max_i \|\delta_i(t)\|$  for each  $\sigma_\eta$ , showing the transient is essentially noise-independent while the settled band widens with  $\sigma_\eta$ .

The tracking error with and without  $d_{\text{ext}}$  is graphically indistinguishable for  $t < 35$  s and separates only modestly thereafter: the settled residual (mean over  $t > 25$  s) rises from 0.026 m to 0.045 m, and the disturbed trajectory does not grow further after the step at  $t = 20$  s or the oscillation’s subsequent cycles – it remains bounded, as (14) requires. Quantitatively, evaluating the two sides of (14) directly from the disturbed run over  $t > 25$  s (past the initial transient, so the  $\beta_i(\delta_i(0))$  term is not dominant) gives  $\sum_i \int \|z_i\|^2 dt = 1.13$  against  $\sum_i \int \|\omega_i\|^2 dt = 450.3$ , a ratio of 0.0025 – comfortably below  $\gamma^2 = 1$ . The margin is large rather than tight, which is expected: as Remark 2 notes,  $\gamma$  is deliberately chosen conservatively in the absence of a closed-form optimal value, and the design weights  $R_{ii} = 0.3I$  were chosen (Sec. 5, first paragraph) for the tracking/stability trade-off of Sec. 5.3, not to tighten this particular margin.

### 5.3 Contribution of the learning component

A natural question is how much of the reported performance is actually attributable to the reinforcement-learning machinery, as opposed to the model-based terms  $v_a, \tau_a$  or to the particular

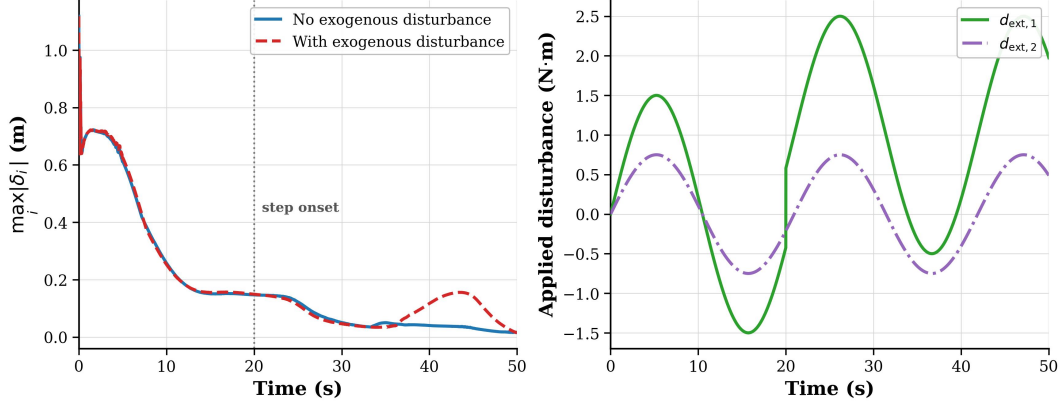


Figure 8: Robustness to a genuinely exogenous disturbance. Left: worst-agent position-error norm with and without the applied disturbance  $d_{\text{ext}}$  (vertical line: step onset). Right: the applied disturbance itself, on the two components of the dynamic channel.

initial weight guess. We therefore repeat the experiment with individual components of the scheme disabled, keeping every other setting (topology, reference, noise realisation, seed) fixed. The results are summarised in Table 1 and Fig. 9.

Table 1: Ablation of the proposed scheme.  $J = \sum_i \int_5^{50} (e_i^T Q_i e_i + u_i^{rT} R_{ii} u_i^r) dt$  is evaluated after the excitation window;  $\|\bar{\delta}\|$  is the mean position-error norm over  $t > 25$  s.

| Variant                                   | $J$               | $\ \bar{\delta}\ $ (m) |
|-------------------------------------------|-------------------|------------------------|
| Proposed (full)                           | 4.38              | 0.026                  |
| Without Bellman-residual / replay terms   | 4.49              | 0.025                  |
| Without stabilizing term $\xi_i \Theta_i$ | diverges          | —                      |
| Critic weights frozen at $\hat{W}_i(0)$   | diverges          | —                      |
| Without learned policy $u_i^r$            | $1.5 \times 10^8$ | 14.11                  |

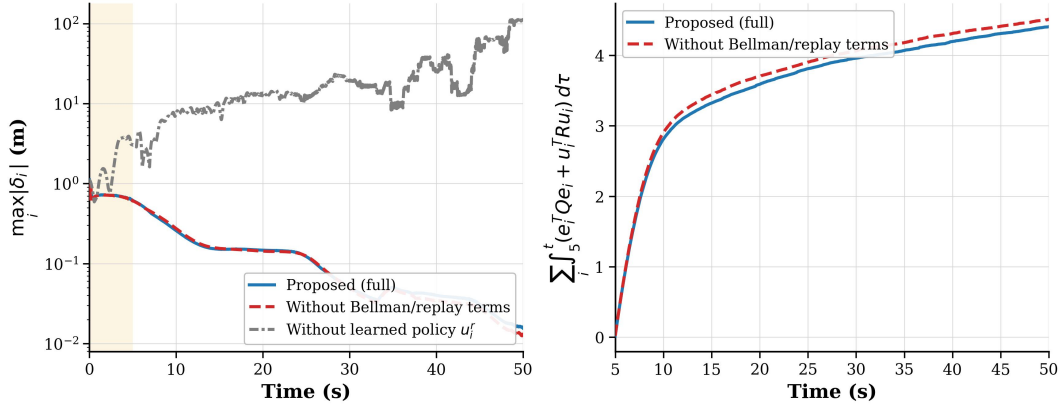


Figure 9: Contribution of the learning component. Left: worst-agent formation-error norm; removing the learned policy  $u_i^r$  leaves the model-based terms unable to stabilise the formation, whereas the full scheme and the variant without the Bellman/replay terms are graphically indistinguishable. Right: accumulated cost after the excitation window, where the two curves separate slightly — the Bellman-residual and experience-replay terms lower the cost functional by a modest  $\sim 2\%$  at equal tracking accuracy.

Three points follow. First, the learned policy is not a refinement of an already-working controller: with  $u_i^r$  removed the neighbourhood error grows past 14 m, so the model-based feedforward alone

does not stabilise the formation. Second, the two halves of the tuning law (35) play distinct roles. Removing the stabilizing term  $\xi_i \Theta_i$ , or freezing the weights at  $\hat{W}_i(0) = 0.5 \cdot \mathbf{1}$ , makes the closed loop diverge within about three seconds — unsurprisingly, since that initial guess assigns equal weight to every quadratic monomial and hence induces an indefinite  $\hat{V}_i$  and a non-stabilising policy at the feedback gain  $R_{ii}^{-1}$  used here. The stabilizing term is what carries the state out of that region; as noted above, this demonstration relies on operating close to the stability margin identified by the  $R_{ii}$  sweep; at a looser gain neither ablated variant diverges. Third, and most relevant to the claim of *optimality*, the Bellman-residual and experience-replay terms leave the tracking error essentially unchanged (0.026 versus 0.025 m) while reducing the cost functional from 4.49 to 4.38, a modest but consistent 2.4% improvement, split roughly evenly between the state term (3.32 versus 3.23) and the control-effort term (1.17 versus 1.16) rather than concentrated in either one. The effect is small because, at this operating point, even the frozen initial weights already yield a policy close enough to stabilizing that little further improvement is available before the residual band imposed by the process noise (Theorem 1) is reached; nonetheless the direction of the effect is exactly what the theory predicts:  $\xi_i \Theta_i$  secures the mean-square UUB guarantee of Theorem 1 and is responsible for the difference between stability and divergence, while the reinforcement-learning terms drive  $\hat{W}_i$  towards the solution of the coupled HJI equations and thus towards the optimal policy, lowering the cost at equal tracking accuracy. It also illustrates why tracking-error plots alone are a poor diagnostic for this class of scheme: the error is not the quantity being minimised, and a scheme can track well with a policy that is still measurably sub-optimal in cost.

#### 5.4 Verification of the model-free construction

Every result above uses the model-based scheme of Algorithm 1. This subsection examines the model-free extension of Section 4.3 on its own terms. The object under test is the off-policy Bellman equation (48) and the regressor (51) built from it: the claim is that they carry exactly the same information as their model-based counterparts while evaluating neither  $g_i$  nor  $k_i$ , and that the enlarged rank condition they impose is attainable from data.

A single off-policy data set is collected over 50 s under a fixed behaviour controller  $u_i^b = -(b_i + d_i)g_i^T(x_i)\delta_i$  — the policy induced by the naive guess  $V_i = \frac{1}{2}\|\delta_i\|^2$ , not an optimal one — perturbed by bounded exploration of amplitude 0.5 on both the control and the disturbance channel. Nothing after the collection touches the plant again, which is what the behaviour/target separation of Section 4.3 buys. Windows of length  $T = 0.1$  s are taken at a stride of 0.04 s, giving 1248 windows per agent. With the bases  $\psi_{u_i} = \psi_{ij}^u = \{\delta_i, \delta_i \cos \theta, \delta_i \sin \theta\}$  and  $\psi_{\vartheta_i} = \psi_{ij}^\vartheta = \{\delta_i\}$ , the enlarged parameter has  $\dim(\mathcal{Z}_i) = 95, 255, 175$  for agents 1, 2, 3 against  $L = 15$  for the model-based critic — the neighbourhood-size scaling of Section 4.3 made concrete, agent 2 being the one coupled to both others.

**The actor bases represent the ideal policies exactly.** Comparing  $\hat{u}_i = \hat{W}_{u_i}^T \psi_{u_i}$  against the model-based policy (32) evaluated along the same trajectory gives a maximum discrepancy of  $2.9 \times 10^{-15}$ ,  $1.7 \times 10^{-14}$  and  $3.9 \times 10^{-15}$  for the three agents, and  $\sim 10^{-16}$  for the disturbance policies: the representation is exact to machine precision, not approximate. Repeating the fit with a basis in  $\delta_i$  alone — the parameterisation one would write down by default — leaves relative errors of 24.1%, 29.3% and 17.3%, which would enter the guarantee as a permanent, irreducible actor approximation error. This is the quantitative form of the observation made after (49): for a nonholonomic vehicle the ideal actor carries the heading through  $S(\theta_i)$ , so the basis must be allowed to depend on the measured configuration and not on the neighbourhood error alone.

**The model-free regressor is algebraically identical to the model-based one.** Substituting into (51) the parameter  $\mathcal{Z}_i$  that the identities (46)–(47) claim is equivalent to a given critic weight  $W_i$  — constructed from the model purely so that the claim can be checked — and comparing  $\Lambda_i^T \mathcal{Z}_i$  against  $\Delta \phi_i^T W_i - \int \nabla V_i^T \Delta_i d\tau$  window by window gives a maximum difference of  $5.6 \times 10^{-16}$ ,  $2.9 \times 10^{-15}$  and  $7.2 \times 10^{-16}$  against signal magnitudes of 0.32, 0.66 and 0.51. The lumping of the own-channel products into the agent’s own policies, and of the graphical-game cross-channel terms into  $\Xi_{ij}^u, \Xi_{ij}^\vartheta$ , is therefore exact rather than an approximation licensed by small residuals.

**The rank condition is attainable – but demands more than “some” excitation.** With mutually independent exploration signals on all  $3 \times 4 \times 2$  input channels,  $\text{rank}(\Lambda_i)$  reaches  $\dim(\mathcal{Z}_i)$  exactly for every agent, at condition numbers 166,  $1.9 \times 10^3$  and 368. The corresponding model-based regressors are conditioned at 18.0, 8.8 and 16.0; the two-order-of-magnitude gap is the concrete price of removing the model, and it is precisely the quantity that Corollary 1 and Proposition 1 carry as  $\lambda_{\min}(\Xi_i)$  in the denominator of the ultimate bound. The failure mode is worth recording because it is easy to walk into: if the disturbance channel is probed with a scaled copy of the control probe rather than an independent signal – a natural implementation shortcut, since both are “just exploration noise” – the own-channel blocks of  $\Lambda_i$  become collinear, the rank drops by three for every agent, and the condition number jumps to  $\sim 10^{16}$ , i.e. numerical singularity, no matter how long the run. Figure 10 shows the two singular-value spectra side by side. The lesson is that the enlarged condition of Section 4.3 constrains not only how *much* exploration is injected but how *independently* it is injected across agents and across the control and disturbance channels; on hardware, where exploration competes with actuator limits, this is the binding constraint.

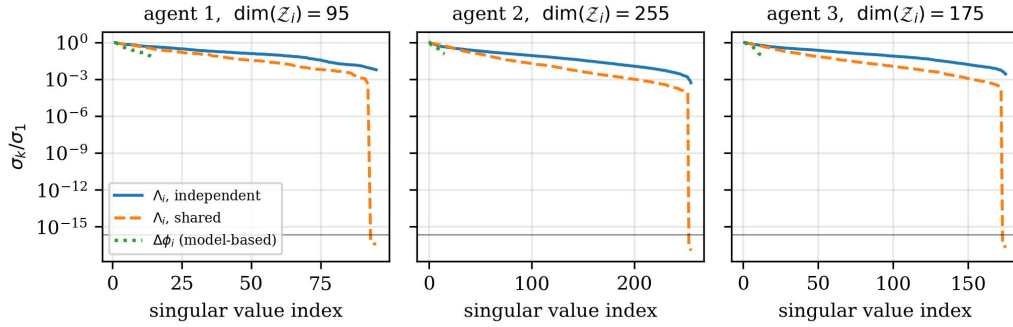


Figure 10: Normalised singular-value spectra of the model-free regressor  $\Lambda_i$  of (51), for the three agents, against that of the model-based regressor  $\Delta\phi_i$ . Solid: independent exploration on every input channel, for which  $\text{rank}(\Lambda_i) = \dim(\mathcal{Z}_i)$  exactly. Dashed: the disturbance channel probed with a scaled copy of the control probe, for which the spectrum collapses to machine precision (horizontal line) in three directions per agent and the rank condition fails. Dotted: the 15-dimensional model-based regressor on the same data, two orders of magnitude better conditioned – the price of removing  $g_i, k_i$ .

**What the three checks do and do not establish.** The checks above establish that the model-free construction is exact and that its excitation requirement is satisfiable, which are the claims specific to Section 4.3. They do not by themselves say what happens once the construction is placed in a loop, and one quantity makes clear why the distinction matters. On the same data, the relative residual of the *model-based* off-policy Bellman equation at the evaluation policy  $V_i = \frac{1}{2} \|\delta_i\|^2$  is 25.4%, 40.7% and 31.7%: the 15-monomial quadratic critic simply cannot satisfy (24) closely on this nonholonomic graphical game, which is the same finite-basis limitation that makes Theorem 1 a boundedness rather than a convergence statement. Proposition 1 then says exactly what to expect: an error of that size, divided by a  $\lambda_{\min}(\Xi_i)$  two orders of magnitude smaller than the model-based one, bounds a correspondingly larger parameter error, so a batch least-squares fit of the 255-dimensional  $\mathcal{Z}_2$  is not expected to land on the model-based fixed point – and, in our experiments, does not. Corollary 1 claims something weaker than that fixed point, namely mean-square boundedness of  $\delta_i$  and  $\tilde{\mathcal{Z}}_i$  under the enlarged rank condition, and that claim is tested directly in the next subsection.

## 5.5 Online closed-loop run of the model-free algorithm

Algorithm 2 is now run online on the same three-agent plant, with  $f_{ei}, g_i, k_i$  and  $\Sigma_i$  all unavailable to the learner: the applied control is  $u_j = \hat{u}_j + e_j$  with the actors  $\hat{u}_i = \hat{W}_{u_i}^T \psi_{u_i}$ ,  $\hat{\vartheta}_i = \hat{W}_{\vartheta_i}^T \psi_{\vartheta_i}$  of (49), and the enlarged parameter  $\mathcal{Z}_i$  is adapted by (35) under the substitution (53) on windows of the same length  $T = 0.1$  s used above. The stabilizing term  $\Theta_i^{\text{mf}}$  is driven by the auxiliary regression

(55), whose parameter is adapted by a normalised gradient step without a history stack of its own – legitimate because the proof uses  $\Theta_i^{\text{mf}}$  only through the bound  $\|\Theta_i\| \leq \theta_{\max_i}$ , so boundedness rather than accuracy of that estimate is what is required. Exploration is injected independently on all  $3 \times 4 \times 2$  channels, as Check 3 above requires, and is withdrawn after 25 s; nothing probes the plant thereafter.

Figure 11 shows the result. The formation error contracts from an initial 1.49 m to  $\|\delta_i\| = 0.014, 0.009, 0.006$  m at 50 s, averaging 0.041, 0.061 and 0.014 m over the post-exploration phase; the residual level is set by the process noise and by the withdrawal of the exploration signal, not by a drift in the parameters. The enlarged rank condition holds for the entire run –  $\text{rank}(\Lambda_i) = \dim(\mathcal{Z}_i) = 95, 255, 175$  at 25 s and at 50 s – which is what licenses Corollary 1 throughout rather than only while the probes are active.

That the loop is stable is necessary but not sufficient to claim the learning does the work, since the run must start from an admissible policy in any case. The comparison that isolates it is against the same initialization with the adaptation frozen,  $\alpha_i = 0$ : the cost  $\int (\delta_i^T Q_{ii} \delta_i + \hat{u}_i^T R_{ii} \hat{u}_i) dt$  accumulated over  $t > 5$  s falls from 98.1 to 79.4, a 19% reduction, and it falls in *both* terms – the state part from 49.2 to 35.1 and the control part from 48.9 to 44.3 – so the improvement is not a reallocation between tracking and effort. Repeating over three exploration seeds gives 98.1/100.2/105.8 frozen against 79.4/84.2/81.8 learned, with no divergent run.

Two caveats are worth stating. First,  $\|\mathcal{Z}_i(t)\|$  is still rising slowly at 50 s rather than settling, which is the expected signature of the 25–41% Bellman residual recorded above: with a critic basis that cannot represent  $V_i$  exactly, the guarantee available is the uniform ultimate boundedness of Theorem 1, not convergence to  $\mathcal{Z}_i^*$ , and the run exhibits precisely that. Second, agent 2 – coupled to both neighbours, and carrying the 255-dimensional parameter – learns the most ( $\|\mathcal{Z}_2\|$  grows by 49% against 10% for agent 1) and also retains the largest residual error, consistent with the  $|\mathcal{N}_i|$ -scaling of the excitation requirement being the dominant practical cost of removing the model.

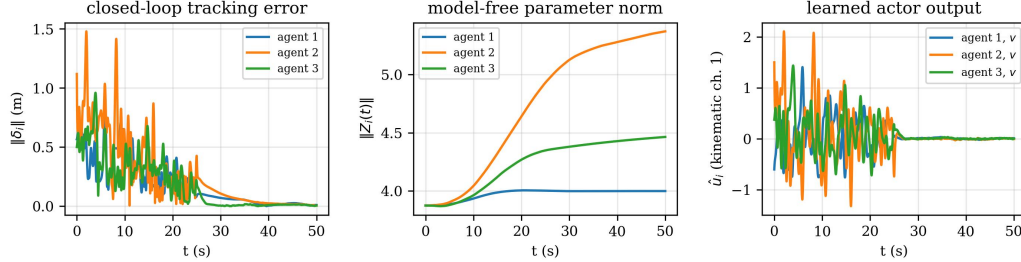


Figure 11: Online closed-loop run of Algorithm 2 with  $f_{ei}, g_i, k_i, \Sigma_i$  all unknown. Left: formation tracking error; exploration is withdrawn at 25 s, after which the error settles to the centimetre level. Centre: norm of the enlarged model-free parameter  $\mathcal{Z}_i$  of (50), which grows most for agent 2, the agent coupled to both neighbours; it has not plateaued by 50 s, the expected signature of uniform ultimate boundedness under a finite critic basis rather than convergence. Right: the learned actor output  $\hat{u}_i$  on the first kinematic channel.

**Two implementation conditions that turn out to be binding.** Both concern hypotheses that are stated in the theory and are easy to satisfy only nominally. The first is admissibility of the initialization (Remark 4). Because the certificate available without a model is an empirical one – candidate gains applied to the plant and retained if the trajectories stay bounded – its value depends entirely on the horizon over which it is scored, and the relevant instability here is slow: a candidate whose worst tracking error over a 4 s rollout is 1.17 m, an unremarkable figure, drives agent 2 past  $10^3$  m by 38 s with no learning, no exploration and no noise. The certificate must therefore be scored over the full run horizon; a short screen silently admits policies for which the standing hypothesis of Theorem 1 fails from  $t = 0$ . The second concerns the history stack. Once exploration is withdrawn,  $u_j^b - \hat{u}_j$  vanishes identically, so every actor and cross-channel block of  $\Lambda_i$  is exactly zero and the sample retains only its 15-dimensional critic block. Such samples are not degenerate in norm and

are admitted by any magnitude-based test, so a first-in-first-out stack is refilled with them within one window of the exploration ending, and  $\text{rank}(\Lambda_i)$  collapses from  $\dim(\mathcal{Z}_i)$  to  $L$  — the rank condition that Corollary 1 assumes is then violated for the remainder of the run, precisely when no further information can be acquired. Admitting by a singular-value criterion instead, so that a stored sample is replaced only when the replacement increases  $\lambda_{\min}$  of the stack, retains the exploration-era samples permanently and is what keeps the rank at  $\dim(\mathcal{Z}_i)$  in Figure 11. In the off-policy setting this is not a refinement but a requirement: the excitation is available only during a bounded initial window, and the stack is the only thing that carries it forward.

## 5.6 Computational cost

The claim that the scheme is usable online is a claim about cost per sampling instant, so it is measured rather than asserted. What is timed is the control-plus-learning block that a robot would have to execute at each instant — reference evaluation, neighbourhood errors, the feedforward  $u_a$ , policy evaluation, the Bellman residual and the weight update — excluding the recording arrays and the Euler–Maruyama integration of the plant, neither of which the controller performs. All figures are for the three-agent run of Section 5 at Step = 10 ms on one core of an Intel Core Ultra 7 155H, in unoptimised NumPy; a compiled implementation would be substantially faster, so these are upper bounds.

The model-based scheme of Algorithm 1 costs 1.72–1.81 ms per step in the mean over three runs (5001 steps each), with a median of 1.65–1.68 ms and a 99th percentile of 3.0–3.6 ms: between a fifth and a third of the 10 ms sampling period, so the loop closes in real time with roughly a five-fold margin.

The model-free law of Algorithm 2 is, perhaps counter-intuitively, *cheaper* per update at 0.89 ms in the mean (0.84 ms median, 1.65 ms at the 99th percentile), despite carrying  $\dim(\mathcal{Z}_i) = 95, 255, 175$  parameters against  $L = 15$ . The enlarged regressor is built from outer products of measured mismatches and the experience-replay sum is a single mat-vec against the stored stack, whereas the model-based running cost  $\varrho_i$  of (35) must form the products  $\nabla\phi_i g_i R_{ii}^{-1} g_i^T \nabla\phi_i^T$  and their disturbance counterparts for the agent *and* each neighbour at every step. Removing the model removes those products; the extra parameters cost less than the matrix triples they replace.

One qualification must be stated plainly, because it is the one place where the present implementation does exceed the budget. The history stack of Section 5.5 admits samples by the singular-value criterion, and evaluating that criterion as written — two full singular-value decompositions of the whole stack, before and after the trial replacement — costs 9.7, 28.7 and 18.3 ms for agents 1, 2, 3 respectively. On the steps where all three agents attempt an admission simultaneously this adds some 57 ms, which is why the end-to-end mean over the model-free run is 18–21 ms while its median is only 1.4–3.1 ms. The cost is an artefact of recomputing the decomposition from scratch rather than an intrinsic property of the criterion: the test fires at most once per admission gap, never after exploration is withdrawn (no informative sample can arrive once  $u_j - \hat{u}_j \equiv 0$ ), and a rank-one update of the stored Gram matrix would replace the decomposition entirely. We report the figure as measured rather than as it could be, and note that the 0.89 ms above is what the learning law itself costs once the admission test is excluded.

## 6 Conclusions

This study introduces a novel control strategy for achieving synchronization in multi-wheel mobile robot systems subject to both stochastic process noise and worst-case external disturbances. The proposed framework incorporates elements of distributed control, optimal control, and stochastic differential game theory to tackle the multi-agent zero-sum graphical game problem under Itô stochastic dynamics. The control scheme is developed based on a set of coupled stochastic HJI equations — which gain a genuine Itô-correction term over their deterministic counterpart and reduce to them exactly as the diffusion intensity vanishes — and the resulting solution ensures that the global synchronization error remains bounded in the  $L_2$ -norm in the mean-square sense.

Furthermore, an innovative online reinforcement learning (RL) algorithm is developed based on neural networks (NNs) to achieve optimal formation tracking under process noise. Each agent employs a single-layer NN to approximate its value function, while a concurrent learning mechanism utilizing historical data is introduced to relax the need for traditional persistent excitation (PE) conditions. This facilitates the adaptive tuning of NN parameters in real time, and the resulting closed-loop tracking error and weight estimation error are shown to be ultimately uniformly bounded in the mean-square sense, with an explicit noise-dependent ultimate bound.

Finally, Section 4.3 removes the remaining model dependence. Whereas the IRL-based Bellman formulation already eliminates the drift, the policies and the stabilizing term still evaluate the input channels  $g_i, k_i$  explicitly. An off-policy reformulation, in which the plant is driven by exploration-perturbed behaviour signals while the target policies are evaluated and improved, lumps the unknown products  $\nabla V_i^T g_i$  and  $\nabla V_i^T k_i$  into the agent's own policies through the stationarity conditions, and identifies the graphical-game cross-channel terms  $\nabla V_i^T g_j$  from data. Because the diffusion of (5) does not depend on the control or disturbance channels, no diffusion-dependent parameter has to be identified. Remark 3 isolates the property actually responsible for this: it is control-independence, not additivity – the same cancellation holds for the state-dependent multiplicative noise of [22], whose regression vector likewise contains no diffusion block, and fails only in the control-dependent case [21], where an extra lumped unknown is forced into the least-squares problem. The extension is a change of regressor rather than a change of algorithm, so the mean-square UUB guarantee carries over unchanged (Corollary 1), at the price of a correspondingly richer excitation requirement.

Section 4.5 then closes the gap between the idealized recursion and the one an implementation actually executes. The sampling error incurred in realizing the conditional expectations of (24) and (48), the quadrature error incurred on Hölder- $\frac{1}{2}$  Itô paths, the least-squares error in the auxiliary identification of (54), and communication error are aggregated into a single exogenous signal  $\Delta_i$ , and Proposition 1 shows that the parameter-error ultimate bound inflates by a class- $\mathcal{K}$  function of  $\|\Delta_i\|_\infty$ , that the tracking-error bound  $\hat{\chi}_{2i}$  is unaffected to first order, and that the nominal guarantee is recovered exactly as  $\|\Delta_i\|_\infty \rightarrow 0$ . This is the continuous-time, single-trajectory analogue of the input-to-state stability analysis of inexact policy iteration in [22]; unlike that result it imposes no proximity of the initialization to the optimal solution, at the price of concluding ultimate boundedness rather than a convergence rate.

Section 5.4 verifies the construction numerically: the actor and cross-channel bases represent the ideal quantities to machine precision, the model-free regressor is algebraically identical to the model-based one window by window, and the enlarged rank condition is attained exactly – provided the exploration signals are mutually independent across agents and across the control and disturbance channels, failing outright when the disturbance channel is probed with a scaled copy of the control probe. Section 5.5 then runs Algorithm 2 online on the full three-agent plant with the drift, both input channels and the diffusion all unknown: the formation error contracts to the centimetre level and stays there after the exploration is withdrawn, the enlarged rank condition holds for the whole run rather than only while the probes are active, and the adaptation accounts for a 19% reduction in accumulated cost relative to the same initialization held frozen, in both the state and the control term. Consistent with the finite critic basis,  $\mathcal{Z}_i$  remains bounded without settling, which is the uniform ultimate boundedness that Corollary 1 asserts and not convergence to  $\mathcal{Z}_i^*$ . Two hypotheses that the theory states but that are easy to satisfy only nominally turn out to be binding in that run: an admissibility certificate for the initialization is worthless unless scored over the full horizon, since the instability it must catch is slow; and the history stack must be admitted by a singular-value criterion rather than first-in-first-out, because once exploration stops every sample carries only its critic block and a magnitude-based test refills the stack with rank-deficient data exactly when no further information can be acquired.

Section 5.6 closes the loop on the practicality claim by measuring it: the model-based update costs 1.7 ms per sampling instant and the model-free one 0.89 ms – the latter cheaper despite carrying an order of magnitude more parameters, because the matrix triples  $\nabla \phi_i g_i R_{ii}^{-1} g_i^T \nabla \phi_i^T$  that the model-

based running cost must form for the agent and each neighbour disappear with the model – against a 10 ms period. The one component that does exceed the budget is the singular-value admission test as implemented, which recomputes two full decompositions of the stack and would be replaced by a rank-one update of its Gram matrix.

Three directions remain open. The first is the sample complexity of the enlarged excitation condition, which grows with the neighbourhood size  $|\mathcal{N}_i|$  and is the dominant practical cost of removing the model. The second is a converse to Remark 3: whether a stationarity-type identity can absorb a *control-dependent* diffusion into the policies themselves, which would extend the model-free construction to noise entering through the actuation channel. The third is critic expressiveness: the run of Section 5.5 is bounded but does not settle, and the 25–41% Bellman residual of the 15-monomial basis is what separates the boundedness obtained here from convergence to  $\mathcal{Z}_i^*$ ; a basis rich enough to close that gap, without inflating a parameter that already scales with  $|\mathcal{N}_i|$ , is the natural next step. Beyond these lies experimental validation on hardware, where the exploration signals required by the enlarged rank condition compete with actuator limits in a way simulation does not expose, and where the admissibility certificate of Remark 4 must be obtained without the luxury of rolling the real plant out to divergence.

## References

- [1] Q. Wu, X. Wang, and X. Qiu. Embedded technique-based formation control of multiple wheeled mobile robots with application to cooperative transportation. *Control Engineering Practice*, 150:106002, 2024.
- [2] K. Oh, M. Park, and H. Ahn. A survey of multi-agent formation control. *Automatica*, 53: 424–440, 2015.
- [3] F. Golmisheh and S. Shamaaghdari. Heterogeneous optimal formation control of nonlinear multi-agent systems with unknown dynamics by safe reinforcement learning. *Applied Mathematics and Computation*, 460:128302, 2024.
- [4] S. Yoo and B. Park. Distributed adaptive formation tracking using quantized feedback communication for networked mobile robots with unknown wheel slippage. *Nonlinear Analysis: Hybrid Systems*, 47:101294, 2023.
- [5] Y. Li, S. Dong, and K. Li. Fuzzy adaptive finite-time event-triggered control of time-varying formation for nonholonomic multirobot systems. *IEEE Transactions on Intelligent Vehicles*, 9 (1):725–737, 2024.
- [6] P. Sanila, A. Pradeep, J. Jacob, and R. Ramchand. Leader–follower target interception control of multi-robotic vehicles with holonomic dynamics based on unscented kalman filter. *Nonlinear Dynamics*, 111:11171–11190, 2023.
- [7] H. Pang, M. Liu, C. Hu, and F. Zhang. Adaptive sliding mode attitude control of two-wheel mobile robot with an integrated learning-based rbfn approach. *Neural Computing and Applications*, 34:14959–14969, 2022.
- [8] H. Qin, J. Si, N. Wang, and L. Gao. Fast fixed-time nonsingular terminal sliding-mode formation control for autonomous underwater vehicles based on a disturbance observer. *Ocean Engineering*, 270:113423, 2023.
- [9] H. Wang, N. Li, and Q. Luo. Adaptive fractional-order nonsingular fast terminal sliding mode formation control of multiple quadrotor UAVs-based distributed estimator. *Asian Journal of Control*, 25(5):3671–3686, 2023.
- [10] J. Long, D. Yu, G. Wen, L. Li, Z. Wang, and C. Philip Chen. Game-based backstepping design for strict-feedback nonlinear multi-agent systems based on reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 35(1):817–830, 2024.

- [11] C. Mu, Q. Zhao, Z. Gao, and C. Sun. Q-learning solution for optimal consensus control of discrete-time multiagent systems using reinforcement learning. *Journal of the Franklin Institute*, 356(13):6946–6967, 2019.
- [12] C. Chen, F. Lewis, K. Xie, Y. Lyu, and S. Xie. Distributed output data-driven optimal robust synchronization of heterogeneous multi-agent systems. *Automatica*, 153:111030, 2023.
- [13] J. Li, H. Modares, T. Chai, F. Lewis, and L. Xie. Off-policy reinforcement learning for synchronization in multiagent graphical games. *IEEE Transactions on Neural Networks and Learning Systems*, 28(10):2434–2445, 2017.
- [14] J. Zhu, G. Wen, and K. Veluvolu. Optimized backstepping consensus control using adaptive observer-critic-actor reinforcement learning for strict-feedback multi-agent systems. *Journal of the Franklin Institute*, 361(6):106693, 2024.
- [15] D. Zhang, Y. Yao, and Z. Wu. Reinforcement learning based optimal synchronization control for multi-agent systems with input constraints using vanishing viscosity method. *Information Sciences*, 637:118949, 2023.
- [16] T. Başar and P. Bernhard. *H-Infinity Optimal Control and Related Minimax Design Problems: A Dynamic Game Approach*. Springer Science & Business Media, 2008.
- [17] K. G. Vamvoudakis and F. L. Lewis. Online actor-critic algorithm to solve the continuous-time infinite horizon optimal control problem. *Automatica*, 46(5):878–888, 2010.
- [18] K. G. Vamvoudakis and F. L. Lewis. Online solution of nonlinear two-player zero-sum games using synchronous policy iteration. *International Journal of Robust and Nonlinear Control*, 22(13):1460–1483, 2012.
- [19] J. Yong and X. Y. Zhou. *Stochastic Controls: Hamiltonian Systems and HJB Equations*, volume 43. Springer Science & Business Media, 1999.
- [20] J. Sun, J. Yong, and S. Zhang. Linear quadratic stochastic two-person zero-sum differential games in an infinite horizon. *ESAIM: Control, Optimisation and Calculus of Variations*, 22(3):743–769, 2016.
- [21] Z. Sun and G. Jia. Robust policy iteration for continuous-time stochastic  $H_\infty$  control problem with unknown dynamics. Preprint, 2024.
- [22] W. Zhang, J. Guo, and X. Jiang. Model-free  $H_\infty$  control of itô stochastic system via off-policy reinforcement learning. *Automatica*, 174:112144, 2025.
- [23] F. Lewis, C. Abdallah, and D. Dawson. *Control of Robot Manipulators*. 1993.
- [24] H. Modares, F. Lewis, and Z. Jiang.  $H_\infty$  tracking control of completely unknown continuous-time systems via off-policy reinforcement learning. *IEEE Transactions on Neural Networks and Learning Systems*, 26(10):2550–2562, 2015.
- [25] Q. Jiao, H. Modares, S. Xu, F. Lewis, and K. Vamvoudakis. Multi-agent zero-sum differential graphical games for disturbance rejection in distributed control. *Automatica*, 69:24–34, 2016.
- [26] B. Pang, T. Bian, and Z.-P. Jiang. Robust policy iteration for continuous-time linear quadratic regulation. *IEEE Transactions on Automatic Control*, 67(1):504–511, 2022.

## Appendix

### A Appendix

*Proof of Lemma 1.* Let  $\bar{e}_i \triangleq [(p_i + \Delta_i - p_0)^T, (\eta_i - \eta_0)^T]^T \in \mathbb{R}^5$  denote the actual, leader-relative formation tracking error of agent  $i$  under the original dynamics (5): driving  $\mathbb{E}\|\bar{e}_i(t)\|^2 \rightarrow 0$  (or keeping it mean-square bounded) is precisely the objective of the distributed  $H_\infty$  optimal *formation tracking* problem stated for (5). Write  $\bar{e} \triangleq [\bar{e}_1^T, \dots, \bar{e}_n^T]^T$ .

Expanding the definitions of  $\delta_{pi}, \delta_{\eta i}$  preceding (7),

$$\delta_{pi} = \sum_{j \in \mathcal{N}_i} a_{ij} [(p_i + \Delta_i - p_0) - (p_j + \Delta_j - p_0)] + b_i(p_i + \Delta_i - p_0) = (d_i + b_i) \bar{e}_{pi} - \sum_{j \in \mathcal{N}_i} a_{ij} \bar{e}_{pj},$$

which is exactly row  $i$  of  $(\mathcal{D} + \mathcal{B} - \mathcal{A})\bar{e}_p = (\mathcal{L} + \mathcal{B})\bar{e}_p$ , where  $\bar{e}_{pi} \triangleq p_i + \Delta_i - p_0$ ; the identical computation for  $\delta_{\eta i}$  with  $\bar{e}_{\eta i} \triangleq \eta_i - \eta_0$  gives row  $i$  of  $(\mathcal{L} + \mathcal{B})\bar{e}_\eta$ . Stacking both channels,

$$\delta = T\bar{e}, \quad T \triangleq (\mathcal{L} + \mathcal{B}) \otimes I_5, \quad (86)$$

with  $\delta$  as defined before (10).

$T$  is invertible. By the standing assumptions of Section 2 the communication graph is strongly connected and at least one follower has  $a_{i0} = b_i > 0$ ; under these conditions  $\mathcal{L} + \mathcal{B}$  is a nonsingular  $M$ -matrix, a standard fact in the leader-follower consensus literature underlying graphical-game formulations such as [25] (every eigenvalue of  $\mathcal{L} + \mathcal{B}$  has strictly positive real part, since  $\mathcal{L}$  has a simple zero eigenvalue with eigenvector  $\mathbf{1}$  that is destroyed once  $\mathcal{B} \neq 0$  couples at least one node to the leader through a strongly connected graph). Hence  $\mathcal{L} + \mathcal{B}$ , and therefore  $T = (\mathcal{L} + \mathcal{B}) \otimes I_5$ , is invertible, with

$$\underline{\kappa}\|\bar{e}\| \leq \|\delta\| \leq \bar{\kappa}\|\bar{e}\| \quad (87)$$

for constants  $0 < \underline{\kappa} \leq \bar{\kappa} < \infty$  given by the extreme singular values of  $T$  and  $T^{-1}$ .

$T$  is exact, not merely infinitesimal. Because  $T$  is a *constant* matrix,  $\delta(t) = T\bar{e}(t)$  holds pathwise at every  $t$ , not only to first order: differentiating (86) requires no Itô correction (unlike a nonlinear change of variables), so  $\bar{e}$  driven by (5) and  $\delta$  driven by (10) are related by the *same* linear map  $T$  at every sample path and every time, for matching initial conditions  $\delta(0) = T\bar{e}(0)$  and identical control, disturbance and Brownian-motion realizations  $(u, \vartheta, W)$ . In particular (87) holds at every  $t$  along every trajectory, so  $\mathbb{E}\|\delta(t)\|^2 \rightarrow 0$  (or remains ultimately bounded by  $\chi$ ) if and only if  $\mathbb{E}\|\bar{e}(t)\|^2 \rightarrow 0$  (respectively, remains ultimately bounded by a constant in  $[\underline{\kappa}^{-2}\chi, \bar{\kappa}^{-2}\chi]$ ); i.e. admissibility of a control law  $u_i \in \Psi(\Omega)$  for (10) in the sense of Definition 1 is equivalent to mean-square stabilization of the original formation error  $\bar{e}_i$  under (5).

*The cost and the saddle point transfer unchanged.* The performance index (13) is, by definition, a functional of  $\delta_i(\cdot)$  and of the *same* input signals  $u_i, u_{-i}, \vartheta_i, \vartheta_{-i}$  that drive both (5) and (10);  $T$  acts only on the state and does not touch the input channels  $g_i, k_i$ , which enter (10) exactly as they enter (5) (11) after the same computation used above. Consequently, for any admissible inputs,  $J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$  evaluated along a trajectory of (10) equals  $J_i(T\bar{e}_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$  evaluated along the *corresponding* trajectory of (5) (same inputs, same noise, initial conditions related by  $\delta_i(0) = (T\bar{e}(0))_i$ ), because the two trajectories coincide under  $T$  by the previous paragraph. The zero-sum game  $\min_{u_i} \max_{\vartheta_i} J_i$  therefore has the same value and the same saddle-point input trajectories  $(u_i^*, \vartheta_i^*)$  whether posed on (10) or, after composing with the fixed invertible map  $T$ , on (5).

Combining the three paragraphs above: a policy solves the distributed  $H_\infty$  optimal control problem of Section 3 for (10) if and only if it solves, in the mean-square sense, the distributed  $H_\infty$  optimal formation tracking problem for (5), which is the claim of Lemma 1.  $\square$